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(54) **COMPOSITION FOR INDUCING
PROLIFERATION OR ACCUMULATION OF
REGULATORY T CELLS**

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None

See application file for complete search history.

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(57) **ABSTRACT**

It was found that bacteria belonging to the genus
Clostridium induce accumulation of regulatory T cells (Treg
cells) in the colon. Moreover, the present inventors found
that regulatory T cells (Treg cells) induced by from these
bacteria suppressed proliferation of effector T-cells. From
these findings, the present inventors found that the use of
bacteria belonging to the genus *Clostridium* or a physiologi-
cally active substance derived therefrom made it possible to
induce proliferation or accumulation of regulatory T cells
(Treg cells), and further to suppress immune functions.

20 Claims, 41 Drawing Sheets

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FIG.1

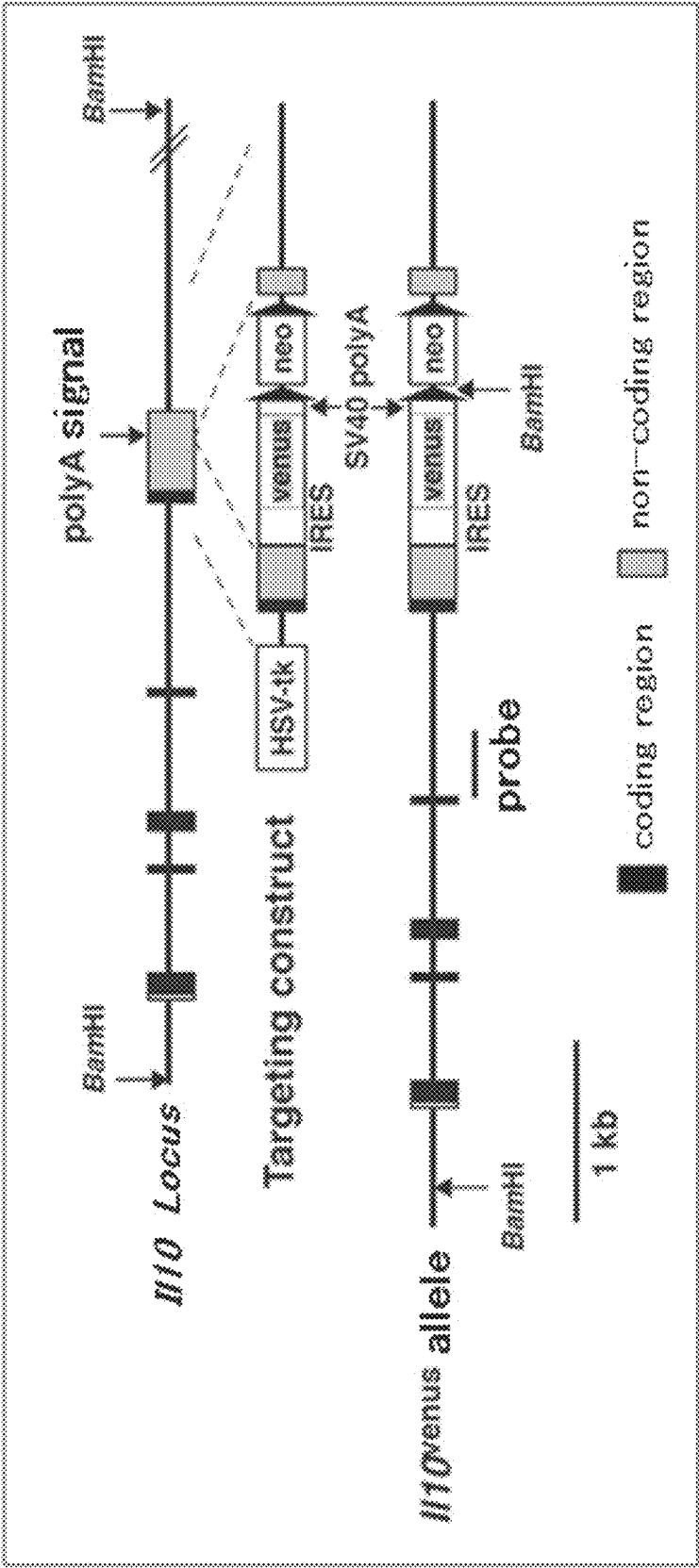


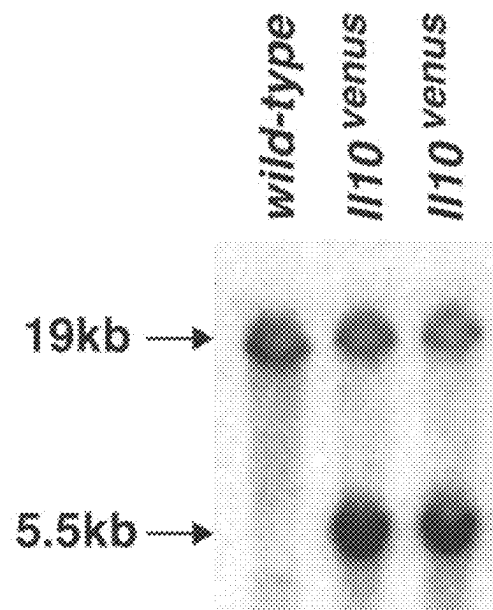
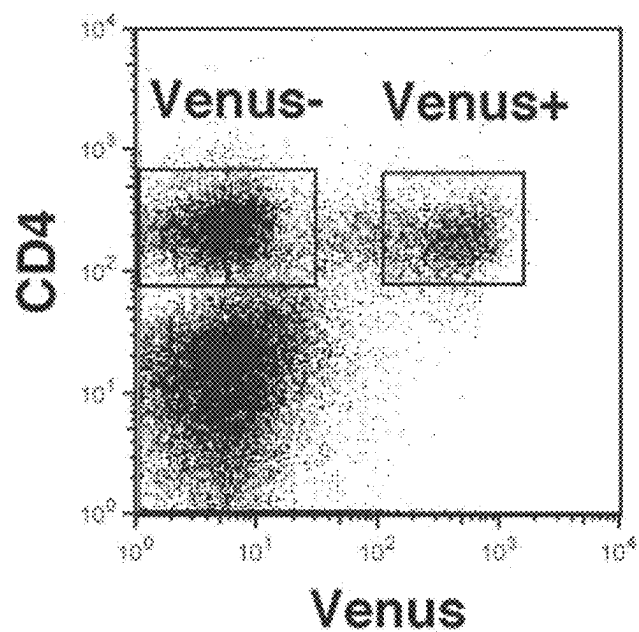
FIG. 2**FIG. 3**

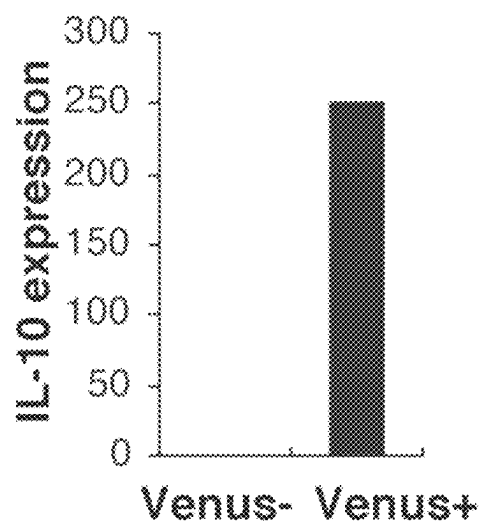
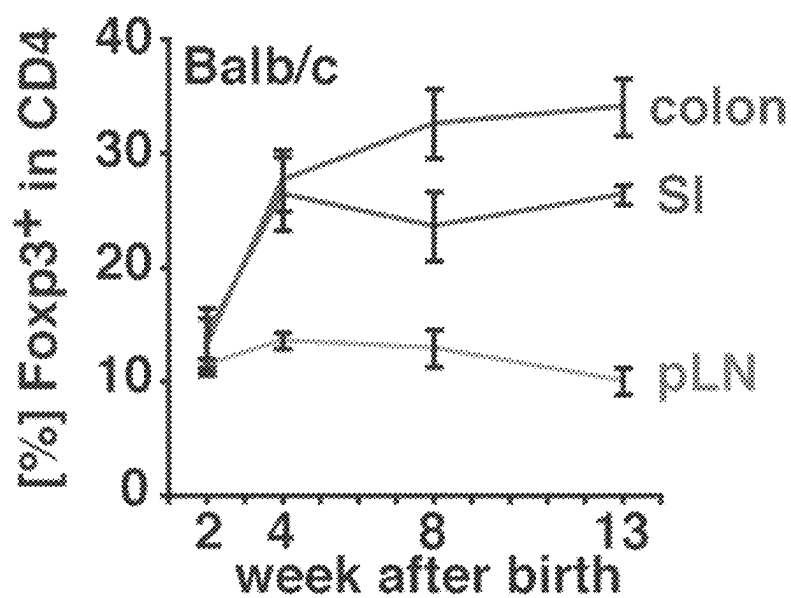
FIG. 4**FIG. 5**

FIG. 6

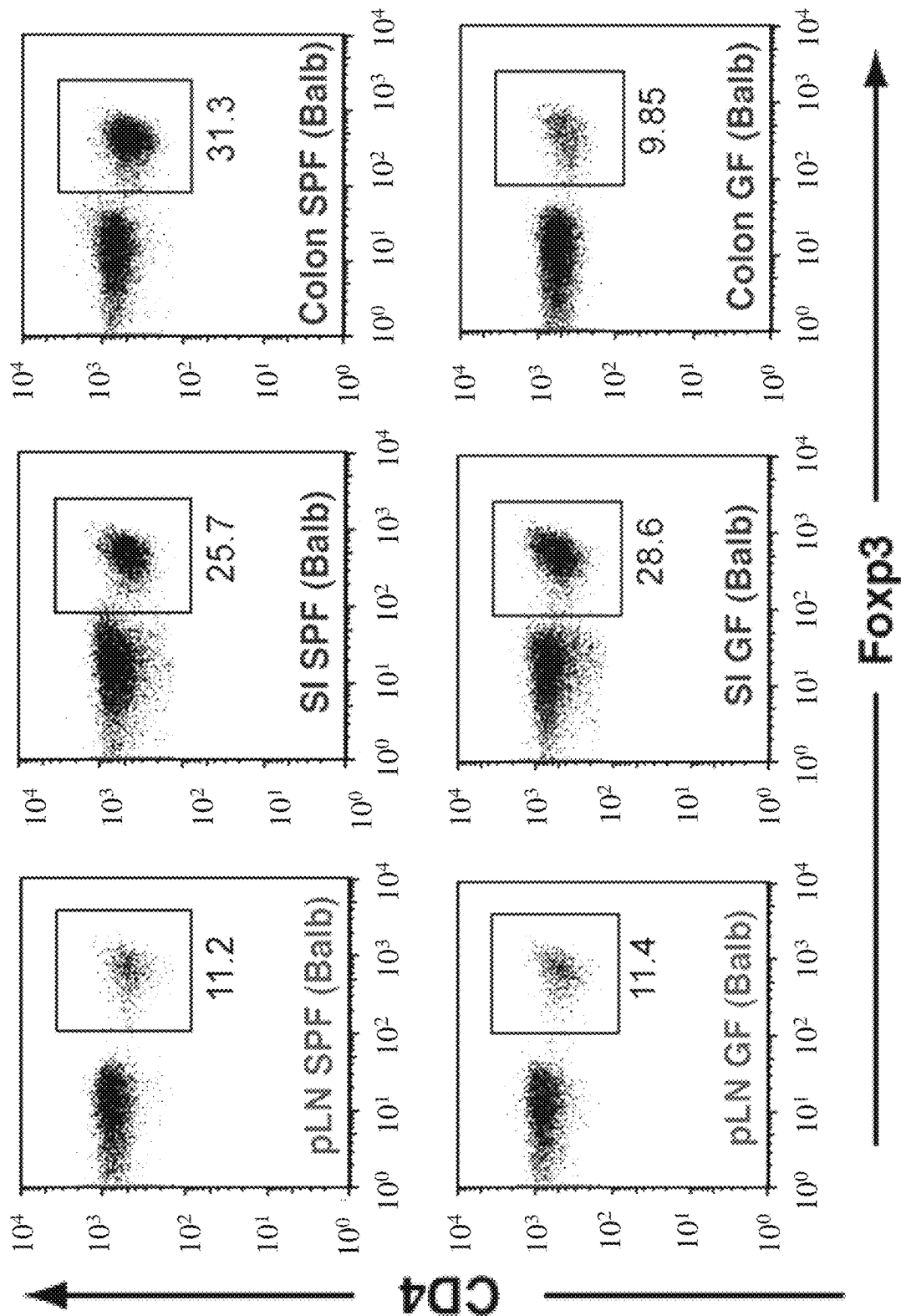


FIG. 7

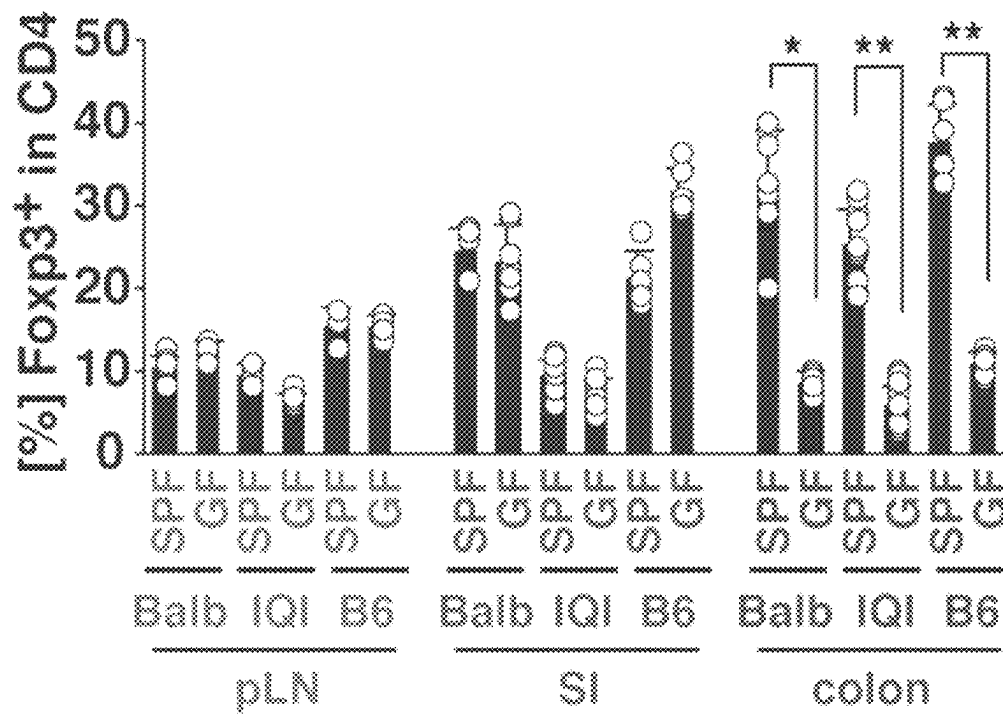


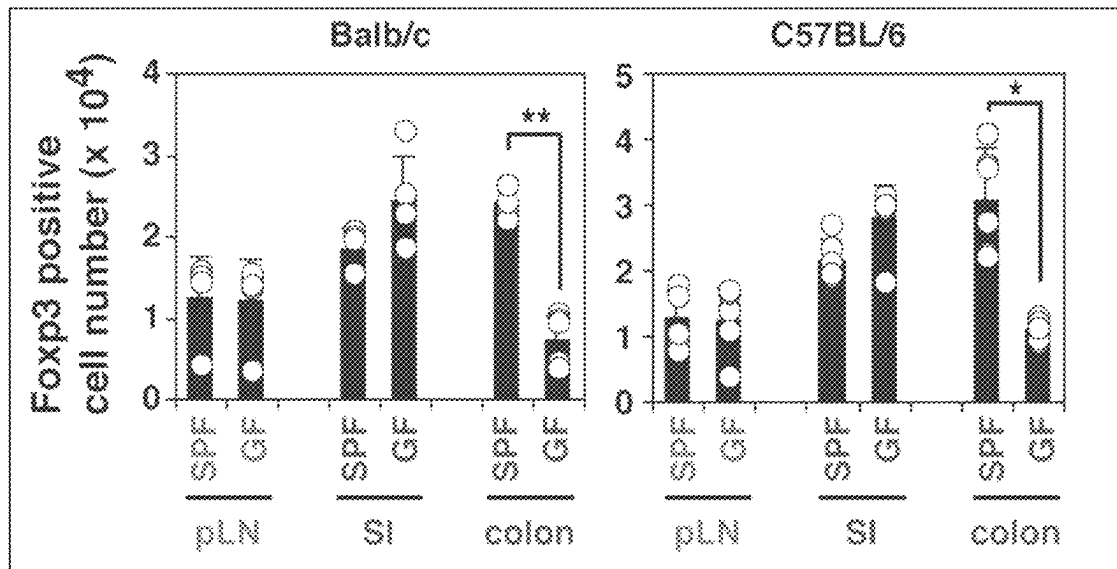
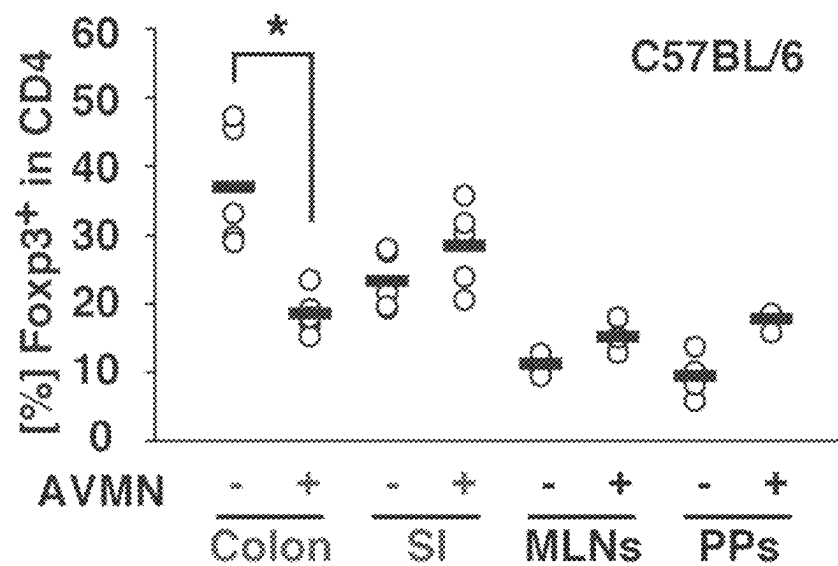
FIG. 8**FIG. 9**

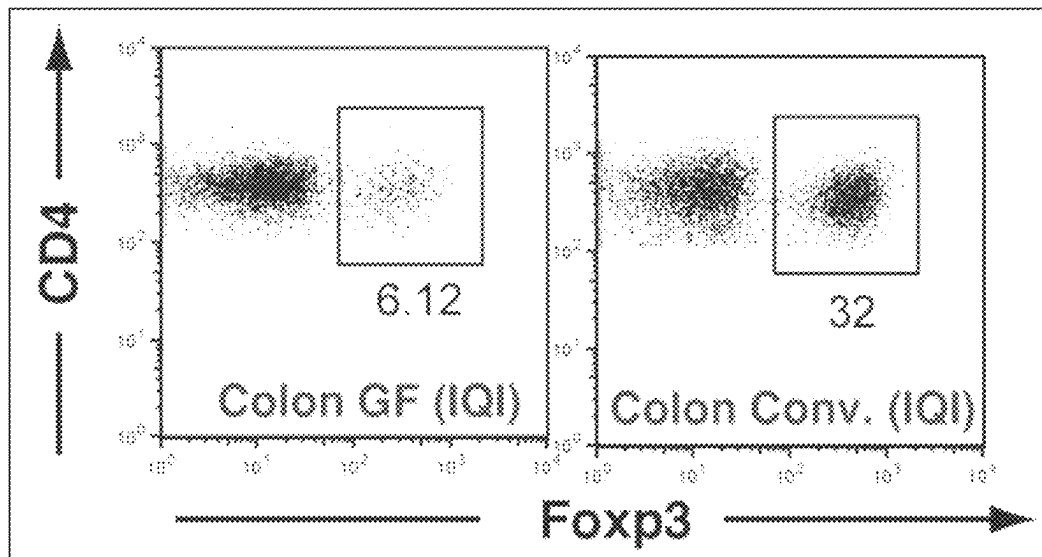
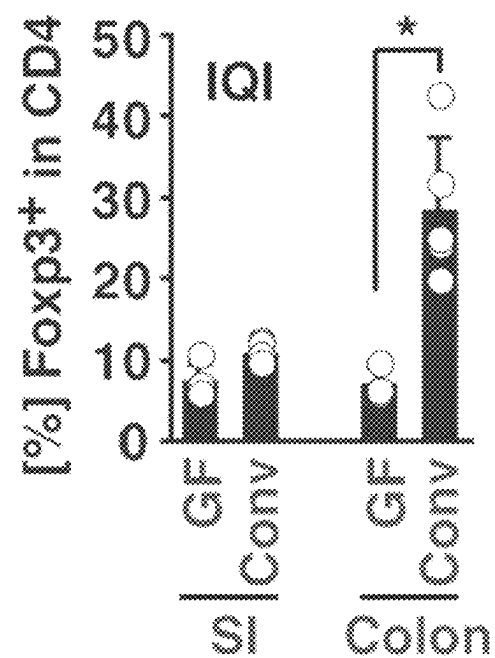
FIG. 10**FIG. 11**

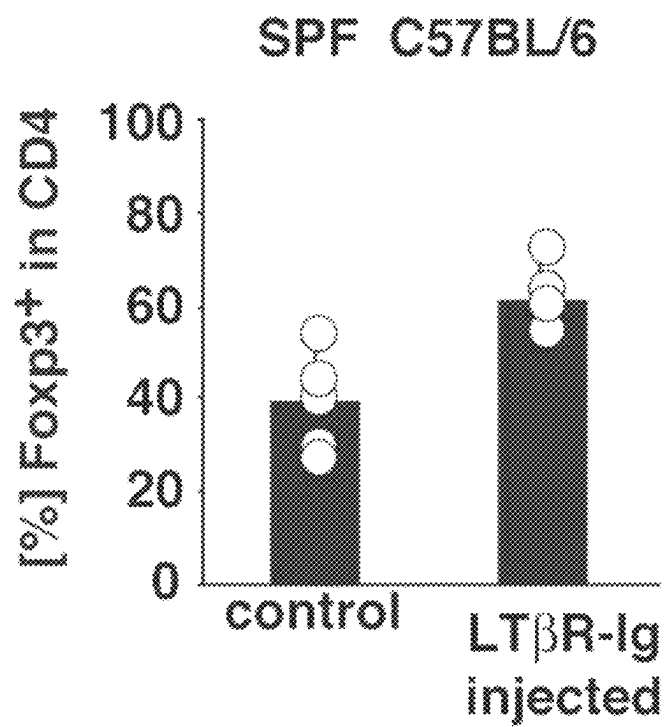
FIG. 12

FIG. 13

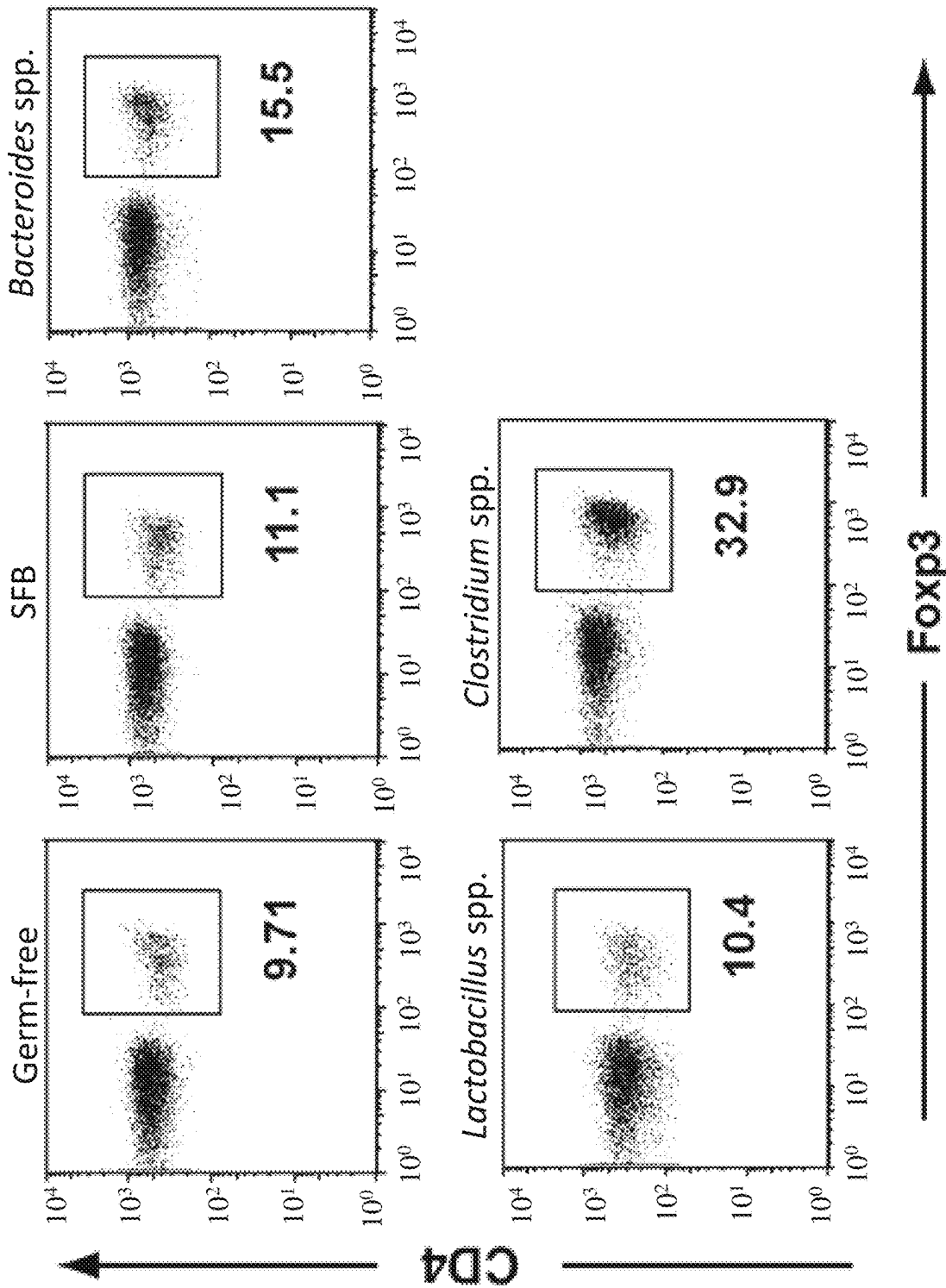


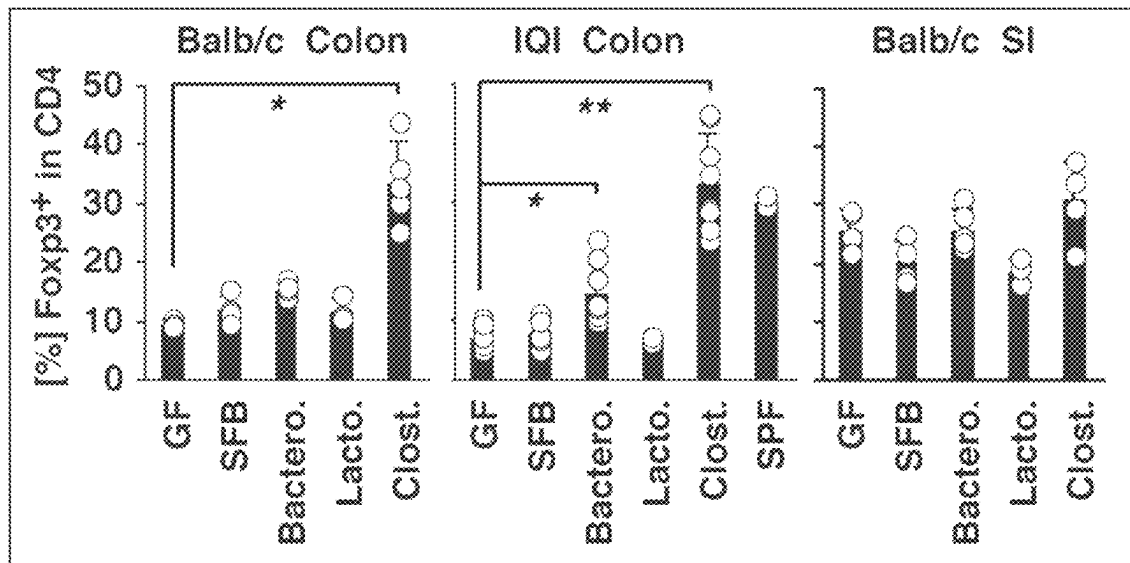
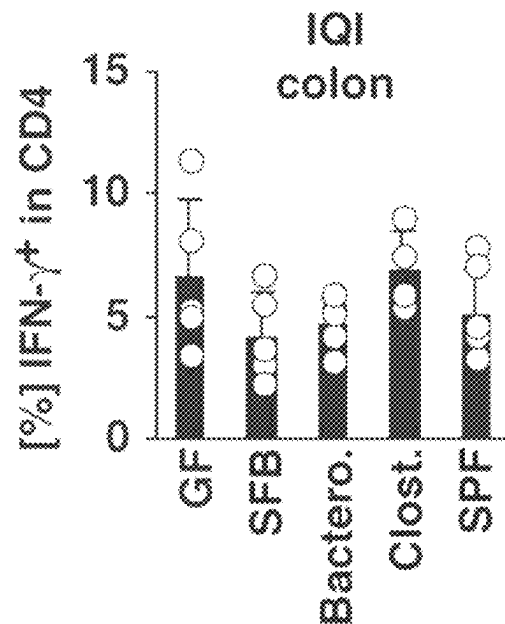
FIG. 14**FIG. 15**

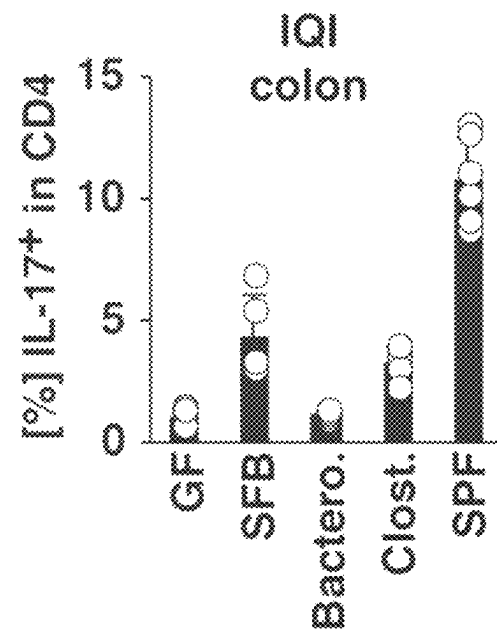
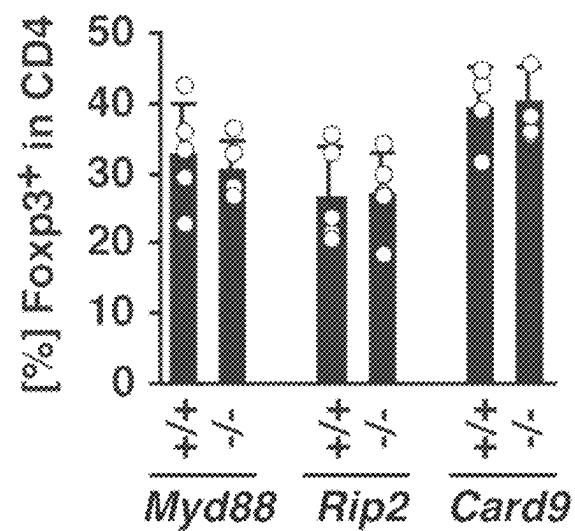
FIG. 16**FIG. 17**

FIG. 18

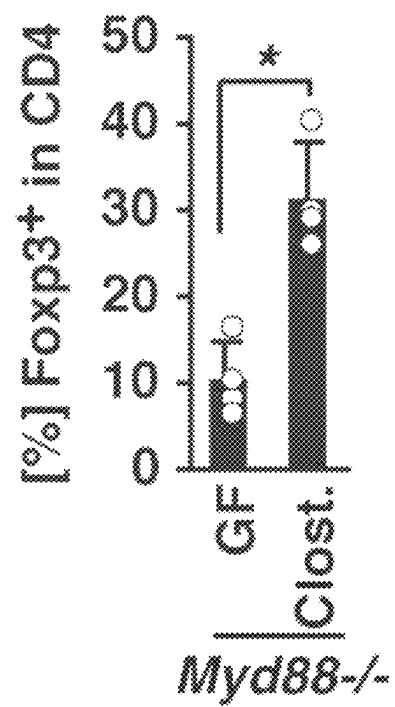


FIG. 19

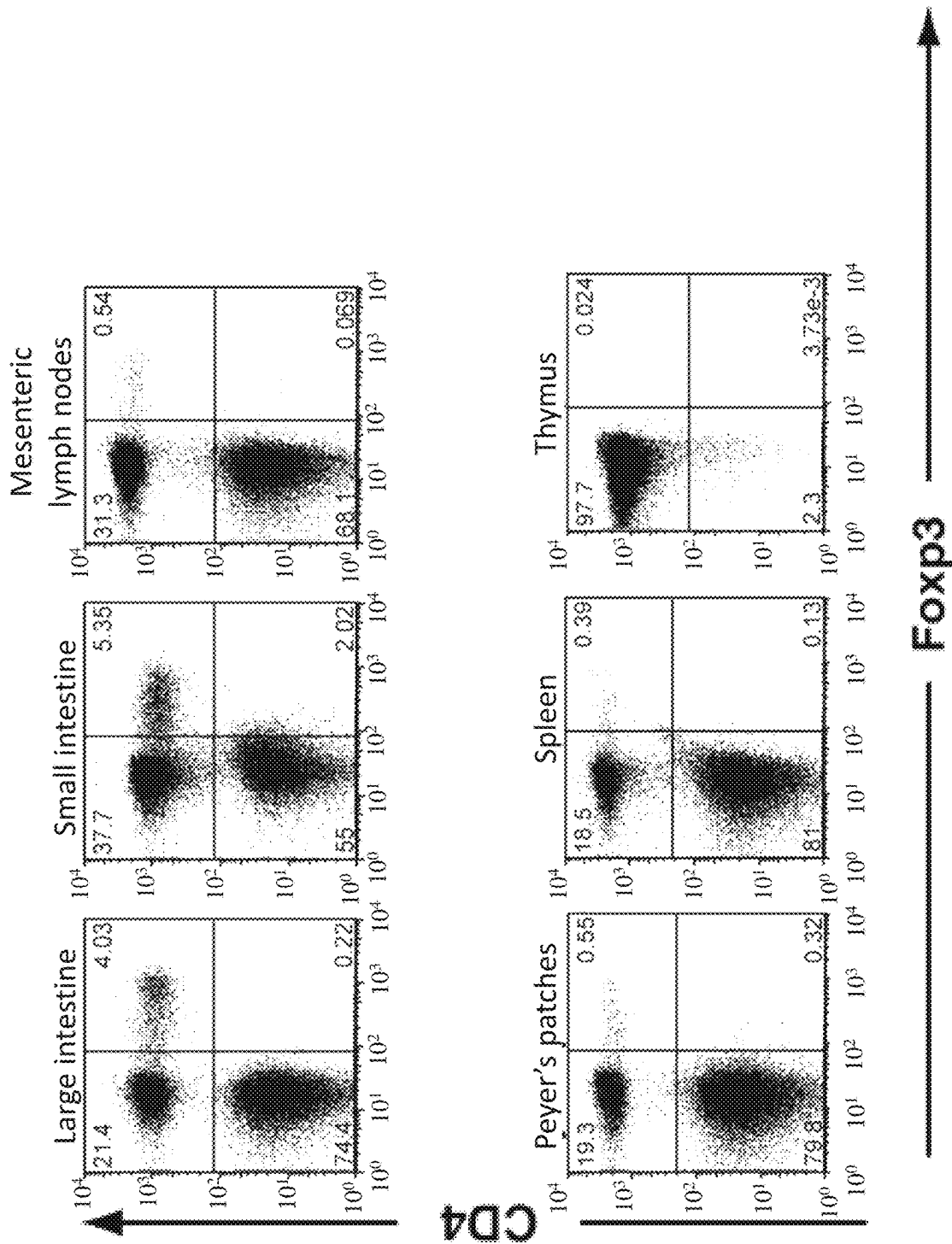


FIG. 19 cont.

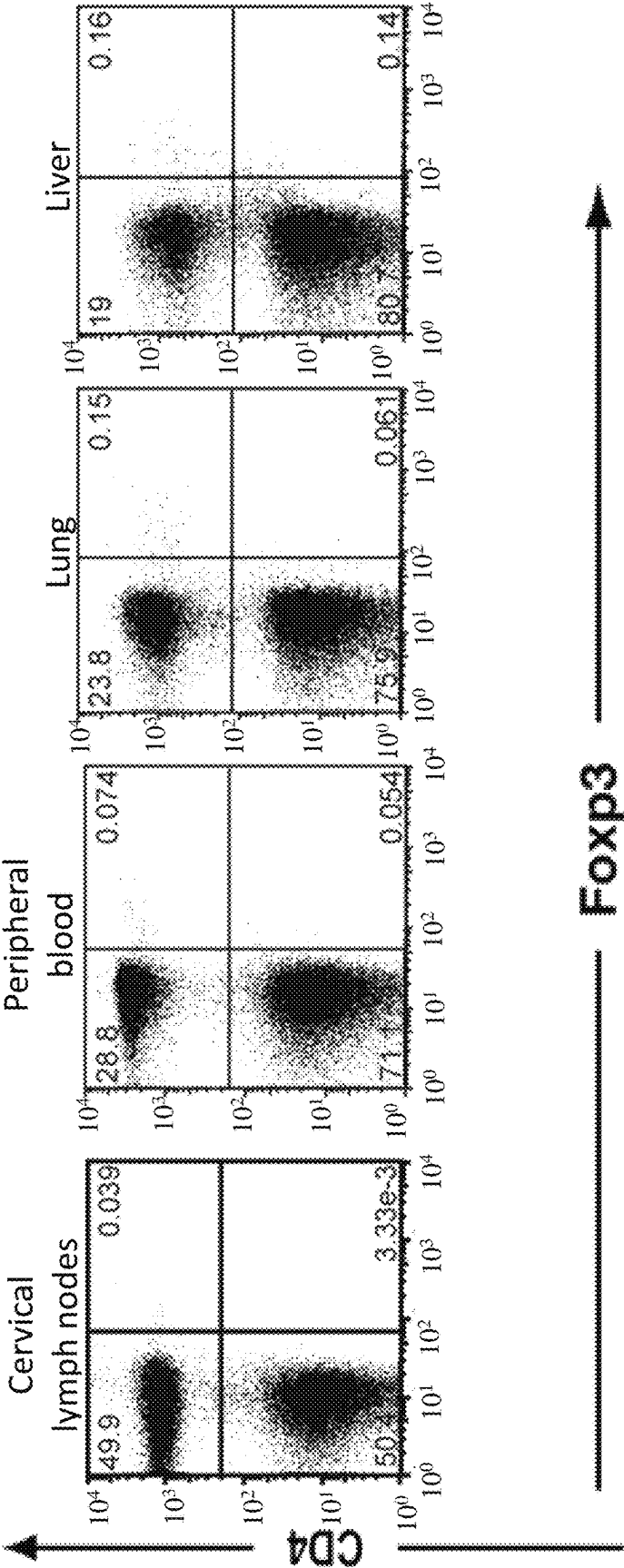


FIG. 20

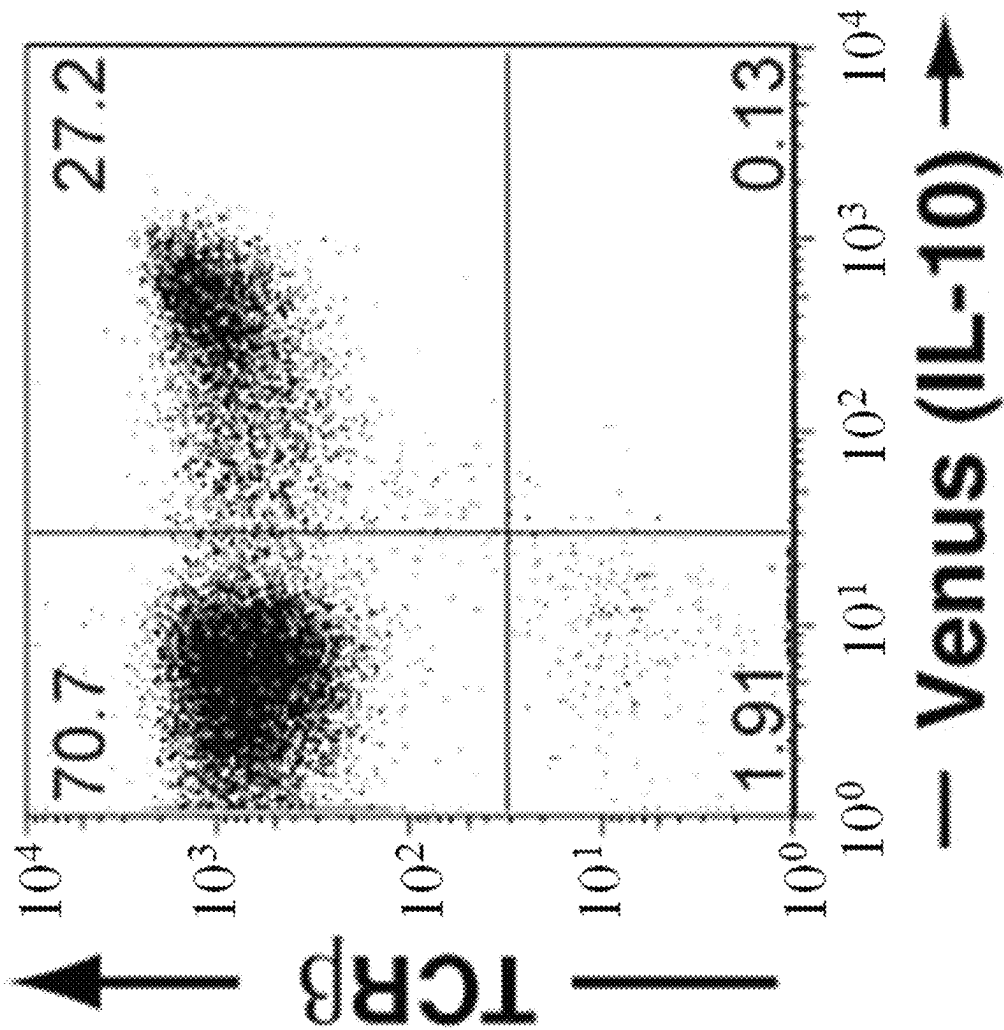


FIG. 21

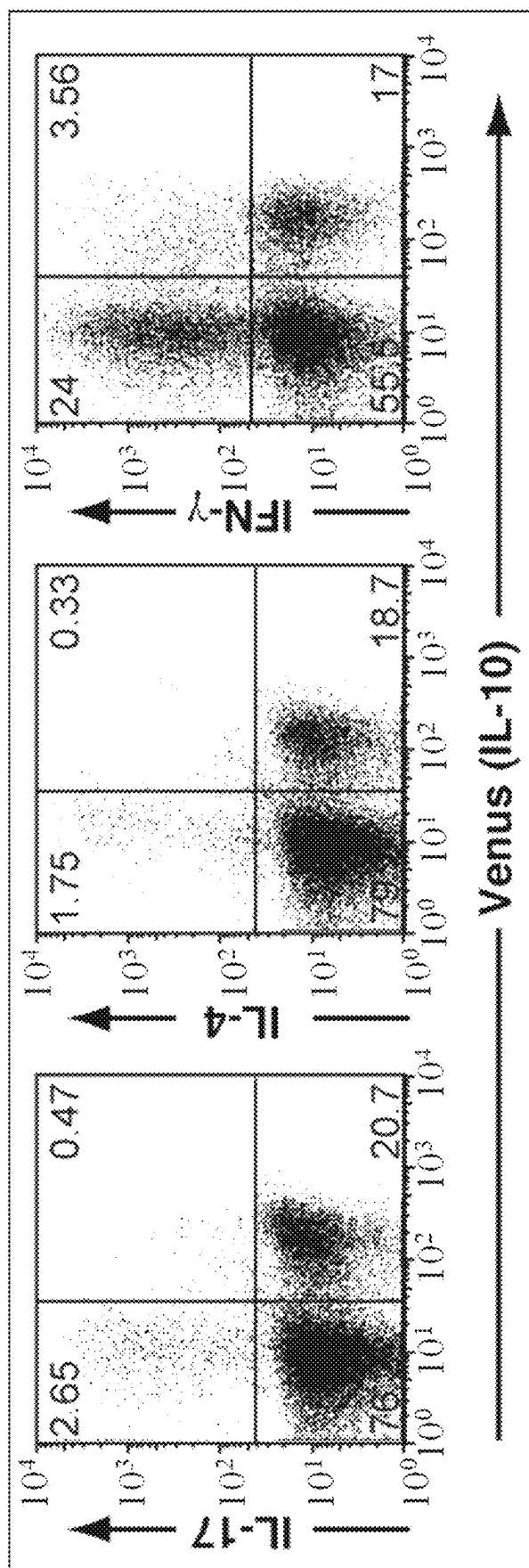


FIG. 22

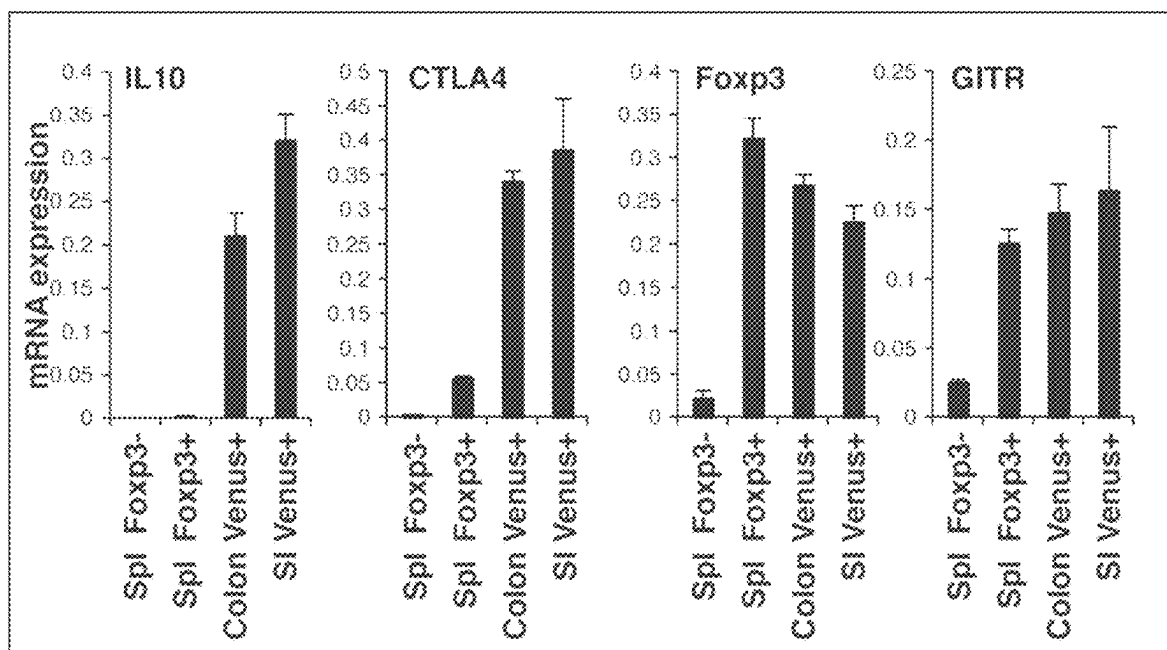


FIG. 23

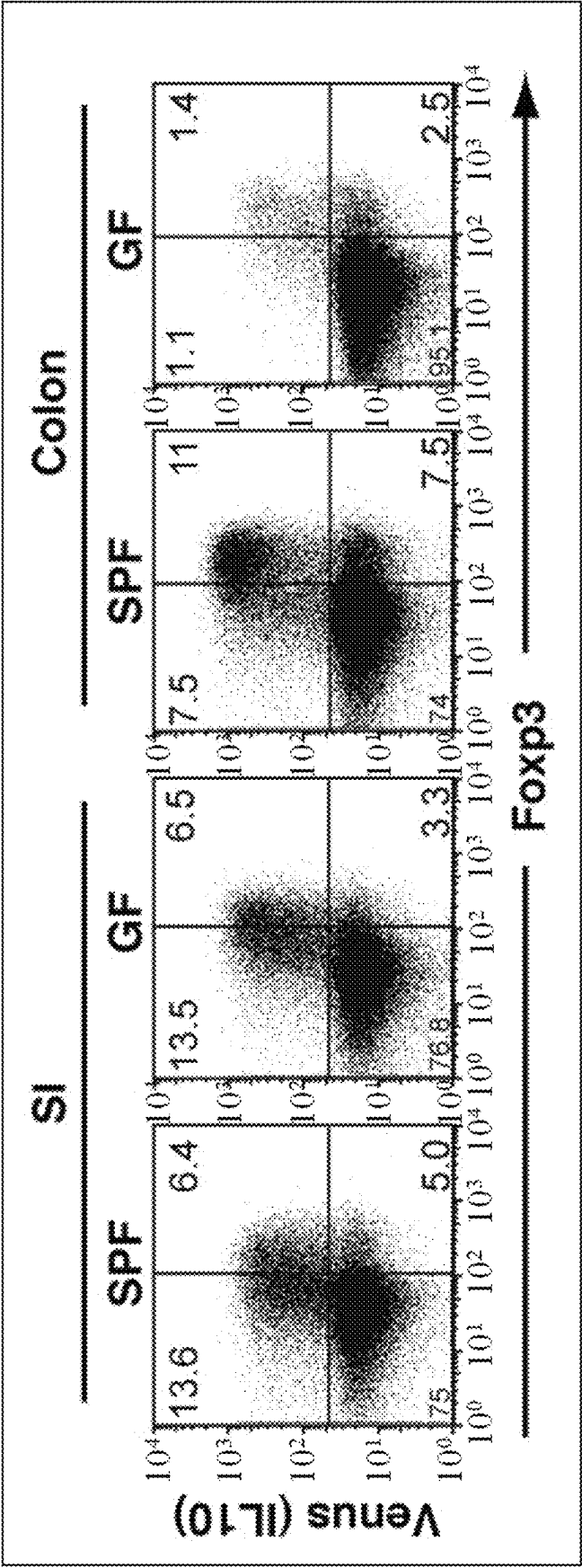


FIG. 24

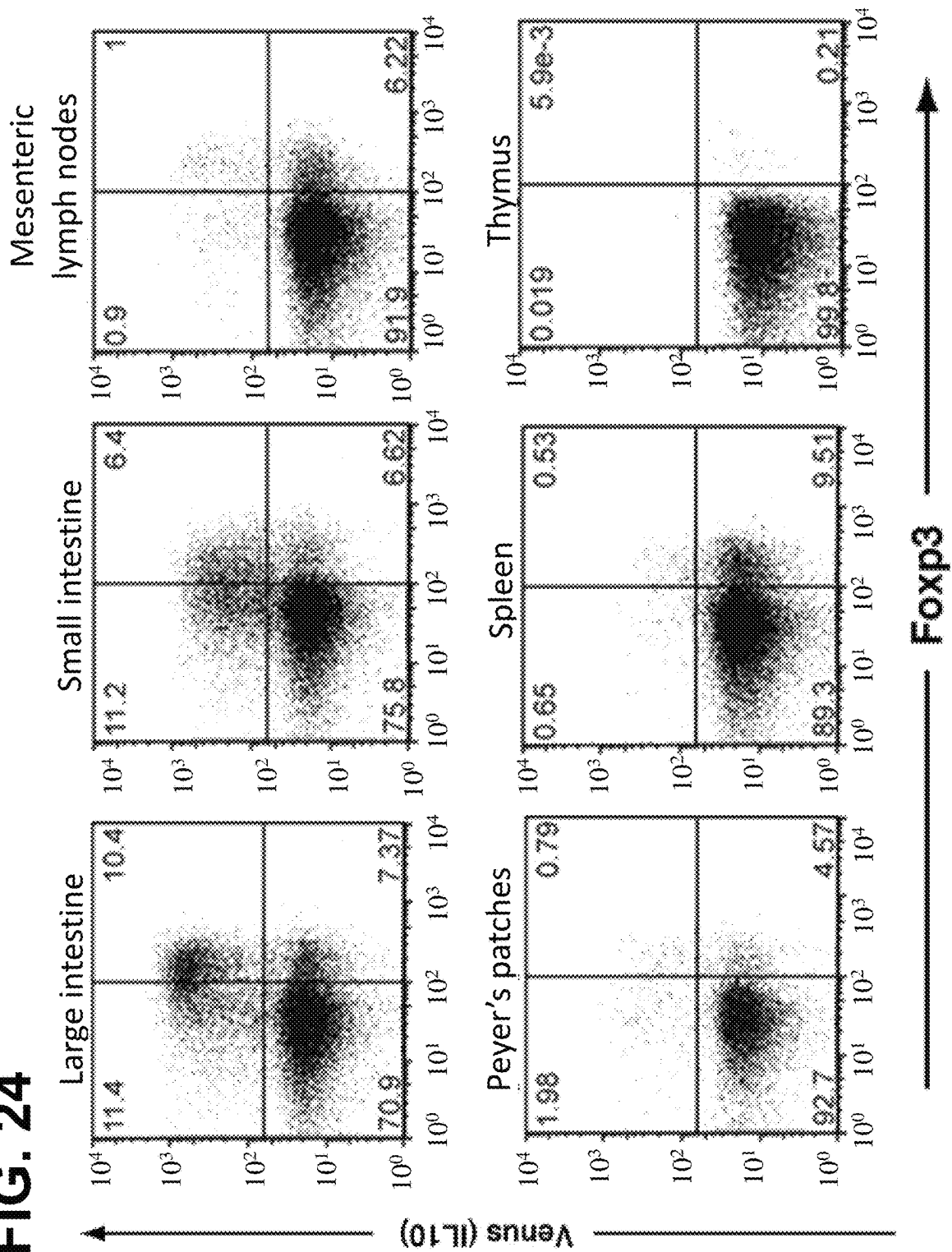


FIG. 24 cont.

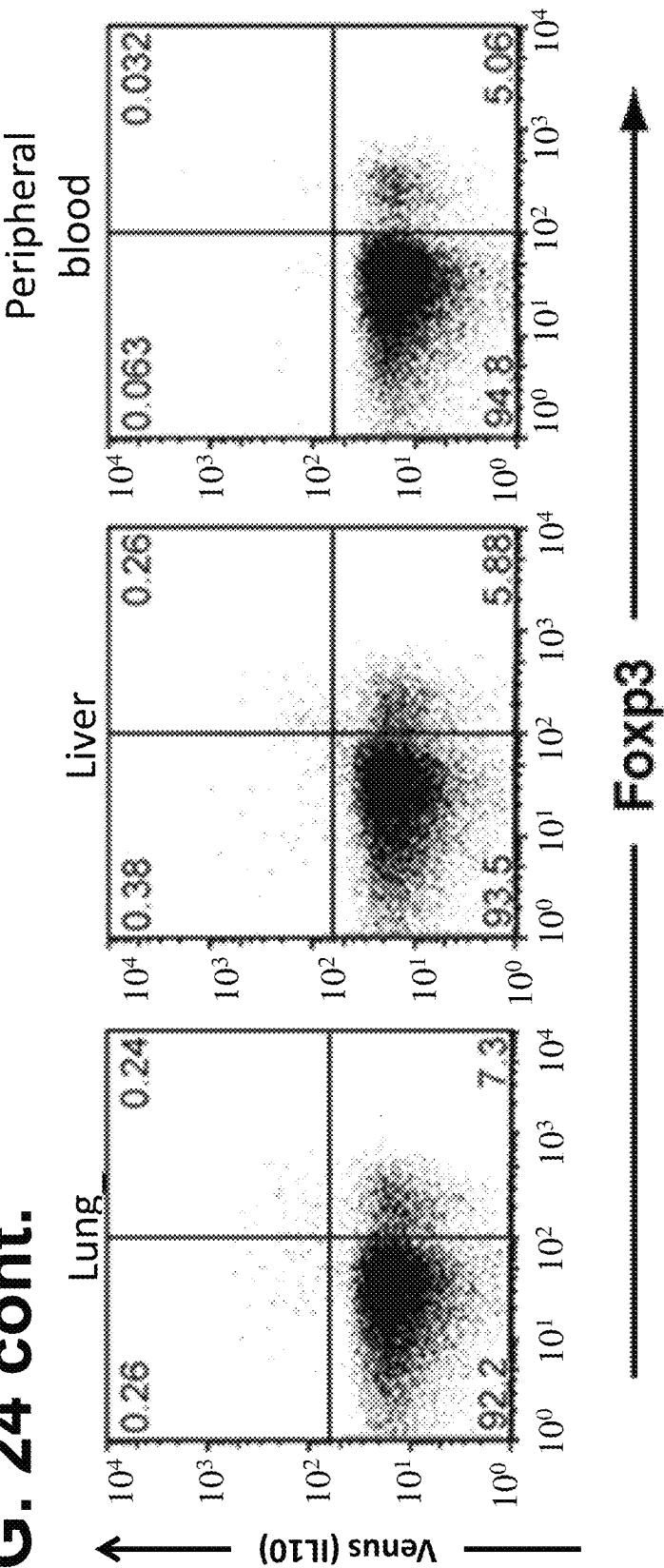


FIG. 25

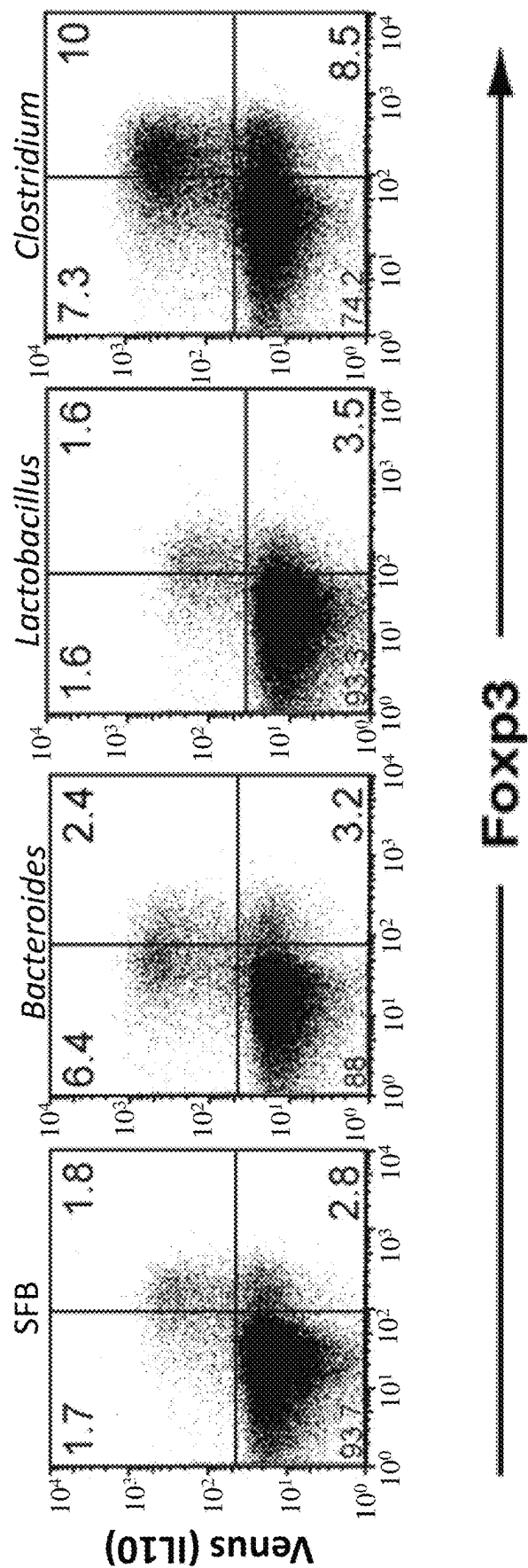


FIG. 26

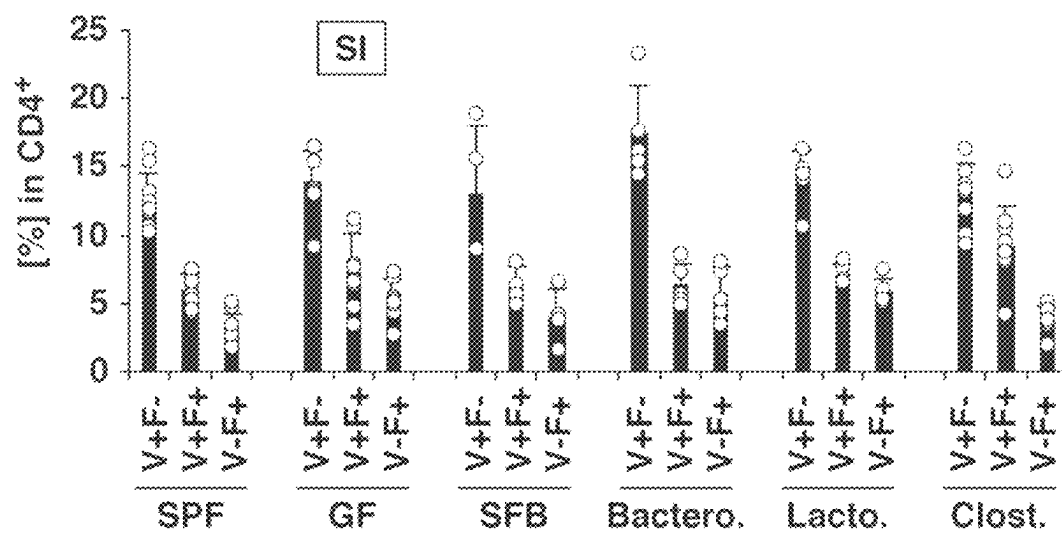


FIG. 27

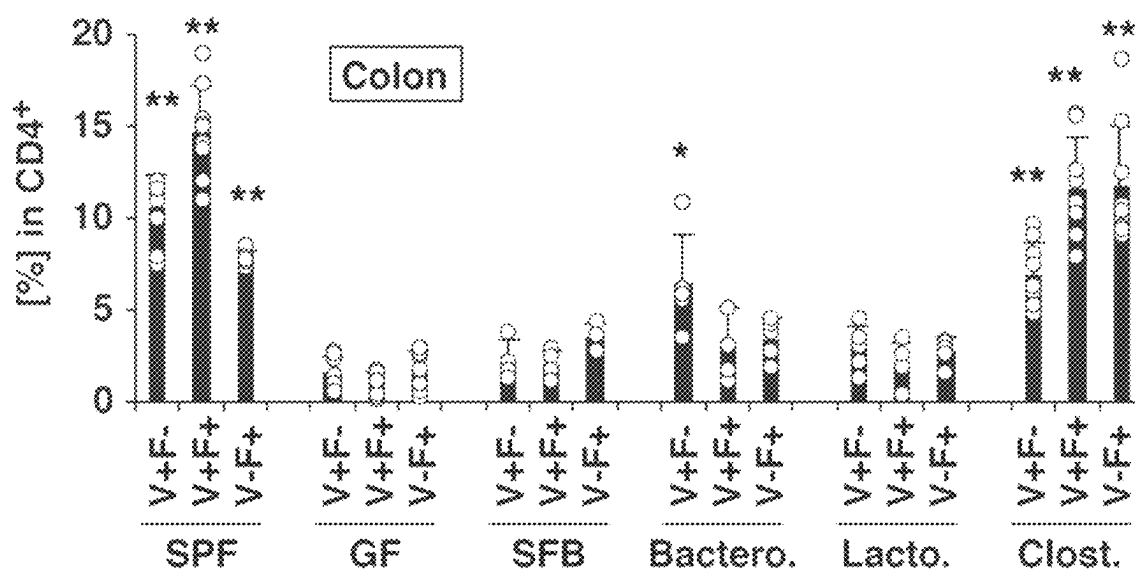


FIG. 28

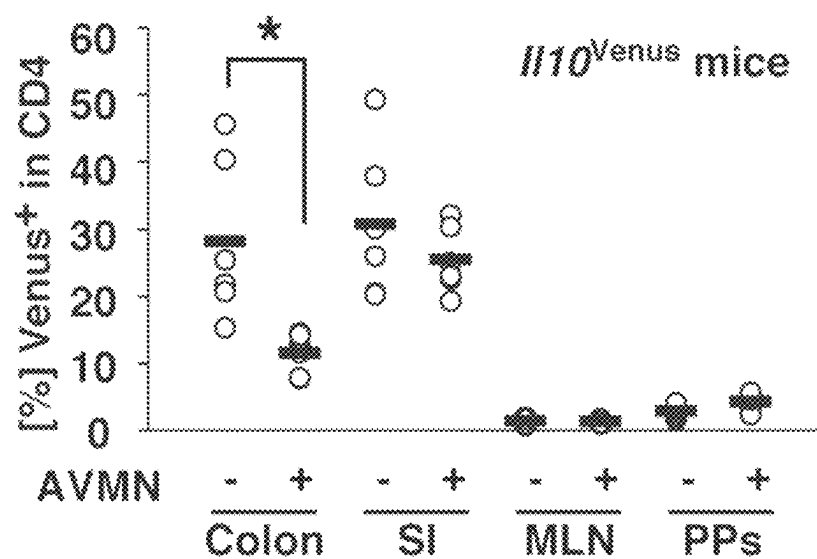


FIG. 29

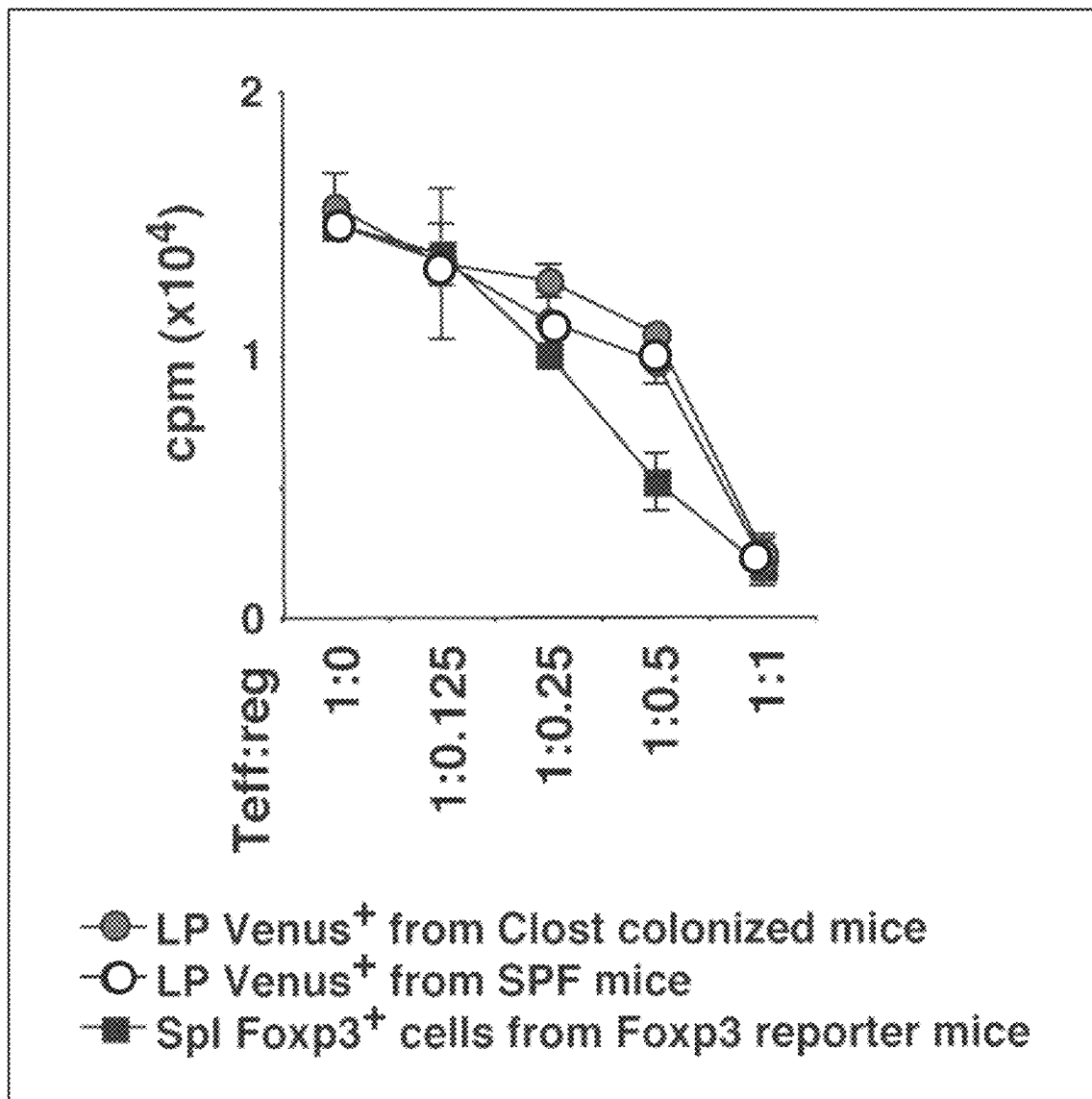


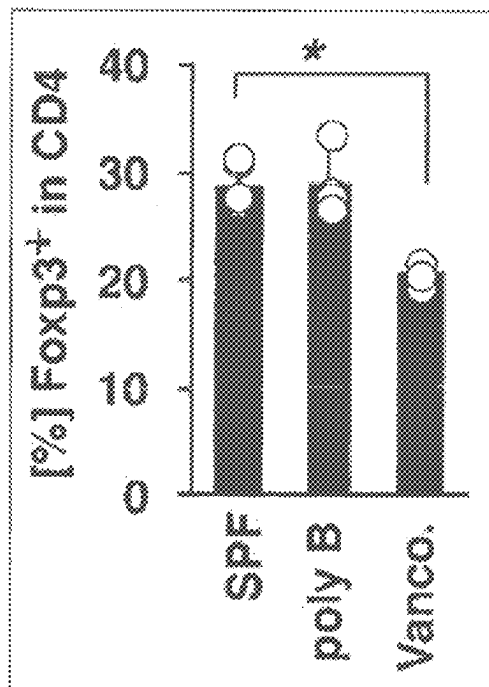
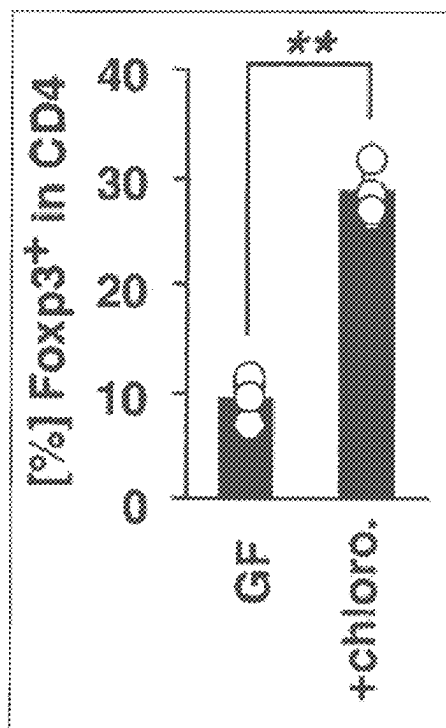
FIG. 30**FIG. 31**

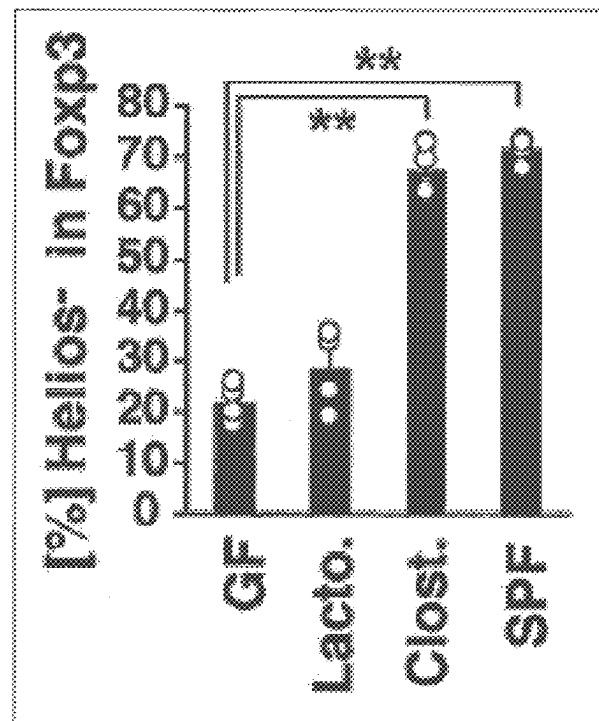
FIG. 32

FIG. 33

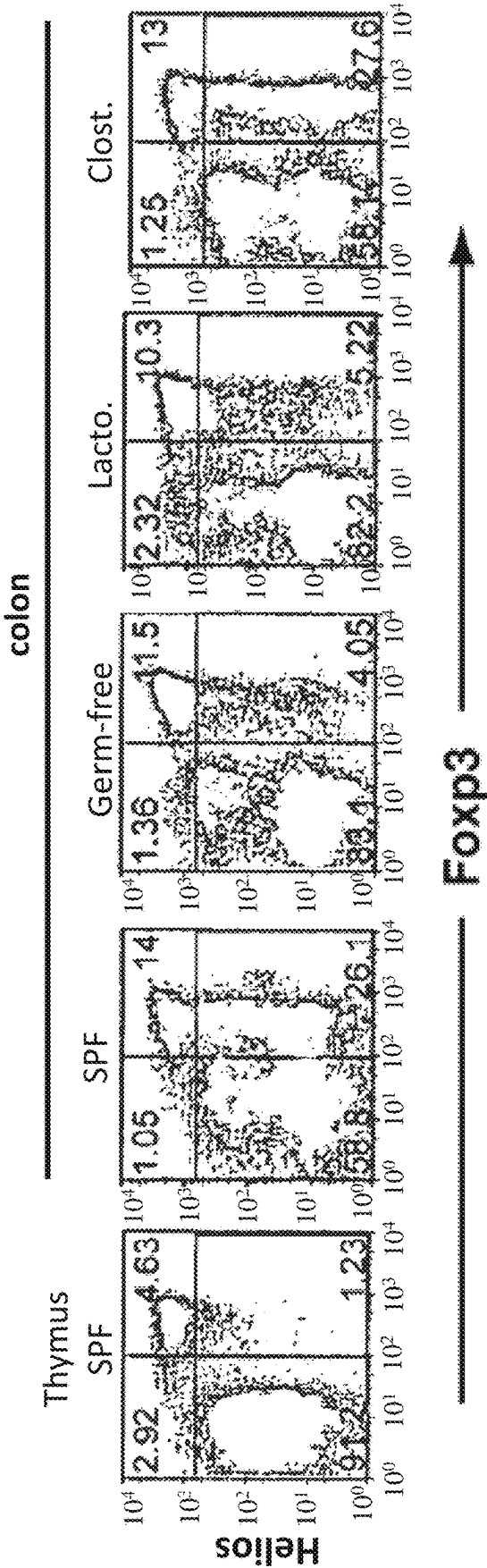


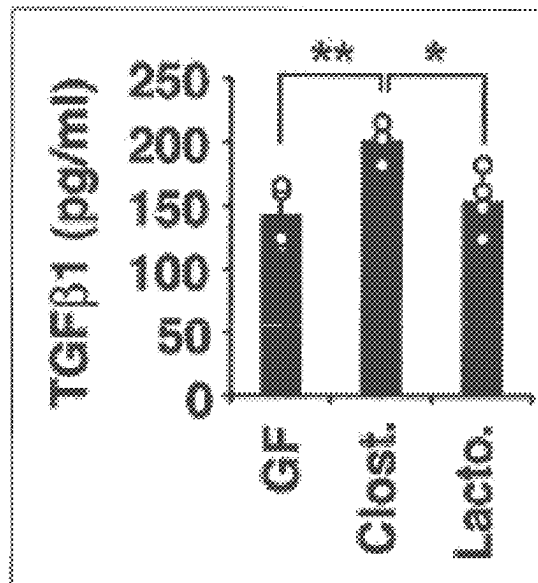
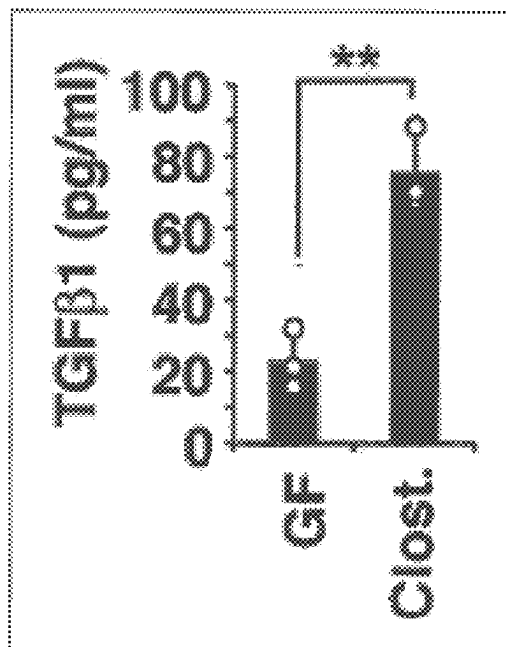
FIG. 34**FIG. 35**

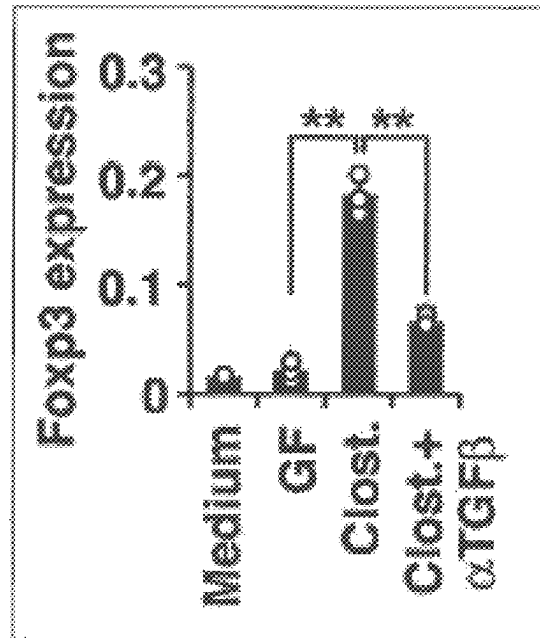
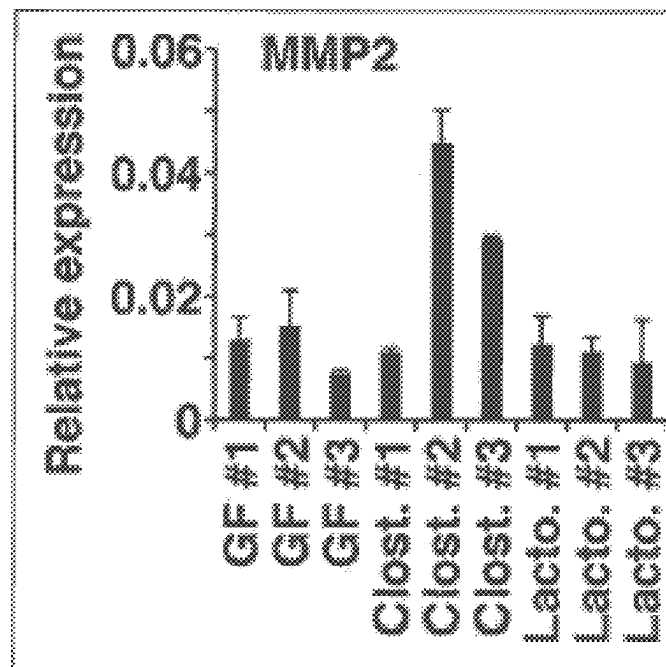
FIG. 36**FIG. 37**

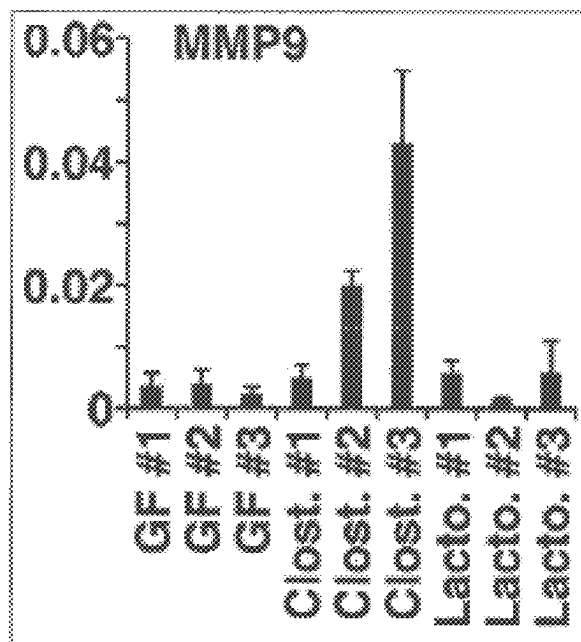
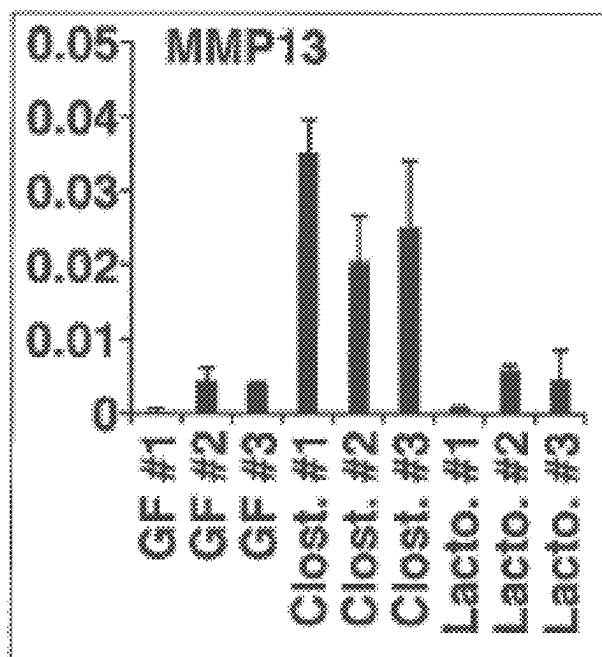
FIG. 38**FIG. 39**

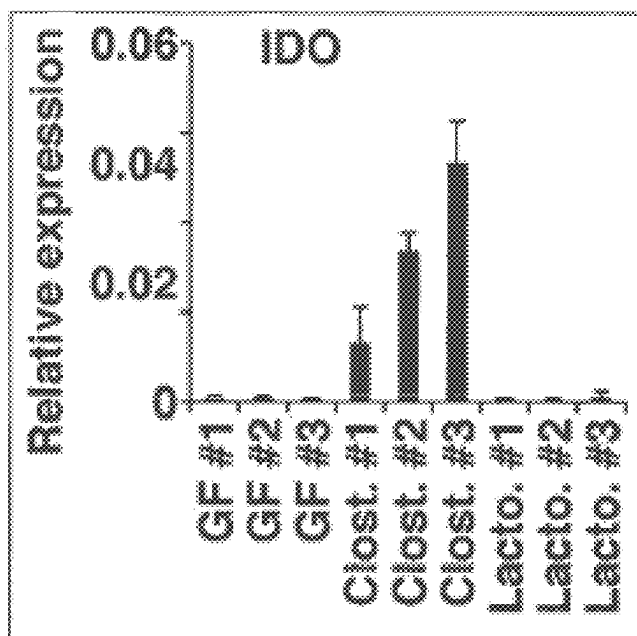
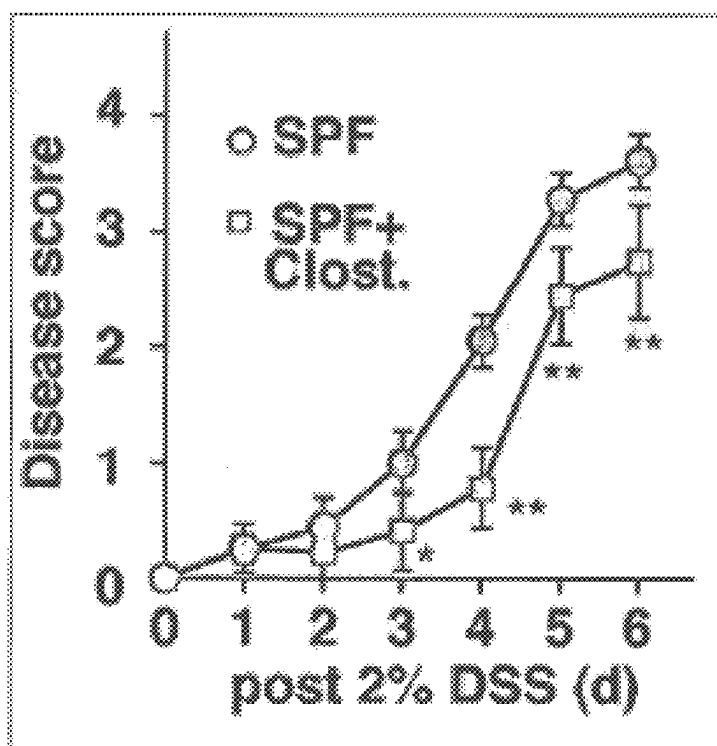
FIG. 40**FIG. 41**

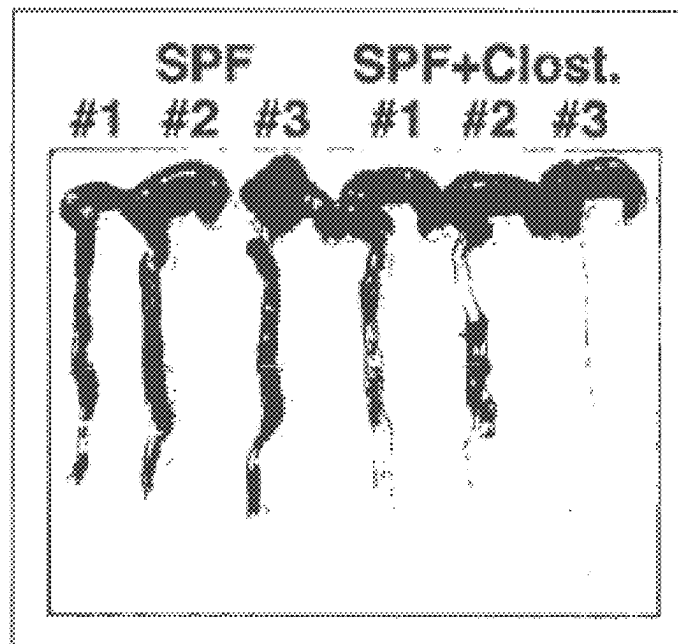
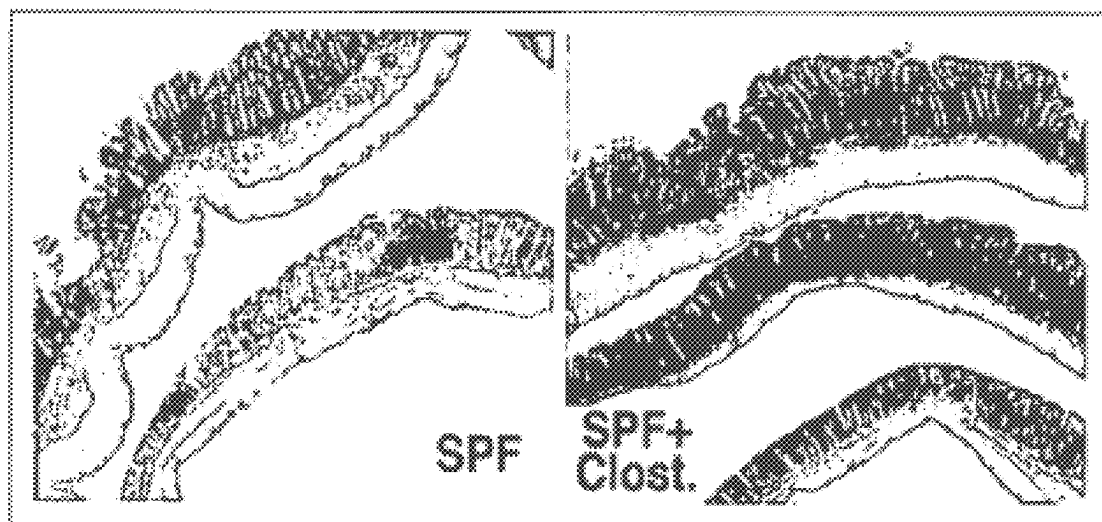
FIG. 42**FIG. 43**

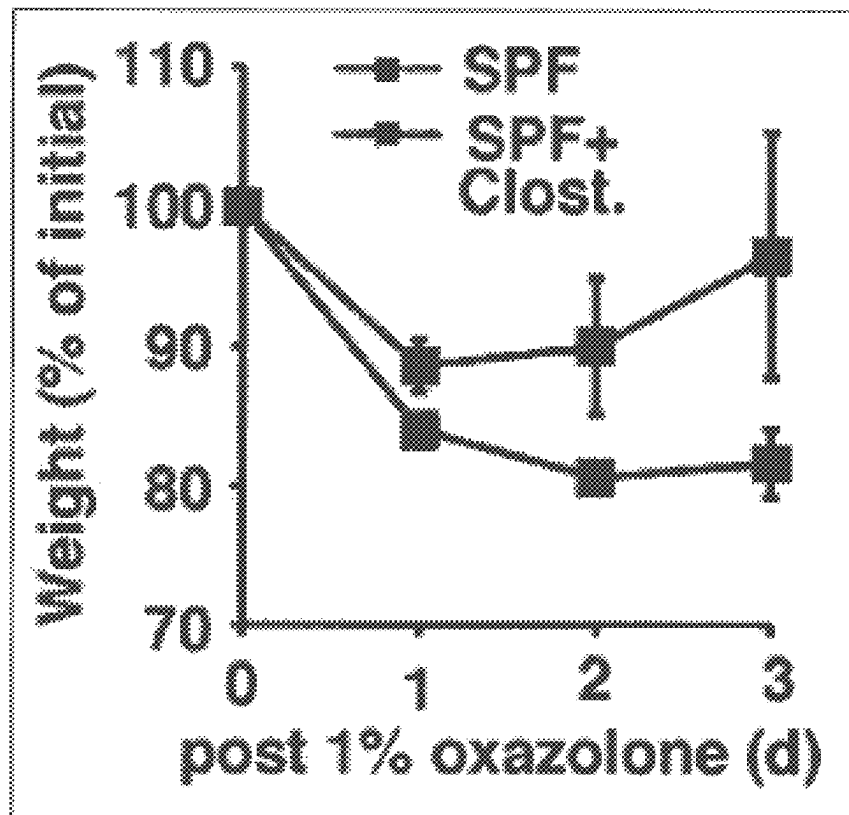
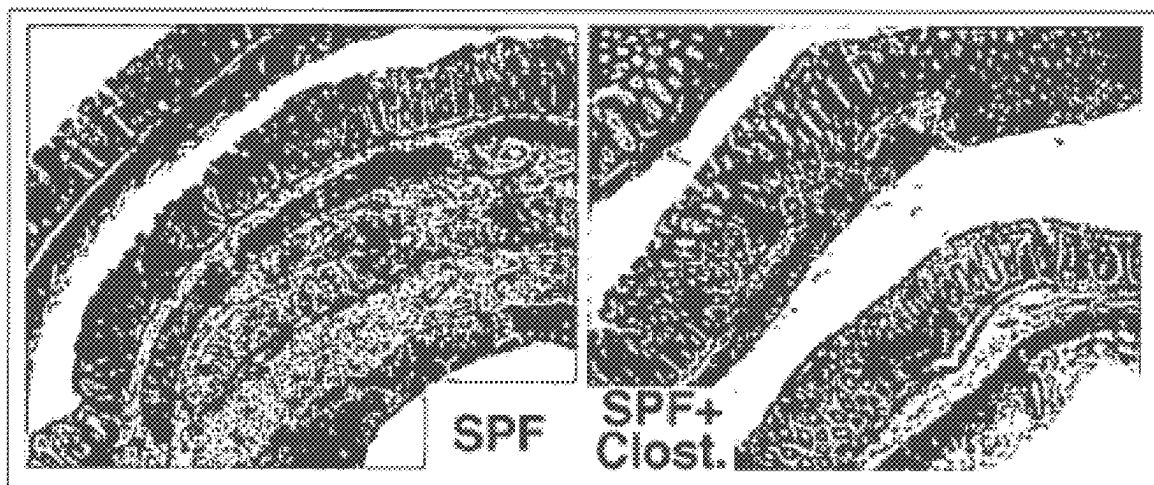
FIG. 44**FIG. 45**

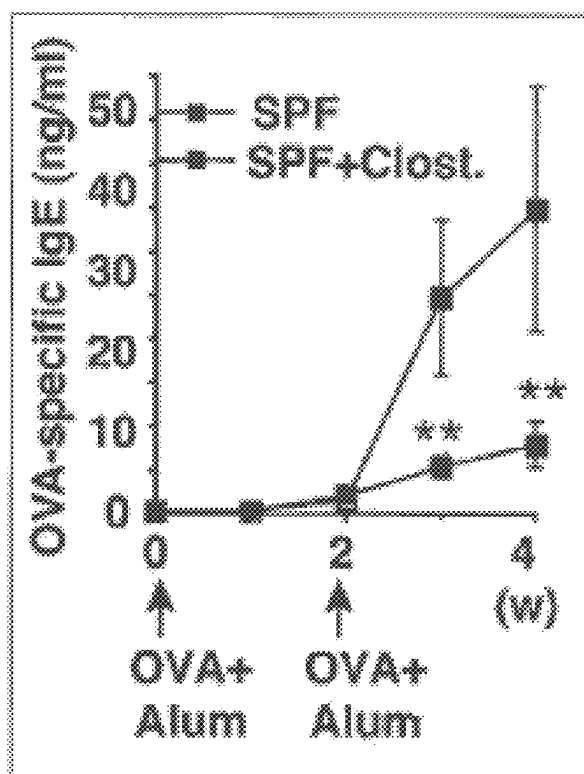
FIG. 46

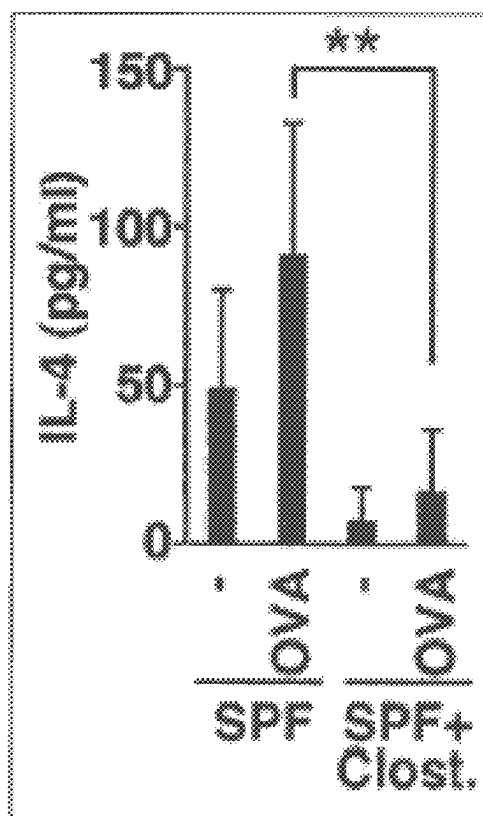
FIG. 47

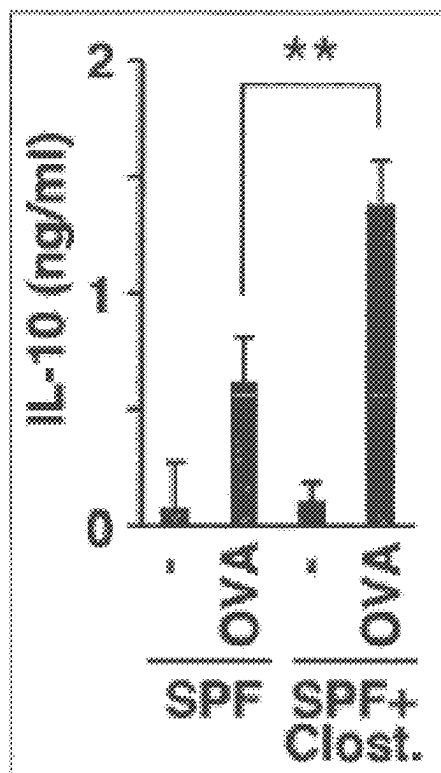
FIG. 48

FIG. 49

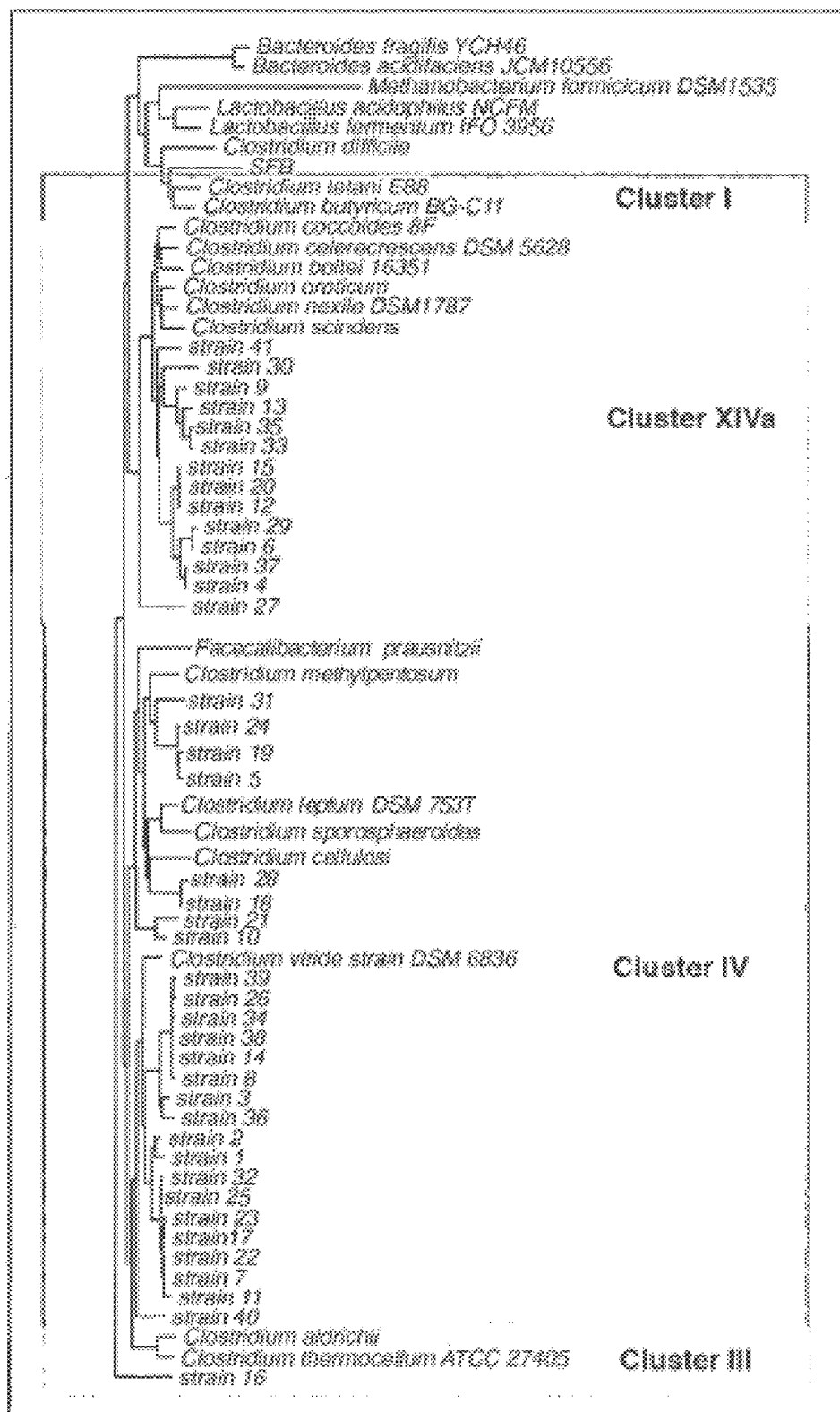


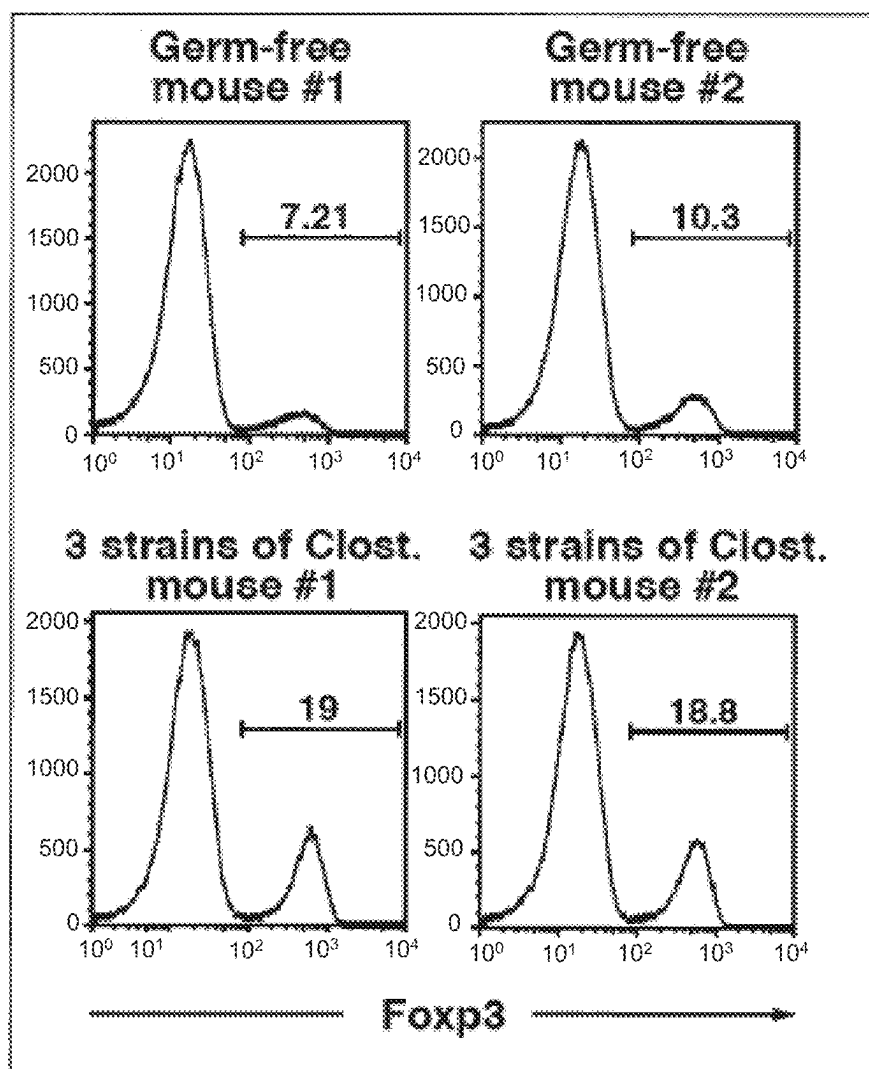
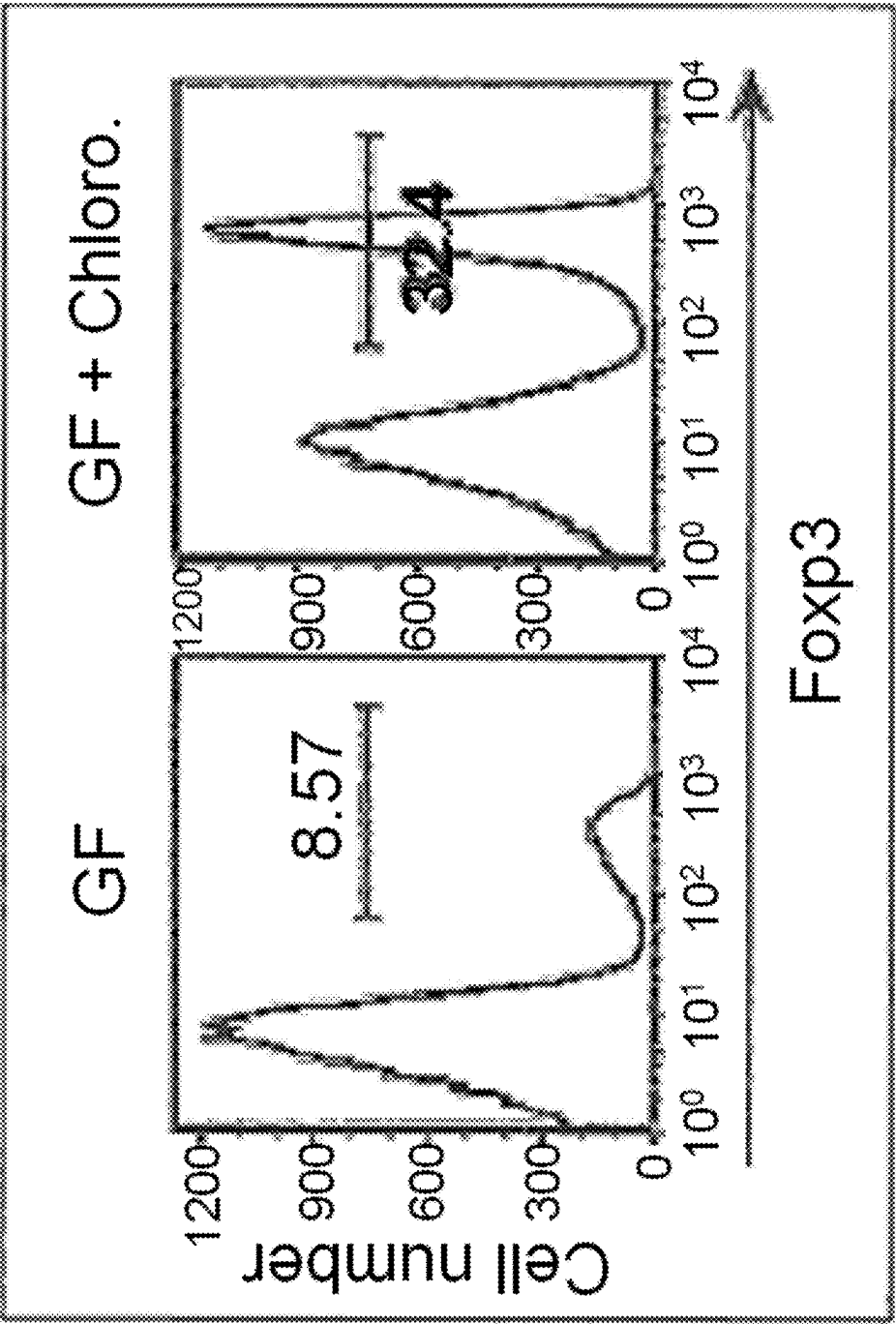
FIG. 50

FIG. 51



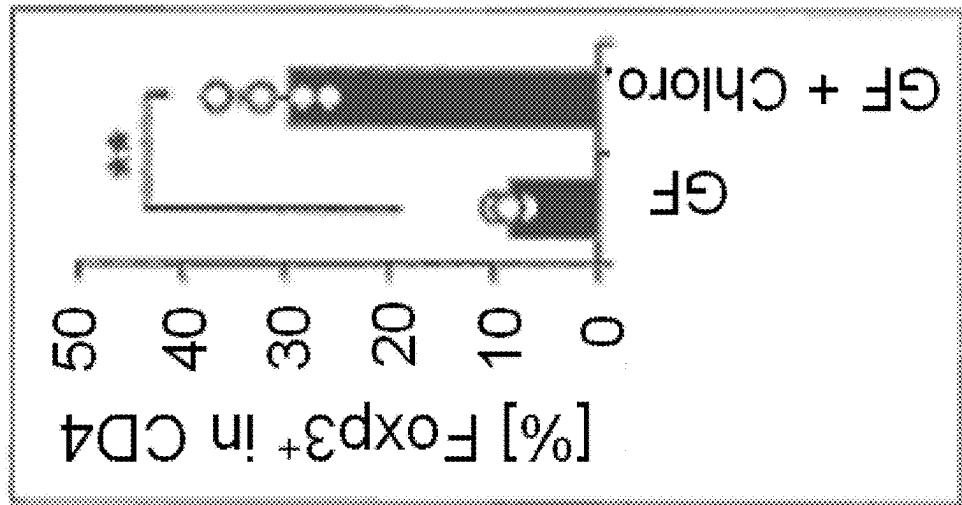
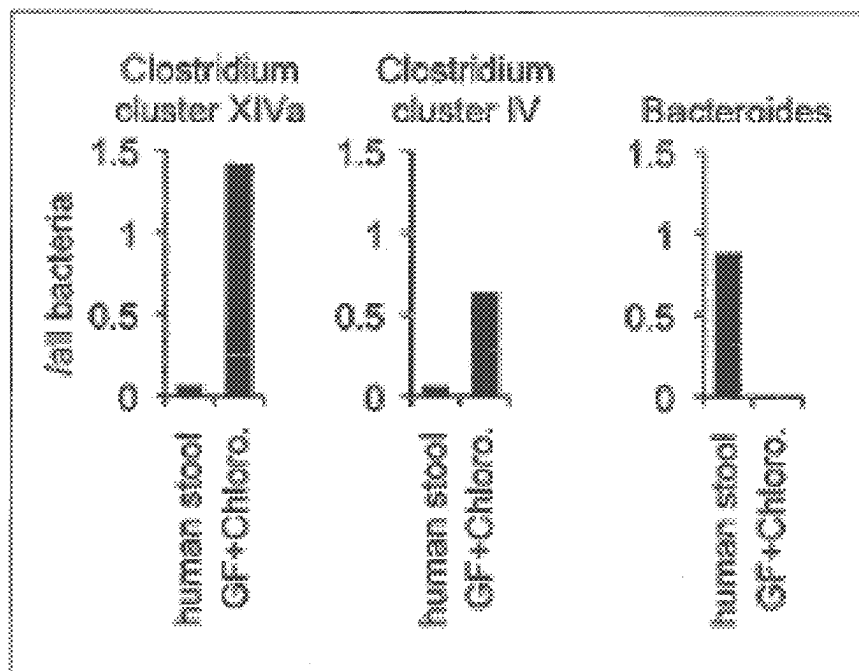
**FIG. 52**

FIG. 53

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COMPOSITION FOR INDUCING PROLIFERATION OR ACCUMULATION OF REGULATORY T CELLS

RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 16/780,116, filed Feb. 3, 2020, which is a continuation of U.S. application Ser. No. 16/425,030, filed May 29, 2019, now issued as U.S. Pat. No. 10,555,978, which is a continuation of U.S. application Ser. No. 16/389,380, filed Apr. 19, 2019, now issued as U.S. Pat. No. 10,588,925, which is a continuation of U.S. application Ser. No. 16/171,558, filed Oct. 26, 2018, now issued as U.S. Pat. No. 10,328,108, which is a continuation of U.S. application Ser. No. 16/117,054, filed Aug. 30, 2018, now issued as U.S. Pat. No. 10,322,150, which is a continuation of U.S. application Ser. No. 15/730,203, filed Oct. 11, 2017, now issued as U.S. Pat. No. 10,092,603, which is a continuation of U.S. application Ser. No. 15/216,015, filed Jul. 21, 2016, now issued as U.S. Pat. No. 9,801,933, which is a continuation of U.S. application Ser. No. 14/492,850, filed Sep. 22, 2014, now issued as U.S. Pat. No. 9,433,652, which is a continuation of U.S. application Ser. No. 13/701,467, filed Feb. 11, 2013, which is a national stage filing under 35 U.S.C. 371 of International Application PCT/JP2011/063302, filed Jun. 3, 2011, which claims the benefit of and priority to JP 2010-129134, filed Jun. 4, 2010 and PCT/JP2010/071746, filed Dec. 3, 2010. The entire teachings of the referenced applications are incorporated by reference herein.

TECHNICAL FIELD

The present invention relates to a composition which has an effect of inducing proliferation or accumulation of regulatory T cells, and which comprises, as an active ingredient, bacteria belonging to the genus *Clostridium*, a physiologically active substance derived from the bacteria, bacterial spores, or the like. The present invention also relates to a method for inducing proliferation or accumulation of regulatory T cells, as well as a method for inhibiting such proliferation or accumulation. Moreover, the present invention relates to a vaccine composition containing at least one strain of bacteria belonging to the genus *Clostridium* or a spore of bacteria, as well as a method for treating or preventing at least one disease or condition selected from infectious diseases and autoimmune diseases by administering the vaccine composition to an individual in need thereof. The present invention also relates to a method for screening for a compound that promotes proliferation or accumulation of regulatory T cells, as well as a non-human mammal which is used in this method, and in which a reporter gene is expressed under control of IL-10 gene expression.

BACKGROUND ART

Hundreds of species of commensal microorganisms are harbored in gastrointestinal tracts of mammals, and intimately interact with the host immune systems. Results of researches using germ-free (GF) animals have shown that the commensal microorganisms exert great influences on the development of mucosal immune systems such as histogenesis of Peyer's patches (PPs) and isolated lymphoid follicles (ILFs), secretion of antimicrobial peptides from epithelium, and accumulation of unique lymphocytes in mucosal tissues, the unique lymphocytes including immunoglobulin A-producing plasma cells, intraepithelial lymphocytes, IL-17-

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producing CD4-positive T cells (Th 17), and IL-22-producing NK-like cells (Non-Patent Documents 1 to 7). Consequently, the presence of intestinal bacteria enhances protective functions of the mucous membranes, providing the hosts with robust immune responses against pathogenic microbes invading the bodies. On the other hand, the mucosal immune systems maintain unresponsiveness to dietary antigens and harmless microbes (Non-Patent Document 3). For this reason, abnormality in the regulation of cross-talk between commensal bacteria and an immune system (intestinal dysbiosis) may lead to overly robust immune response to environmental antigens, so that inflammatory bowel disease (IBD) is caused (Non-Patent Documents 8 to 10).

Results of recent studies have shown that individual commensal bacteria control differentiation of their specific immune cells in the mucosal immune system. For example, *Bacteroides fragilis*, which is a commensal bacterium in humans, specifically induces a systemic Th1 cell response and a mucosal IL-10-producing T cell response in mice, and plays a role in protecting the host from colitis, which would otherwise be caused by a pathogen (Non-Patent Document 3). Segmented filamentous bacteria, which are intestinal commensal bacteria in mice, are shown to induce mucosal Th17 cell response and thereby to enhance resistance against infection of gastrointestinal tracts of the host with a pathogen (Non-Patent Documents 11 to 13). In addition, short-chain fatty acids derived from several commensal bacteria are known to suppress intestinal inflammation (Non-Patent Document 14). Moreover, it is presumed that the presence of some species of intestinal microbiota exerts a great influence on the differentiation of regulatory T cells (hereafter referred to as "Treg cells") which maintain homeostasis of the immune system.

Meanwhile, regulatory T cells which have been identified as a subset suppressing immunity are CD4⁺ T cells in which a transcription factor Foxp3 is expressed, and are known to play an important role in maintaining immunological homeostasis (Non-Patent Documents 8, 9, 15, and 16). Moreover, it has been known that the Foxp3-expressing cells are present in a large number especially in the colon, and only Treg cells present locally in the colon constantly expresses IL-10, which is an immunosuppressive cytokine, at a high level (Non-Patent Document 17). It is also known that animals having CD4⁺ Foxp3⁺ cells from which IL-10 is specifically removed develop inflammatory bowel disease (Non-Patent Document 18).

Accordingly, if the mechanism of the induction of Treg cells which produce IL-10 in the colon at a high level is elucidated, immunosuppression can be enhanced, which in turn can be applied to treatment of autoimmune diseases such as inflammatory bowel disease, as well as to organ transplantation.

However, mechanisms of how a large number of Treg cells come to be present in the colon and how the Treg cells produce IL-10 in the colon at a high level are still unclear. Moreover, it is also still unclear what species of bacteria constituting the intestinal commensal bacterial flora exerts the influence on the induction of regulatory T cells.

CITATION LIST

Non Patent Literature

- [NPL 1] J. J. Cebra, "Am J Clin Nutr", May, 1999, 69, 1046S
[NPL 2] A. J. Macpherson, N. L. Harris, "Nat Rev Immunol", June 2004, 4, 478

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- [NPL 3] J. L. Round, S. K. Mazmanian, "Nat Rev Immunol", May 2009, 9, 313
- [NPL 4] D. Bouskra et al., "Nature", Nov. 27, 2008, 456, 507
- [NPL 5] K. Atarashi et al., "Nature", Oct. 9, 2008, 455, 808
- [NPL 6] Ivanov, II et al., "Cell Host Microbe", Oct. 16, 2008, 4, 337
- [NPL 7] S. L. Sanos et al., "Nat Immunol", January 2009, 10, 83
- [NPL 8] M. A. Curotto de Lafaille, J. J. Lafaille, "Immunity", May 2009, 30, 626
- [NPL 9] M. J. Barnes, F. Powrie, "Immunity", Sep. 18, 2009, 31, 401
- [NPL 10] W. S. Garrett et al., "Cell", Oct. 5, 2007, 131, 33
- [NPL 11] Ivanov, II et al., "Cell", Oct. 30, 2009, 139, 485.
- [NPL 12] V. Gaboriau-Routhiau et al., "Immunity", Oct. 16, 2009, 31, 677
- [NPL 13] N. H. Salzman et al., "Nat Immunol", 11, 76.
- [NPL 14] K. M. Maslowski et al., "Nature", Oct. 29, 2009, 461, 1282
- [NPL 15] L. F. Lu, A. Rudensky, "Genes Dev", Jun. 1, 2009, 23, 1270
- [NPL 16] S. Sakaguchi, T. Yamaguchi, T. Nomura, M. Ono, "Cell", May 30, 2008, 133, 775
- [NPL 17] C. L. Maynard et al., "Nat Immunol", September 2007, 8, 931
- [NPL 18] Y. P. Rubtsov et al., "Immunity", April 2008, 28, 546

SUMMARY OF INVENTION

Technical Problem

The present invention has been made in view of the above-described problems of the conventional techniques. Accordingly, an object of the present invention is to identify intestinal commensal bacteria which induce the proliferation or accumulation of regulatory T cells. Another object of the present invention is to provide compositions or the like which comprise the identified intestinal commensal bacteria or a physiologically active substance derived therefrom, and which thus has have induce the proliferation or accumulation of regulatory T cells (Treg cells).

Solution to Problem

The present inventors have made earnest studies to solve the above-described problems. As a result, the present inventors have found that a chloroform-treated fraction and a spore-forming fraction of a fecal sample obtained from a mammal induces accumulation of regulatory T cells (Treg cells) in the colon. Moreover, the present inventors have found that bacteria belonging to the genus *Clostridium* induce proliferation or accumulation of regulatory T cells in the colon. The present inventors have also found that the regulatory T cells induced by these bacteria suppress proliferation of effector T cells. Furthermore, the present inventors have also found that colonization of bacteria belonging to the genus *Clostridium* and resultant proliferation or accumulation of Treg cells regulate local and systemic immune responses.

From these findings, the present inventors have found that the use of bacteria belonging to the genus *Clostridium*, spores thereof, or a physiologically active substance derived therefrom makes it possible to induce the proliferation or accumulation of regulatory T cells (Treg cells), and further to suppress immune functions.

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More specifically, the present invention has the following aspects:

- (1) A composition that induces proliferation or accumulation of regulatory T cells, the composition comprising, as an active ingredient, at least one substance selected from the group consisting of the following (a) to (c):
 - (a) bacteria belonging to the genus *Clostridium* or a physiologically active substance derived from the bacteria;
 - (b) a spore-forming fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction; and
 - (c) a chloroform-treated fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction.
- (2) Any of the compositions described herein wherein the regulatory T cells are transcription factor Foxp3-positive regulatory T cells or IL-10-producing regulatory T cells.
- (3) The composition according to any one of (1) and (2), wherein the composition has an immunosuppressive effect.
- (4) The composition according to any one of (1) to (3), wherein the composition is a pharmaceutical composition.
- (5) A method for inducing proliferation or accumulation of regulatory T cells in an individual (e.g., an individual in need thereof, such as an individual in need of induction of proliferation or accumulation of regulatory T cells), the method comprising a step of administering, to the individual, at least one substance selected from the group consisting of the following (a) to (c):
 - (a) bacteria belonging to the genus *Clostridium* or a physiologically active substance derived from the bacteria;
 - (b) a spore-forming fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction; and
 - (c) a chloroform-treated fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction.
- (6) A method for inducing proliferation or accumulation of regulatory T cells in an individual (e.g., an individual in need thereof, such as an individual in need of induction of proliferation or accumulation of regulatory T cells), the method comprising a step of administering an antibiotic against Gram-negative bacteria to the individual. And the antibiotic can be administered alone or in combination with at least one substance selected from the group consisting of the following (a) to (c):
 - (a) bacteria belonging to the genus *Clostridium* or a physiologically active substance derived from the bacteria;
 - (b) a spore-forming fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction; and
 - (c) a chloroform-treated fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction.
- (7) A method for inducing proliferation or accumulation of regulatory T cells in an individual, the method comprising a step of administering, to the individual, at least one substance selected from the group consisting of almond skin, inulin, oligofructose, raffinose, lactu-

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lose, pectin, hemicellulose, amylopectin, acetyl-Co A, biotin, beet molasses, yeast extracts, and resistant starch.

- (8) The method according to any one of (5) to (7), wherein a therapeutic composition is further administered to the individual.

Note that, the “therapeutic composition” here is meant to be something other than (a)-(c) described in (5) and (6), the antibiotic against Gram-negative bacteria described in (6), or the substances described in (7).

- (9) The method according to (8), wherein the therapeutic composition is at least one composition selected from the group consisting of corticosteroids, mesalazine, mesalamine, sulfasalazine, sulfasalazine derivatives, immunosuppressive drugs, cyclosporin A, mercaptopurine, azathiopurine, prednisone, methotrexate, antihistamines, glucocorticoids, epinephrine, theophylline, cromolyn sodium, anti-leukotrienes, anti-cholinergic drugs for rhinitis, anticholinergic decongestants, mast-cell stabilizers, monoclonal anti-IgE antibodies, vaccines, and combinations thereof.

- (10) The method according to any one of (5) to (9), wherein one measurement selected from the group consisting of promotion of IL-10 expression, promotion of CTLA4 expression, promotion of IDO expression, and suppression of IL-4 expression is used as an index of the induction of proliferation or accumulation of regulatory T cells in the individual.

- (11) A method for inhibiting proliferation or accumulation of regulatory T cells in an individual (e.g., an individual thereof), the method comprising a step of administering an antibiotic against Gram-positive bacteria to the individual.

- (12) The composition according to any one of (5) to (11), wherein the regulatory T cells are transcription factor Foxp3-positive regulatory T cells or IL-10-producing regulatory T cells.

- (13) A vaccine composition comprising at least one substance selected from the group consisting of the following (a) to (c):

- (a) bacteria belonging to the genus *Clostridium*;
(b) a spore of bacteria in a spore-forming fraction of a fecal sample obtained from a mammal; and
(c) bacteria in a chloroform-treated fraction of a fecal sample obtained from a mammal.

- (14) A method for treating aiding in treating, reducing the severity of, or preventing at least one disease selected from infectious diseases and autoimmune diseases in an individual (e.g., an individual in need thereof, such as an individual in need of treatment, reduction in the severity of or prevention of at least one such disease), the method comprising administering the vaccine composition according to (13) to the individual.

- (15) A method for screening for a compound having an activity to promote proliferation or accumulation of regulatory T cells, the method comprising:

- (I) preparing a test substance from at least one substance selected from the group consisting of the following (a) to (c):
(a) bacteria belonging to the genus *Clostridium* or a physiologically active substance derived from the bacteria;

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- (b) a spore-forming fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction; and

- (c) a chloroform-treated fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction.

- (II) preparing non-human mammals in which a reporter gene is to be expressed under control of IL-10 gene expression;

- (III) bringing the test substance into contact with the non-human mammal;

- (IV) after the contact with the test substance, detecting cells expressing the reporter gene in a CD4⁺ Foxp3⁺ cell group of the non-human mammal, and determining the number of cells in the CD4⁺ Foxp3⁺ cell group expressing the reporter gene or a ratio of cells in the CD4⁺ Foxp3⁺ cell group expressing the reporter gene to cells in the CD4⁺ Foxp3⁺ cell group not expressing the reporter gene;

- (V) detecting cells expressing the reporter gene in a CD4⁺ Foxp3⁺ cell group of the non-human mammal which has not been in contact with the test substance, and determining the number of cells in the CD4⁺ Foxp3⁺ cell group expressing the reporter gene or a ratio of cells in the CD4⁺ Foxp3⁺ cell group expressing the reporter gene to cells in the CD4⁺ Foxp3⁺ cell group not expressing the reporter gene; and

- (VI) comparing the number or the ratio determined in (IV) with the number or the ratio determined in (V), and determining, when the number or the ratio determined in (IV) is greater than that determined in (V), that the test substance is a compound that promotes proliferation or accumulation of Treg cells.

- (16) A non-human mammal which is used for the method according to (15), and in which the reporter gene is expressed under the control of the IL-10 gene expression.

- (17) A method for isolating, from a sample of bacteria belonging to the genus *Clostridium*, a compound having an activity to promote proliferation or accumulation of regulatory T cells, the method comprising (I) to (III):

- (I) preparing a genomic DNA from the sample of bacteria belonging to the genus *Clostridium*;

- (II) inserting the genomic DNA into a cloning system, and preparing a gene library derived from the sample of bacteria belonging to the genus *Clostridium*; and

- (III) isolating a compound having an activity to promote proliferation or accumulation of regulatory T cells, by use of the gene library obtained in step (II).

- (18) A method of treatment comprising (I) to (III):

- (I) measuring the percentage and/or absolute amounts of *Clostridium* Clusters IV and XIV in the microbiota of a subject,

- (II) comparing them to the same measurements in a healthy individual; and

- (III) administering a substance to the subject, if a statistically significant decrease in the number/amounts of *Clostridium* cluster IV, XIV in the subject compared to the healthy individual is detected, wherein the substance is at least one substance selected from the group consisting of the following (a) to (c):

- (a) any of the substances described herein;
(b) an antibiotic against Gram-negative bacteria; and
(c) the substance selected from the group consisting of almond skin, inulin, oligofructose, raffinose, lactu-

lose, pectin, hemicellulose, amylopectin, acetyl-CoA, biotin, beet molasses, yeast extracts, and resistant starch.

(19) A method of monitoring, comprising (I) to (II):

(I) measuring level of *Clostridium* cluster IV, XIV in a subject after administering at least one substance selected from the group consisting of the following (a) to (c):

- (a) any of the substances described herein;
- (b) an antibiotic against Gram-negative bacteria; and
- (c) the substance selected from the group consisting of almond skin, inulin, oligofructose, raffinose, lactulose, pectin, hemicellulose, amylopectin, acetyl-CoA, biotin, beet molasses, yeast extracts, and resistant starch; and

(II) if the level increases, it is judged to be a sign that the subject is responding favorably.

Advantageous Effects of Invention

The compositions of the present invention containing as an active ingredient bacteria belonging to the genus *Clostridium* or a physiologically active substance derived from the bacteria serves as an excellent composition for inducing the proliferation or accumulation of regulatory T cells (Treg cells). Immunity in a living organism can be suppressed through administration of the composition of the present invention as a pharmaceutical product or ingestion of the composition as a food or beverage. Accordingly, the composition of the present invention can be used, for example, to prevent or treat autoimmune diseases or allergic diseases, as well as to suppress immunological rejection in organ transplantation or the like. In addition, if a food or beverage such as a health food comprises the composition of the present invention, healthy individuals can ingest the composition easily and routinely. As a result, it is possible to induce the proliferation or accumulation of regulatory T cells and thereby to improve immune functions.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a schematic diagram showing a method of producing $\text{Il10}^{\text{venus}}$ mouse.

FIG. 2 is a diagram showing results of Southern blotting performed for analysis as to whether or not the $\text{Il10}^{\text{venus}}$ mice have an $\text{Il10}^{\text{venus}}$ allele.

FIG. 3 is a FACS dot-plot diagram showing results obtained when Venus-positive cells and Venus-negative cells from the $\text{Il10}^{\text{venus}}$ mice were sorted.

FIG. 4 is a graph showing the results obtained when the amounts of IL-10 mRNA expressed in Venus positive-cells and Venus-negative cells of the $\text{Il10}^{\text{venus}}$ mice were analyzed by real-time RT-PCR.

FIG. 5 is a graph showing change in the ratio of Foxp3⁺ cells in CD4⁺ lymphocytes of SPF mice.

FIG. 6 shows FACS dot-plot diagrams showing analysis results of the ratios of Foxp3⁺ cells in CD4⁺ lymphocytes isolated from the small intestine, the colon, and the peripheral lymph nodes of GF mice and SPF mice.

FIG. 7 is a graph showing analysis results of the ratios of Foxp3⁺ cells in CD4⁺ lymphocytes isolated from the small intestine, the colon, and the peripheral lymph nodes of GF mice and SPF mice.

FIG. 8 shows graphs showing analysis results of the numbers of CD4⁺ Foxp3⁺ cells isolated from the small intestine, the colon, and the peripheral lymph nodes of GF mice and SPF mice.

FIG. 9 is a plot diagram showing analysis results of the ratios of Venus⁺ cells in CD4⁺ cells in various tissues of SPF mice treated with antibiotics.

FIG. 10 shows FACS dot-plot diagrams showing analysis results of the ratio of Foxp3⁺ cell in CD4⁺ lymphocytes isolated from the colonic lamina propria of GF mice to which a fecal suspension of SPF mice was administered.

FIG. 11 is a graph showing analysis results of the ratios of Foxp3⁺ cells in CD4⁺ lymphocytes isolated from the lamina propria of the colon and the lamina propria of the small intestine of GF mice to which a fecal suspension of SPF mice was administered.

FIG. 12 is a graph showing analysis results of the ratio of Foxp3⁺ cells in CD4⁺ lymphocytes isolated from the lamina propria of mice deficient in ILFs, PPs, and colonic-patches.

FIG. 13 shows FACS dot-plot diagrams showing analysis results of the ratios of Foxp3⁺ cells in CD4⁺ lymphocytes isolated from the colonic lamina propria of GF mice to which specific commensal bacteria were administered.

FIG. 14 shows graphs showing analysis results of the ratios of Foxp3⁺ cells in CD4⁺ lymphocytes isolated from the colonic lamina propria of GF mice to which specific commensal bacteria were administered.

FIG. 15 is a graph showing analysis results of the ratios of IFN- γ ⁺ cells in CD4⁺ lymphocytes isolated from the colonic lamina propria of mice in which specific commensal bacteria were colonized.

FIG. 16 is a graph showing analysis results of the ratios of IL-17⁺ cells in CD4⁺ lymphocytes isolated from the colonic lamina propria of mice in which specific commensal bacteria were colonized.

FIG. 17 is a graph showing analysis results of the ratios of Foxp3⁺ cells in CD4⁺ lymphocytes isolated from the colon of kinds of SPF mice each being deficient in a pathogen-associated molecular pattern recognition receptor-associated factor.

FIG. 18 is a graph showing analysis results of the ratios of Foxp3⁺ cells in CD4⁺ lymphocytes isolated from the colonic lamina propria of Myd88^{-/-} mice in which the *Clostridium* was colonized.

FIG. 19 shows FACS dot-plot diagrams showing analysis results of the ratios of Venus⁺ cells in lymphocytes isolated from various tissues of $\text{Il10}^{\text{venus}}$ mice.

FIG. 20 is a FACS dot-plot diagram showing analysis results of the expression of a T cell receptor β chain on cell surfaces of lymphocytes isolated from the colonic lamina propria of $\text{Il10}^{\text{venus}}$ mice.

FIG. 21 shows FACS dot-plot diagrams showing analysis results of the expression of IL-17, IL-4, and IFN- γ in lymphocytes isolated from the colonic lamina propria of $\text{Il10}^{\text{venus}}$ mice.

FIG. 22 shows graphs showing analysis results of the amounts of mRNAs of IL-10, CTLA4, Foxp3, and GITR expressed in spleen Foxp3⁺CD4⁺ cells, spleen Foxp3⁺CD4⁺ cells, colonic lamina propria Venus⁺ cells, and small intestinal lamina propria Venus⁺ cells.

FIG. 23 shows FACS dot-plot diagrams showing analysis results of the expression of CD4, Foxp3, and Venus in the lamina propria of the small intestine and the lamina propria of the colon of GF $\text{Il10}^{\text{venus}}$ mice and SPF $\text{Il10}^{\text{venus}}$ mice.

FIG. 24 shows FACS dot-plot diagrams showing analysis results of the expression of Venus and Foxp3 of CD4 cells in various tissues of SPF $\text{Il10}^{\text{venus}}$ mice.

FIG. 25 shows FACS dot-plot diagrams showing analysis results of the expression of Foxp3 and Venus in $\text{Il10}^{\text{venus}}$ mice in which specific commensal bacteria were colonized.

FIG. 26 is a graph showing analysis results of the expression of Foxp3 and/or Venus of CD4⁺ cells in the small intestine of *Il10^{venus}* mice in which specific commensal bacteria were colonized.

FIG. 27 is a graph showing analysis results of the expression of Foxp3 and/or Venus of CD4⁺ cells in the colon of *Il10^{venus}* mice in which specific commensal bacteria were colonized.

FIG. 28 is a plot diagram showing analysis results of the ratios of Venus⁺ cells in CD4⁺ cells isolated from various tissues of *Il10^{venus}* mice treated with antibiotics.

FIG. 29 is a graph showing analysis results of immunoregulatory functions of CD4⁺ Venus⁺ cells from the colonic lamina propria of GF *Il10^{venus}* mice in which the genus *Clostridium* was colonized, CD4⁺ Venus⁺ cells from the colonic lamina propria of SPF *Il10^{venus}* mice, and CD4⁺ GFP⁺ cells from the spleen of Foxp3^{eGFP} reporter mice.

FIG. 30 is a graph showing the results obtained when SPF B6 mice were treated with polymyxin B or vancomycin for 4 weeks, and then analyzed for the ratio of Foxp3⁺ cells in the CD4⁺ cell group.

FIG. 31 is a graph showing the results obtained when SPF mice-derived chloroform-treated feces were orally administered to GF mice, and then the ratio of Foxp3⁺ cells in the CD4⁺ cell group was analyzed.

FIG. 32 is a graph showing the general results of flow cytometry analysis on Helios expression in LP lymphocytes in the thymuses or the colons of SPF mice, GF mice, *Lactobacillus*-colonized mice, or *Clostridium*-colonized mice.

FIG. 33 shows plot diagrams showing representative results of flow cytometry analysis on CD4 expression, Foxp3 expression, and Helios expression in the LP lymphocytes in the thymuses or the colons of the SPF mice, the GF mice, the *Lactobacillus*-colonized mice, or the *Clostridium*-colonized mice.

FIG. 34 is a graph showing the results obtained when the whole colons derived from GF mice, *Lactobacillus*-colonized mice, or *Clostridium*-colonized mice were cultured, and the culture supernatants thereof were analyzed for the TGF- β 1 concentration by ELISA.

FIG. 35 is a graph showing the results obtained when intestinal epithelial cells (IECs) derived from GF mice or *Clostridium*-colonized mice were cultured, and the culture supernatants thereof were analyzed for the TGF- β 1 concentration by ELISA.

FIG. 36 is a graph showing the results obtained when splenic CD4⁺ T cells were cultured together with an anti-CD3 antibody and with a culture supernatant of IECs isolated from GF mice or mice colonized with 46 bacterial strains of the genus *Clostridium* (Clost.) in the presence or absence of an anti-TGF- β antibody, and the T cells were collected on day 5 of the culture and analyzed for Foxp3 expression by real-time RT-PCR.

FIG. 37 is a graph showing the results obtained when C57BL/6 GF mice were orally inoculated with 46 bacterial strains of the genus *Clostridium* (Clost.) or three bacterial strains of the genus *Lactobacillus* (Lacto.), and IECs were collected three weeks after the inoculation and analyzed for the relative mRNA expression level of the MMP2 gene by real-time RT-PCR.

FIG. 38 is a graph showing the results obtained when C57BL/6 GF mice were orally inoculated with 46 bacterial strains of the genus *Clostridium* (Clost.) or three bacterial strains of the genus *Lactobacillus* (Lacto.), and IECs were

collected three weeks after the inoculation and analyzed for the relative mRNA expression level of the MMP9 gene by real-time RT-PCR.

FIG. 39 is a graph showing the results obtained when C57BL/6 GF mice were orally inoculated with 46 bacterial strains of the genus *Clostridium* (Clost.) or three bacterial strains of the genus *Lactobacillus* (Lacto.), and IECs were collected three weeks after the inoculation and analyzed for the relative mRNA expression level of the MMP13 gene by real-time RT-PCR.

FIG. 40 is a graph showing the results obtained when C57BL/6 GF mice were orally inoculated with 46 bacterial strains of the genus *Clostridium* (Clost.) or three bacterial strains of the genus *Lactobacillus* (Lacto.), and IECs were collected three weeks after the inoculation and analyzed for the relative mRNA expression level of the IDO gene by real-time RT-PCR.

FIG. 41 is a graph showing the results obtained when control mice (SPF) and *Clostridium*-administered mice (SPF+Clost.) were treated with 2% DSS, observed and measured for the body weight loss, the hardness of stool, and bleeding for six days, and then evaluated numerically.

FIG. 42 is a photograph showing the state of the colons collected on day 6 after the control mice (SPF) and the *Clostridium*-administered mice (SPF+Clost.) were treated with 2% DSS.

FIG. 43 shows photomicrographs showing the results obtained when the control mice (SPF) and the *Clostridium*-administered mice (SPF+Clost.) were treated with 2% DSS, and the colons thereof were collected on day 6 and analyzed histologically by HE staining.

FIG. 44 is a graph showing the results obtained when control mice (SPF) and *Clostridium*-administered mice (SPF+Clost.) were sensitized with oxazolone, and subsequently the inside of each rectum was treated with a 1% oxazolone/50% ethanol solution, and the body weight loss was measured.

FIG. 45 shows photomicrographs showing the results obtained when the control mice (SPF) and the *Clostridium*-administered mice (SPF+Clost.) were sensitized with oxazolone, and subsequently the inside of each rectum was treated with a 1% oxazolone/50% ethanol solution, and the colons obtained by the treatment were analyzed histologically by HE staining.

FIG. 46 is a graph showing the results obtained when control mice (SPF) and *Clostridium*-administered mice (SPF+Clost.) were immunized by administering alum-absorbed ovalbumin (OVA) twice at a 2-week interval, and the sera were collected therefrom and analyzed for the concentration of OVA-specific IgE in these sera by ELISA.

FIG. 47 is a graph showing the results obtained when the control mice (SPF) and the *Clostridium*-administered mice (SPF+Clost.) were immunized by administering the alum-absorbed OVA twice at a 2-week interval, and splenic cells were collected and analyzed for IL-4 production of these splenic cells by in-vitro OVA restimulation.

FIG. 48 is a graph showing the results obtained when the control mice (SPF) and the *Clostridium*-administered mice (SPF+Clost.) were immunized by administering the alum-absorbed OVA twice at a 2-week interval, and the splenic cells were collected and analyzed for IL-10 production of these splenic cells by the in-vitro OVA restimulation.

FIG. 49 is Phylogenetic tree constructed by the neighbor-joining method with the resulting sequences of the 41 strains of *Clostridium* and those of known bacteria obtained from Genbank database using Mega software.

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FIG. 50 is histograms showing Foxp3 expression gated CD4 cells from GF mice (Germ-free mouse #1 and #2) or GF mice colonized with three strains of *Clostridium* belonging to cluster IV (3 strains of Clost. mouse #1 and #2).

FIG. 51 is histograms showing Foxp3 expression by CD4 positive lymphocytes from GF mice (GF) or GF mice gavaged with chloroform-treated human stool (GF+ Chloro.).

FIG. 52 is a graph showing Foxp3 expression by CD4 positive lymphocytes from GF mice (GF) or GF mice gavaged with chloroform-treated human stool (GF+ Chloro.).

FIG. 53 is a graph showing amounts of *Clostridium* and *Bacteroides* in feces of mice gavaged with chloroform-treated human stool

DESCRIPTION OF EMBODIMENTS

Composition Having Effect of Inducing Proliferation or Accumulation of Regulatory T Cells

The present invention provides a composition that induces proliferation or accumulation of regulatory T cells, the composition comprising, as an active ingredient, at least one substance selected from the group consisting of the following (a) to (c):

- (a) bacteria belonging to the genus *Clostridium* or a physiologically active substance derived from the bacteria;
- (b) a spore-forming fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction; and
- (c) a chloroform-treated fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction.

In the present invention “regulatory T cells” mean T cells which have a function of suppressing an abnormal or excessive immune response, and which play a role in immune tolerance. The regulatory T cells are typically transcription factor Foxp3-positive CD4-positive T cells. However, the regulatory T cells of the present invention also include transcription factor Foxp3-negative regulatory T cells, as long as the regulatory T cells are IL-10-producing CD4-positive T cells.

The meaning of the “induces proliferation or accumulation of regulatory T cells” in the present invention includes an effect of inducing the differentiation of immature T cells into regulatory T cells, which differentiation leads to the proliferation or the accumulation of regulatory T cells. In addition, the meaning of the “induces proliferation or accumulation of regulatory T cells” in the present invention includes in-vivo effects, in vitro effects, and ex vivo effects. Accordingly, all of the following effects are included: an effect of inducing in vivo proliferation or accumulation of regulatory T cells through administration or ingestion of the bacteria belonging to the genus *Clostridium* or the physiologically active substance or the like derived from the bacteria; an effect of inducing proliferation or accumulation of cultured regulatory T cells by causing the bacteria belonging to the genus *Clostridium* or the physiologically active substance or the like derived from the bacteria to act on the cultured regulatory T cells; and an effect of inducing proliferation or accumulation of regulatory T cells which are collected from a living organism and which are intended to be subsequently introduced into a living organism, such as the organism from which they were obtained or another organism, by causing the bacteria belonging to the genus *Clostridium* or the physiologically active substance or the like derived from the bacteria to act on the regulatory T cells. The effect of inducing proliferation or accumulation of

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regulatory T cells can be evaluated, for example, as follows. Specifically, the bacteria belonging to the genus *Clostridium* or the physiologically active substance or the like derived from the bacteria is orally administered to an experimental animal such as a germ-free mouse, then CD4-positive cells in the colon are isolated, and the ratio of regulatory T cells contained in the CD4-positive cells is measured by flow cytometry (refer to Example 7).

The regulatory T cells of which proliferation or accumulation is induced by the composition of the present invention are preferably transcription factor Foxp3-positive regulatory T cells or IL-10-producing regulatory T cells.

The “bacteria belonging to the genus *Clostridium*,” which are the active ingredient in the composition of the present invention, is not particularly limited as long as the bacteria have the effect of inducing proliferation or accumulation of regulatory T cells. The bacteria preferably belong to the cluster XIVa or the cluster IV. One strain of the bacteria alone can be used for the composition of the present invention, but two or more strains of the bacteria can be used together for the composition of the present invention. The use of multiple strains of bacteria belonging to the cluster XIVa or the cluster IV in combination can bring about an excellent effect on regulatory T cells. In addition to the bacteria belonging to these clusters, bacteria belonging to other clusters (for example, bacteria belonging to the cluster III) can also be used in combination. If more than one strain of bacteria is used (e.g., one or more strain belonging to cluster XIVa, one or more strain belonging to cluster IV, one or more strain belonging to a cluster other than cluster XIVa or cluster IV, such as one or more strain belonging to cluster III), the type and number of strains used can vary widely. The type and number to be used can be determined based on a variety of factors (e.g., the desired effect, such as induction or inhibition of proliferation or accumulation of regulatory T cells; the disease or condition to be treated, prevented or reduced in severity; the age or gender of the recipient) The strains can be present in a single composition, in which case they will be consumed or ingested together, or can be present in more than one composition (e.g., each can be in a separate composition), in which case they can be consumed individually or the compositions can be combined and the resulting combination (combined compositions) consumed or ingested. Any number or combination of strains that proves effective (e.g., any number from one to 200, such as 1 to 100, 1 to 50, 1 to 40, 1 to 30, 1 to 20, 1 to 10, 1 to 5 and any number there between) can be administered. In certain embodiments of the present invention, a combination of some or all of the 46 strains described in Document (Itoh, K., and Mitsuoka, T. Characterization of clostridia isolated from faeces of limited flora mice and their effect on caecal size when associated with germ-free mice. Lab. Animals 19: 111-118 (1985)) is used. For example, at least one, two or more, three, three or more, four, four or more, five, five or more, six, six or more or any other number of the 46 described strains, including 46 strains, can be used. They can be used in combination with one another and in combination with strains not described in the cited reference (e.g., in combination with one or more strains belonging to cluster III). Note that, the cluster of “bacteria belonging to the genus *Clostridium*” can be identified, for example, as follows. Specifically, the bacteria belonging to the genus *Clostridium* are classified by PCR using a primer set consisting of SEQ ID NOs 64 and 65 (for *Clostridium* spp. belonging to the cluster XIVa) or a primer set consisting of SEQ ID NOs 66 and 67 (for *Clostridium* spp. belonging to the cluster IV) (refer to Example 18). In addition, the bacteria belonging to

the genus *Clostridium* are classified by sequencing of 16S rRNA gene amplified using a primer set consisting of SEQ ID NOs 19 and 20 (refer to Example 7).

Viable cells of the bacteria belonging to the genus *Clostridium* can be used for the composition of the present invention, and killed cells thereof may also be used for the composition. In addition, from the viewpoint of stability to heat, resistance to antibiotics and the like, and long storage period, the bacteria belonging to the genus *Clostridium* are preferably in the form of spore.

The meaning of the “physiologically active substance derived from bacteria belonging to the genus *Clostridium*” of the present invention includes substances contained in the bacteria, secretion products of the bacteria, and metabolites of the bacteria. Such a physiologically active substance can be identified by purifying an active component from the bacteria, a culture supernatant thereof, or intestinal tract contents in the intestinal tract of a mouse in which only bacteria belonging to the genus *Clostridium* are colonized by an already known purification method.

The active ingredient “spore-forming fraction of a fecal sample obtained from a mammal” in the composition of the present invention is not particularly limited, as long as the fraction includes spore-forming bacteria present in feces of a mammal, and has the effect of inducing proliferation or accumulation of regulatory T cells.

The active ingredient “chloroform-treated fraction of a fecal sample obtained from a mammal” in the composition of the present invention is not particularly limited, as long as the fraction is obtained by treating feces of a mammal with chloroform (for example, 3% chloroform), and has the effect of inducing proliferation or accumulation of regulatory T cells.

Note that the “mammal” in the present invention is not particularly limited, and examples thereof include humans, mice, rats, cattle, horses, pigs, sheep, monkeys, dogs, and cats.

Meanwhile, when the “spore-forming fraction of a fecal sample obtained from a mammal” or the “chloroform-treated fraction of a fecal sample obtained from a mammal” is cultured in a medium, substances contained in the bacteria, secretion products of the bacteria, metabolites of the bacteria are released from the bacteria and the like contained in the fraction. The meaning of the active ingredient “culture supernatant of the fraction” in the composition of the present invention includes such substances, secretion products, and metabolites. The culture supernatant is not particularly limited, as long as the culture supernatant has the effect of inducing proliferation or accumulation of regulatory T cells. Examples of the culture supernatant include a protein fraction of the culture supernatant, a polysaccharide fraction of the culture supernatant, a lipid fraction of the culture supernatant, and a low-molecular weight metabolite fraction of the culture supernatant.

The composition of the present invention may be in the form of a pharmaceutical composition, a food or beverage (which may also be an animal feed), or a reagent used for an animal model experiment, the pharmaceutical composition, the food or beverage, and the reagent having the effect of inducing proliferation or accumulation of regulatory T cells. An example of the present invention revealed that regulatory T cells (Treg cells) induced by bacteria or the like belonging to the genus *Clostridium* suppressed the proliferation of effector T-cells. Accordingly, the composition of the present invention can be used suitably as a composition having an immunosuppressive effect. The immunosuppressive effect can be evaluated, for example, as follows. Specifically,

regulatory T cells isolated from an experimental animal, such as a mouse, to which the composition of the present invention is orally administered are caused to act on effector T-cells (CD4⁺ CD25⁻ cells) isolated from the spleen, and then proliferation ability thereof is measured by using the intake amount of [³H]-thymidine as an index (refer to Example 14).

The composition of the present invention can be used, for example, as a pharmaceutical composition for preventing or treating an autoimmune disease such as chronic inflammatory bowel disease, systemic lupus erythematosus, rheumatoid arthritis, multiple sclerosis, or Hashimoto's disease, or an allergic disease such as pollenosis or asthma; a pharmaceutical composition for suppressing rejection in organ transplantation or the like; a food or beverage for improving immune functions; or a reagent for suppressing the proliferation or function of effector T-cells.

More specific examples of target diseases of the composition of the present invention include autoimmune diseases, allergic diseases, and rejection in organ transplantations and the like, such as inflammatory bowel disease (IBD), ulcerative colitis, Crohn's disease, sprue, autoimmune arthritis, rheumatoid arthritis, Type I diabetes, multiple sclerosis, graft vs. host disease following bone marrow transplantation, osteoarthritis, juvenile chronic arthritis, Lyme arthritis, psoriatic arthritis, reactive arthritis, spondyloarthropathy, systemic lupus erythematosus, insulin dependent diabetes mellitus, thyroiditis, asthma, psoriasis, dermatitis scleroderma, atopic dermatitis, graft versus host disease, acute or chronic immune disease associated with organ transplantation, sarcoidosis, atherosclerosis, disseminated intravascular coagulation, Kawasaki's disease, Grave's disease, nephrotic syndrome, chronic fatigue syndrome, Wegener's granulomatosis, Henoch-Schoenlein purpura, microscopic vasculitis of the kidneys, chronic active hepatitis, uveitis, septic shock, toxic shock syndrome, sepsis syndrome, cachexia, acquired immunodeficiency syndrome, acute transverse myelitis, Huntington's chorea, Parkinson's disease, Alzheimer's disease, stroke, primary biliary cirrhosis, hemolytic anemia, polyglandular deficiency type I syndrome and polyglandular deficiency type II syndrome, Schmidt's syndrome, adult (acute) respiratory distress syndrome, alopecia, alopecia areata, seronegative arthropathy, arthropathy, Reiter's disease, psoriatic arthropathy, *Chlamydia*, *Yersinia* and *Salmonella* associated arthropathy, spondyloarthropathy, atheromatous disease/arteriosclerosis, atopic allergy, food allergies, autoimmune bullous disease, pemphigus vulgaris, pemphigus foliaceus, pemphigoid, linear IgA disease, autoimmune haemolytic anaemia, Coombs positive haemolytic anaemia, acquired pernicious anaemia, juvenile pernicious anaemia, myalgic encephalitis/Royal Free Disease, chronic mucocutaneous candidiasis, giant cell arteritis, primary sclerosing hepatitis, cryptogenic autoimmune hepatitis, Acquired Immunodeficiency Disease Syndrome, Acquired Immunodeficiency Related Diseases, Hepatitis C, common varied immunodeficiency (common variable hypogammaglobulinaemia), dilated cardiomyopathy, fibrotic lung disease, cryptogenic fibrosing alveolitis, postinflammatory interstitial lung disease, interstitial pneumonitis, connective tissue disease associated interstitial lung disease, mixed connective tissue disease associated lung disease, systemic sclerosis associated interstitial lung disease, rheumatoid arthritis associated interstitial lung disease, systemic lupus erythematosus associated lung disease, dermatomyositis/polymyositis associated lung disease, Sjogren's disease associated lung disease, ankylosing spondylitis associated lung disease, vasculitic diffuse lung disease, haemosiderosis

associated lung disease, drug-induced interstitial lung disease, radiation fibrosis, bronchiolitis obliterans, chronic eosinophilic pneumonia, lymphocytic infiltrative lung disease, postinfectious interstitial lung disease, gouty arthritis, autoimmune hepatitis, type-1 autoimmune hepatitis (classical autoimmune or lupoid hepatitis), type-2 autoimmune hepatitis (anti-LKM antibody hepatitis), autoimmune mediated hypoglycemia, type B insulin resistance with acanthosis nigricans, hypoparathyroidism, acute immune disease associated with organ transplantation, chronic immune disease associated with organ transplantation, osteoarthritis, primary sclerosing cholangitis, idiopathic leucopenia, autoimmune neutropenia, renal disease NOS, glomerulonephritides, microscopic vasculitis of the kidneys, discoid lupus, erythematosus, male infertility idiopathic or NOS, sperm autoimmunity, multiple sclerosis (all subtypes), insulin-dependent diabetes mellitus, sympathetic ophthalmia, pulmonary hypertension secondary to connective tissue disease, Goodpasture's syndrome, pulmonary manifestation of polyarteritis nodosa, acute rheumatoid fever, rheumatoid spondylitis, Still's disease, systemic sclerosis, Takayasu's disease/arthritis, autoimmune thrombocytopenia, idiopathic thrombocytopenia, autoimmune thyroid disease, hyperthyroidism, goitrous autoimmune hypothyroidism (Hashimoto's disease), atrophic autoimmune hypothyroidism, primary myxoedema, phacogenic uveitis, primary vasculitis, vitiligo, allergic rhinitis (pollen allergies), anaphylaxis, pet allergies, latex allergies, drug allergies, allergic rhinoconjunctivitis, eosinophilic esophagitis, hypereosinophilic syndrome, eosinophilic gastroenteritis cutaneous lupus erythematosus, eosinophilic esophagitis, hypereosinophilic syndrome, and eosinophilic gastroenteritis.

The composition of the present invention can also be used as a pharmaceutical composition for preventing or treating infectious diseases in an individual whose resistance to the infectious diseases is impaired because of damage due to excessive inflammation caused by the immunity.

Example of infectious pathogens which impair maintenance or recovery of homeostasis of a host, and which eventually bring about such immunopathological tissue damage include *Salmonella*, *Shigella*, *Clostridium difficile*, *Mycobacterium* (which cause the disease tuberculosis), protozoa (which cause the disease malaria), filarial nematodes (which cause the disease filariasis), *Schistosoma* (which cause the disease schistosomiasis), *Toxoplasma* (which cause the disease toxoplasmosis), *Leishmania* (which cause the disease leishmaniasis), HCV and HBV (which cause the disease hepatitis C and hepatitis B), and herpes simplex viruses (which cause the disease herpes).

Pharmaceutical preparations can be formulated from the composition of the present invention by already known drug formulation methods. For example, the composition of the present invention can be used orally or parenterally in the forms of capsules, tablets, pills, liquids, powders, granules, fine granules, film-coated preparations, pellets, troches, sublingual preparations, chewables, buccal preparations, pastes, syrups, suspensions, elixirs, emulsions, liniments, ointments, plasters, cataplasms, transdermal absorption systems, lotions, inhalations, aerosols, injections, suppositories, and the like.

For formulating these preparations, the composition of the present invention can be used in appropriate combination with carriers acceptable pharmacologically or acceptable for a food or beverage, specifically, with sterile water, physiological saline, vegetable oil, solvent, a base material, an emulsifier, a suspending agent, a surfactant, a stabilizer, a flavoring agent, an aromatic, an excipient, a vehicle, a

preservative, a binder, a diluent, a tonicity adjusting agent, a soothing agent, a bulking agent, a disintegrating agent, a buffer agent, a coating agent, a lubricant, a colorant, a sweetener, a thickening agent, a flavor corrigent, a solubilizer, other additives, or the like.

Meanwhile, for formulating a pharmaceutical preparation thereof, and particularly for formulating a pharmaceutical preparation for oral administration, it is preferable to use in combination a composition which enables an efficient delivery of the composition of the present invention to the colon, from the viewpoint of more efficiently inducing the proliferation or accumulation of regulatory T cells in the colon.

Such a composition or method which enables the delivery to the colon is not particularly limited, and known compositions or methods can be employed as appropriate. Examples thereof include pH sensitive compositions, more specifically, enteric polymers which release their contents when the pH becomes alkaline after the enteric polymers pass through the stomach. When a pH sensitive composition is used for formulating the pharmaceutical preparation, the pH sensitive composition is preferably a polymer whose pH threshold of the decomposition of the composition is 6.8 to 7.5. Such a numeric value range is a range where the pH shifts toward the alkaline side at a distal portion of the stomach, and hence is a suitable range for use in the delivery to the colon.

Moreover, another example of the composition enabling the delivery to the colon is a composition which ensures the delivery to the colon by delaying the release of the contents by approximately 3 to 5 hours, which corresponds to the small intestinal transit time. In an example of formulating a pharmaceutical preparation using the composition for delaying the release, a hydrogel is used as a shell. The hydrogel is hydrated and swells upon contact with gastrointestinal fluid, so that the contents are effectively released. Furthermore the delayed release dosage units include drug-containing compositions having a material which coats or selectively coats a drug. Examples of such a selective coating material include in vivo degradable polymers, gradually hydrolyzable polymers, gradually water-soluble polymers, and/or enzyme degradable polymers. A preferred coating material for efficiently delaying the release is not particularly limited, and examples thereof include cellulose-based polymers such as hydroxypropyl cellulose, acrylic acid polymers and copolymers such as methacrylic acid polymers and copolymers, and vinyl polymers and copolymers such as polyvinylpyrrolidone.

Examples of the composition enabling the delivery to the colon further include bioadhesive compositions which specifically adhere to the colonic mucosal membrane (for example, a polymer described in the specification of U.S. Pat. No. 6,368,586), and compositions into which a protease inhibitor is incorporated for protecting particularly a biopharmaceutical preparation in the gastrointestinal tracts from decomposition due to an activity of a protease.

An example of a system enabling the delivery to the colon is a system of delivering a composition to the colon by pressure change in such a way that the contents are released by utilizing pressure change caused by generation of gas in bacterial fermentation at a distal portion of the stomach. Such a system is not particularly limited, and a more specific example thereof is a capsule which has contents dispersed in a suppository base and which is coated with a hydrophobic polymer (for example, ethyl cellulose).

Another example of the system enabling the delivery to the colon is a system of delivering a composition to the colon, the system being specifically decomposed by an

enzyme (for example, a carbohydrate hydrolase or a carbohydrate reductase) present in the colon. Such a system is not particularly limited, and more specific examples thereof include systems which use food components such as non-starch polysaccharides, amylose, xanthan gum, and azopolymers.

When used as a pharmaceutical composition, the composition of the present invention may be used in combination with an already known pharmaceutical composition for use in immunosuppression. Such a known pharmaceutical composition is not particularly limited, and may be at least one therapeutic composition selected from the group consisting of corticosteroids, mesalazine, mesalamine, sulfasalazine, sulfasalazine derivatives, immunosuppressive drugs, cyclosporin A, mercaptopurine, azathiopurine, prednisone, methotrexate, antihistamines, glucocorticoids, epinephrine, theophylline, cromolyn sodium, anti-leukotrienes, anti-cholinergic drugs for rhinitis, anti-cholinergic decongestants, mast-cell stabilizers, monoclonal anti-IgE antibodies, vaccines (preferably vaccines used for vaccination where the amount of an allergen is gradually increased), and combinations thereof. It is preferable to use these therapeutic compositions in combination with the composition of the present invention.

When the composition of the present invention is used as a food or beverage, the food or beverage can be, for example, a health food, a functional food, a food for specified health use, a dietary supplement, a food for patients, or an animal feed. The food or beverage of the present invention can be ingested in the forms of the compositions as described above, and also can be ingested in the forms of various foods and beverages. Specific examples of the foods and beverages include various beverages such as juices, refreshing beverages, tea beverages, drink preparations, jelly beverages, and functional beverages; alcoholic beverages such as beers; carbohydrate-containing foods such as rice food products, noodles, breads, and pastas; paste products such as fish hams, sausages, paste products of seafood; retort pouch products such as curries, food dressed with a thick starchy sauces, and Chinese soups; soups; dairy products such as milk, dairy beverages, ice creams, cheeses, and yogurts; fermented products such as fermented soybean pastes, yogurts, fermented beverages, and pickles; bean products; various confectionery products such as Western confectionery products including biscuits, cookies, and the like, Japanese confectionery products including steamed bean-jam buns, soft adzuki-bean jellies, and the like, candies, chewing gums, gummies, cold desserts including jellies, crème caramels, and frozen desserts; instant foods such as instant soups and instant soy-bean soups; microwavable foods; and the like. Further, the examples also include health foods and beverages prepared in the forms of powders, granules, tablets, capsules, liquids, pastes, and jellies. The composition of the present invention can be used for animals including humans. The animals, other than humans, are not particularly limited, and the composition can be used for various livestock, poultry, pets, experimental animals, and the like. Specific examples of the animals include pigs, cattle, horses, sheep, goats, chickens, wild ducks, ostriches, domestic ducks, dogs, cats, rabbits, hamsters, mice, rats, monkeys, and the like, but the animals are not limited thereto.

Without wishing to be bound by theory, in the present invention, individuals in which the relative abundance of bacteria belonging to the group Firmicutes (the group to which the *Clostridium* clusters IV and XIVa belong) is large gain more body weight than individuals in which the relative

abundance of bacteria belonging to the group Bacteroidetes is large. Accordingly, the composition of the present invention is capable of conditioning absorption of nutrients and improving feed efficiency. From such a viewpoint, the composition of the present invention can be used for promoting body weight gain, or for an animal feed good in feed efficiency.

Moreover, the addition of the composition of the present invention to an antibiotic-free animal feed makes it possible to increase the body weight of a subject that ingests the animal feed to a level equal to or higher than those achieved by antibiotic-containing animal feeds, and also makes it possible to reduce pathogenic bacteria in the stomach to a level equal to those achieved by typical antibiotic-containing animal feeds. Accordingly, the composition of the present invention can be used for an animal feed which does not need the addition of antibiotics.

In addition, unlike conventional bacteria (*Lactobacillus* and *Bifidobacteria*) in commercial use which are not easy to incorporate into the livestock production, the composition of the present invention in the spore form can be pelletized, sprayed, or easily mixed with an animal feed, and also can be added to drinking water.

The feeding of such an animal feed using the composition of the present invention is not particularly limited, and the animal feed may be fed to a subject at regular intervals in a selective manner, or may be fed for a certain period (for example, at its birth, during weaning, or when the subject to be fed is relocated or shipped).

Moreover, from the above-described viewpoint, the composition of the present invention can be preferably used for malnourished humans. In other words, also when the subject who ingests the composition is a human, the composition of the present invention can preferably be used for promoting the body weight gain, and enhancing the energy absorption from foods.

The food or beverage of the present invention can be manufactured by a manufacturing technique which is well known in the technical field. To the food or beverage, one or more components (for example, a nutrient) which are effective for the improvement of an immune function by the immunosuppressive effect may be added. In addition, the food or beverage may be combined with another component or another functional food exhibiting a function other than the function of the improvement of an immune function to thereby serve as a multi-functional food or beverage.

Moreover, the composition of the present invention can be preferably incorporated into foods requiring a processing step which may destroy ordinary probiotic strains. Specifically, most commercially usable probiotic strains cannot be incorporated into foods which need to be processed by any one of a heat treatment, long term storage, a freezing treatment, a mechanical stress treatment, and a high-pressure treatment (for example, extrusion forming or roll forming). On the other hand, because of an advantageous nature of forming spores, the composition of the present invention can be easily incorporated into such processed foods.

For example, the composition of the present invention in the form of spore can survive even in a dried food, and can remain living even after being ingested. Likewise, the composition of the present invention can withstand low-temperature sterilization processes, typically processes at a temperature in a range from 70° C. to the boiling point, both inclusive. Thus, the composition of the present invention can be incorporated into all kinds of dairy products. Furthermore, the composition of the present invention can withstand long-term storage of many years; high-temperature

processing such as baking and boiling; low-temperature processing such as freezing and cold storage; and high-pressure treatments such as extrusion forming and roll forming.

The foods which need to be processed under such harsh conditions are not particularly limited, and examples thereof include foods which need to be processed in a microwave oven to be edible (for example, oatmeal), foods which need to be baked to be edible (for example, muffin), foods which need to be subjected to a sterilization high-temperature treatment for a short period of time to be edible (for example, milk), and foods which need to be heated to be drinkable (for example, hot tea).

When the composition of the present invention is administered or ingested, the amount thereof for the administration or ingestion is selected as appropriate depending on the age, body weight, symptoms, health conditions, of a subject, the kind of the composition (a pharmaceutical product, a food or beverage, or the like), and the like. For example, the amount per administration or ingestion is generally 0.01 mg/kg body weight to 100 mg/kg body weight, and preferably 1 mg/kg body weight to 10 mg/kg body weight. The present invention also provides a method for suppressing the immunity of a subject, the method being characterized in that the bacteria belonging to the genus *Clostridium* or the physiologically active substance derived from the bacteria is administered into or ingested by the subject as described above.

A product of the composition of the present invention (a pharmaceutical product, a food or beverage, or a reagent) or a manual thereof may be provided with a note stating that the product can be used to suppress the immunity (including a note stating that the product has an immunosuppressive effect, and a note stating that the product has an effect of suppressing the proliferation or function of effector T-cells). Here, the "provision to the product or the manual thereof with the note" means that the note is provided to a main body, a container, a package, or the like of the product, or the note is provided to a manual, a package insert, a leaflet, or other printed matters, which disclose information on the product.

Method for Inducing Proliferation or Accumulation of Regulatory T Cells

As described above, and as will be shown in Examples, the administration of the composition of the present invention to an individual makes it possible to induce proliferation or accumulation of regulatory T cells in the individual. Thus, the present invention can provide a method for inducing proliferation or accumulation of regulatory T cells in an individual, the method comprising a step of administering, to the individual, at least one substance selected from the group consisting of the following (a) to (c):

- (a) bacteria belonging to the genus *Clostridium* or a physiologically active substance derived from the bacteria;
- (b) a spore-forming fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction; and
- (c) a chloroform-treated fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction.

Note that, the "individual" in the present invention is not particularly limited, and examples thereof include humans, various kinds of livestock, poultry, pets, experimental animals, and the like. The "individual" may be in a healthy state or a diseased state.

Moreover, as will be shown in Example 5 to be described later, Gram-positive commensal bacteria play principal roles in the proliferation or accumulation of regulatory T cells. Accordingly, the present invention can also provide a

method for inducing proliferation or accumulation of regulatory T cells in an individual, the method comprising a step of administering an antibiotic against Gram-negative bacteria to the individual.

In the present invention, the "antibiotic against Gram-negative bacteria" is not particularly limited, and examples thereof include aminoglycoside antibiotics (amikacin, gentamicin, kanamycin, neomycin, netilmicin, tobramycin, and paromomycin), cephalosporin antibiotics (cefactor, cefamandole, cefoxitin, cefprozil, cefuroxime, cefixime, cefdinir, cefditoren, cefoperazone, cefotaxime, ceftazidime, ceftibuten, ceftizoxime, ceftriaxone, and cefoxotin), sulfonamides, ampicillin, and streptomycin. Without wishing to be bound by theory, the "antibiotic against Gram-negative bacteria" according to the present invention is preferably one which reduces Gram-negative bacteria, and contributes to the colonization of Gram-positive bacteria.

Moreover, a prebiotic composition such as almond skin, inulin, oligofructose, raffinose, lactulose, pectin, hemicellulose (such as xyloglucan and alpha-glucans), amylopectin, and resistant starch which are not decomposed in the upper gastrointestinal tract and promote the growth of intestinal microbes in the intestinal tract, as well as growth factors such as acetyl-Co A, biotin, beet molasses, and yeast extracts, contribute to the proliferation of bacteria belonging to the genus *Clostridium*. Accordingly, the present invention can also provide a method for inducing proliferation or accumulation of regulatory T cells in an individual, the method comprising a step of administering, to the individual, at least one substance selected from the group consisting of these substances.

Meanwhile, in the "method for inducing proliferation or accumulation of regulatory T cells" of the present invention, the composition of the present invention, the above-described "antibiotic against Gram-negative bacteria," and the above-described "prebiotic composition or growth factor" may be used in combination. Such combined use is not particularly limited, and examples of the combined use are as follows: the "antibiotic against Gram-negative bacteria" is administered to an individual in advance, and then the composition of the present invention is administered; the "antibiotic against Gram-negative bacteria" and the composition of the present invention are simultaneously administered to an individual; the "prebiotic composition or growth factor" is administered to an individual in advance, and then the composition of the present invention is administered; the "prebiotic composition or growth factor" and the composition of the present invention are simultaneously administered to an individual; the composition of the present invention, the "antibiotic against Gram-negative bacteria," and the "prebiotic composition or growth factor" are administered to an individual simultaneously or individually at any appropriate time.

Moreover, a therapeutic composition may be administered to an individual together with at least one substance selected from the group consisting of the composition of the present invention, the "antibiotic against Gram-negative bacteria," and the "prebiotic composition or growth factor."

Such a therapeutic composition is not particularly limited, and may be at least one therapeutic composition selected from the group consisting of corticosteroids, mesalazine, mesalamine, sulfasalazine, sulfasalazine derivatives, immunosuppressive drugs, cyclosporin A, mercaptopurine, azathiopurine, prednisone, methotrexate, antihistamines, glucocorticoids, epinephrine, theophylline, cromolyn sodium, anti-leukotrienes, anti-cholinergic drugs for rhinitis, anti-cholinergic decongestants, mast-cell stabilizers, monoclonal

anti-IgE antibodies, vaccines (preferably, vaccines used for vaccination where the amount of an allergen is gradually increased), and combinations thereof. It is preferable to use these therapeutic compositions in combination with the above-described substance.

Moreover, there is no particular limitation imposed on the combined use of the therapeutic composition with at least one substance selected from the group consisting of the composition of the present invention, the “antibiotic against Gram-negative bacteria,” and the “prebiotic composition or growth factor”. For example, the “one substance” and the therapeutic composition are administered orally or parenterally to an individual simultaneously or individually at any appropriate time.

Moreover, in the above-described “method for inducing proliferation or accumulation of regulatory T cells,” whether or not the administration of the composition of the present invention or the like actually induces the proliferation or accumulation of regulatory T cells can be determined by using, as an index, increase or reinforcement of at least one selected from the group consisting of the number of regulatory T cells, the ratio of regulatory T cells in the T cell group of the colon, a function of regulatory T cells, and expression of a marker of regulatory T cells. It is preferable to use one measurement selected from the group consisting of promotion of IL-10 expression, promotion of CTLA4 expression, promotion of IDO expression, and suppression of IL-4 expression, as the index of the induction of proliferation or accumulation of regulatory T cells.

Note that examples of a method for detecting such expression include the northern blotting, the RT-PCR, and the dot blotting for detection of gene expression at the transcription level; and the ELISA, the radioimmunoassay, the immunoblotting, the immunoprecipitation, and the flow cytometry for detection of gene expression at the translation level.

Meanwhile, a sample used for measuring such an index is not particularly limited, and examples thereof include blood sampled from an individual and tissue pieces obtained in a biopsy.

Method for Predicting Response of Individual to Composition of Present Invention and/or Prognosis of Individual

The present invention can provide a method in which the absolute amount or the ratio of bacteria belonging to the genus *Clostridium* in a microbiota of an individual is determined, and, when the ratio or the absolute value of the bacteria belonging to the genus *Clostridium* is reduced in comparison with a base line value obtained by performing a similar determination on an individual in a typical health state, it is determined that the individual is possibly responsive to the composition of the present invention.

In one embodiment, a method to predict a subject's response to a substance and/or the subject's prognosis is provided. The method comprises measuring the percentage or absolute amounts of *Clostridium* clusters IV and XIV in the microbiota of the subject and comparing them to a baseline value of the same measurements in a prototypical healthy subject, wherein a decreased absolute amount or percentage level of *Clostridium* clusters IV and/or XIV indicates that the subject may respond favorably to the compositions of the invention.

In one embodiment, the method further comprises measuring the composition of the microbiota of the subject after administration of the substance, wherein an increase in the percentage or absolute number of *Clostridium* spp. belonging to clusters IV, XIV after administration of the compositions of the present invention relative to prior to the administering is a positive indicator of enhanced immuno-

suppression (or immunoregulation). The measurement of the composition of the subject's microbiota can be made with techniques known in the art, such as 16srRNA sequencing.

Note that, in these embodiments, the substance is at least one substance selected from the group consisting of the following (a) to (e):

- (a) bacteria belonging to the genus *Clostridium* or a physiologically active substance derived from the bacteria;
- (b) a spore-forming fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction;
- (c) a chloroform-treated fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction;
- (d) an antibiotic against Gram-negative bacteria according to the present invention; and
- (e) at least one substance selected from the group consisting of almond skin, inulin, oligofructose, raffinose, lactulose, pectin, hemicellulose (such as xyloglucan and alpha-glucans), amylopectin, acetyl-Co A, biotin, beet molasses, yeast extracts, and resistant starch.

Method for Inhibiting Proliferation or Accumulation of Regulatory T Cells

As will be shown in Example 5 to be described later, Gram-positive commensal bacteria have principal roles in the proliferation or accumulation of regulatory T cells. Accordingly, the present invention can also provide a method for inhibiting proliferation or accumulation of regulatory T cells in an individual, the method comprising a step of administering an antibiotic against Gram-positive bacteria to the individual.

In the present invention, the “antibiotic against Gram-positive bacteria” is not particularly limited, and examples thereof include cephalosporin antibiotics (cephalexin, cefuroxime, cefadroxil, cefazolin, cephalothin, cefaclor, cefamandole, cefoxitin, cefprozil, and ceftobiprole); fluoroquinolone antibiotics (cipro, Levaquin, floxin, tequin, avelox, and norflox); tetracycline antibiotics (tetracycline, minocycline, oxytetracycline, and doxycycline); penicillin antibiotics (amoxicillin, ampicillin, penicillin V, dicloxacillin, carbenicillin, vancomycin, and methicillin); and carbapenem antibiotics (ertapenem, doripenem, imipenem/cilastatin, and meropenem).

As described above, the “individual” in the present invention is not particularly limited, and examples thereof include humans, various kinds of livestock, poultry, pets, experimental animals, and the like. The “individual” may be in a healthy state or a diseased state. Such a diseased state is not particularly limited, and examples thereof include states of being subjected to cancer immunotherapy and of suffering from an infectious disease.

Moreover, as another mode of the “method for inhibiting proliferation or accumulation of regulatory T cells,” the present invention can provide a method for inhibiting proliferation or accumulation of regulatory T cells in an individual, the method comprising a step of administering, to the individual, any one of an antibody, an antibody fragment, and a peptide, which are against an antigen that is at least one substance selected from the group consisting of the following (a) to (c):

- (a) bacteria belonging to the genus *Clostridium* or a physiologically active substance derived from the bacteria;
- (b) a spore-forming fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction; and
- (c) a chloroform-treated fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction.

<Vaccine Composition and Method for Treating or Preventing Infectious Disease or Autoimmune Disease by Using the Vaccine Composition>

As described above, and as will be shown in Example 15 to be described later, the induction of Treg cells in the colon by the *Clostridium* has an important role in local and systemic immune responses. Accordingly, the present invention can also provide a “vaccine composition comprising at least one substance selected from the group consisting of the following (a) to (c): (a) bacteria belonging to the genus *Clostridium*; (b) a spore of bacteria in a spore-forming fraction of a fecal sample obtained from a mammal; and (c) bacteria in a chloroform-treated fraction of a fecal sample obtained from a mammal” and a “method for treating, aiding in treating, reducing the severity of, or preventing at least one disease selected from infectious diseases and autoimmune diseases in an individual, the method comprising administering the vaccine composition to the individual.”

Note that such “autoimmune diseases” are not particularly limited, and examples thereof include those described as the “specific examples of target diseases” in Composition Having Effect of Inducing Proliferation or Accumulation of Regulatory T cells. The “infectious diseases” are also not particularly limited, and examples thereof include infectious diseases associated with “infectious pathogens” described as the “example of infectious pathogens” in Composition Having Effect of Inducing Proliferation or Accumulation of Regulatory T cells.

Method for Screening for Compound Having Activity to Promote Proliferation or Accumulation of Regulatory T Cells

The present invention can also provide a method for screening for a compound having an activity to promote proliferation or accumulation of regulatory T cells, the method comprising:

- (1) preparing a test substance from at least one substance selected from the group consisting of the following (a) to (c):
 - (a) bacteria belonging to the genus *Clostridium* or a physiologically active substance derived from the bacteria;
 - (b) a spore-forming fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction; and
 - (c) a chloroform-treated fraction of a fecal sample obtained from a mammal or a culture supernatant of the fraction.
- (2) preparing non-human mammals in which a reporter gene is to be expressed under control of IL-10 gene expression;
- (3) bringing the test substance into contact with the non-human mammal;
- (4) after the contact with the test substance, detecting cells expressing the reporter gene in a CD4⁺ Foxp3⁺ cell group of the non-human mammal, and determining the number of cells in the CD4⁺ Foxp3⁺ cell group expressing the reporter gene or a ratio of cells in the CD4⁺ Foxp3⁺ cell group expressing the reporter gene to cells in the CD4⁺ Foxp3⁺ cell group not expressing the reporter gene;
- (5) detecting cells expressing the reporter gene in a CD4⁺ Foxp3⁺ cell group of the non-human mammal which has not been in contact with the test substance, and determining the number of cells in the CD4⁺ Foxp3⁺ cell group expressing the reporter gene or a ratio of cells in the CD4⁺ Foxp3⁺ cell group expressing the

reporter gene to cells in the CD4⁺ Foxp3⁺ cell group not expressing the reporter gene; and

- (6) comparing the absolute numbers or the ratios determined in steps (4) with the number or the ratio determined in (5), and determining, when the number or the ratio determined in (4) is greater than that determined in (5), that the test substance is a compound that promotes proliferation or accumulation of Treg cells.

The “test substance” according to the present invention is not particularly limited, as long as the test substance is a substance prepared from at least one substance selected from the group consisting of the substances (a) to (c). Examples of the test substance include proteins, polysaccharides, lipids, and nucleic acids which are derived from at least one substance selected from the group consisting of the above described substances (a) to (c).

The “non-human mammal in which a reporter gene is to be expressed under control of IL-10 gene expression” according to the present invention is not particularly limited, as long as the non-human mammal is a non-human mammal having a reporter gene whose expression is controlled by an IL-10 gene expression control region (for example, a promoter, or an enhancer). Examples of such a reporter gene include genes encoding fluorescent proteins (for example, GFP), and genes encoding luciferase. As the “non-human mammal in which a reporter gene is to be expressed under control of IL-10 gene expression” according to the present invention, an *IL10^{venus}* mouse to be shown later in Examples can be preferably used.

The “contact” according to the present invention is not particularly limited, and examples thereof include administration of the test substance to the non-human mammal orally or parenterally (for example, intraperitoneal injection, or intravenous injection).

The present invention can also provide a non-human mammal which is used for the method, and in which the reporter gene is to be expressed under the control of the IL-10 gene expression.

Furthermore, the present invention can also provide a method for isolating, from a sample of bacteria belonging to the genus *Clostridium*, a compound having an activity to promote proliferation or accumulation of regulatory T cells, the method comprising the following steps (1) to (3):

- (1) preparing a genomic DNA from the sample of bacteria belonging to the genus *Clostridium*;
- (2) inserting the genomic DNA into a cloning system, and preparing a gene library derived from the sample of bacteria belonging to the genus *Clostridium*; and
- (3) isolating a compound having an activity to promote proliferation or accumulation of regulatory T cells, by use of the gene library obtained in step (2).

In such steps, methods for the preparation and the isolation are not particularly limited, and known techniques for an in-vitro or in-vivo system can be used as appropriate. Moreover, the compound isolated by this method is not particularly limited, and examples thereof include nucleic acids (for example, a DNA, a mRNA, and a rRNA) derived from bacteria belonging to the genus *Clostridium*, as well as polypeptides and proteins derived from the bacteria belonging to the genus *Clostridium*.

Other Embodiment Modes According to Present Invention

In addition to the above-described embodiment modes, the present invention can also provide the following embodiment modes.

Specifically, the present invention can also provide a method for determining the composition of a microbiota in an individual, wherein the increase in the ratio or the absolute number of bacteria belonging to the genus *Clostridium* after the administration of the composition of the present invention to the individual with respect to the ratio or the absolute number before the administration is used as an index of increased immunosuppression. In such a method, the method for determining the composition of the microbiota is not particularly limited, and known techniques (for example, 16S rRNA sequencing) can be used as appropriate.

The present invention can also provide a method for measuring differentiation of Treg cells, wherein the increase in differentiation of Treg cells in an individual after administration of the composition of the present invention to the individual with respect to that before the administration is used as an index of increased immunosuppression (or immunoregulation).

Moreover, the composition of the present invention can also be administered to an individual under an antibiotic treatment. The timing of the administration is not particularly limited, and the composition of the present invention can be administered before or simultaneously with the antibiotic treatment, for example. Meanwhile, the composition of the present invention is preferably administered in the spore form from the viewpoint of resistance to antibiotics.

Moreover, in a preferred mode of such administration, the composition of the present invention is administered after or simultaneously with administration of an antibiotic against Gram-positive bacteria, for example. Note that such an "antibiotic against Gram-positive bacteria" is not particularly limited, and examples thereof include cephalosporin antibiotics (cephalexin, cefuroxime, cefadroxil, cefazolin, cephalothin, cefaclor, cefamandole, cefoxitin, cefprozil, and ceftibiprole); fluoroquinolone antibiotics (cipro, Levaquin, floxin, tequin, avelox, and norflox); tetracycline antibiotics (tetracycline, minocycline, oxytetracycline, and doxycycline); penicillin antibiotics (amoxicillin, ampicillin, penicillin V, dicloxacillin, carbenicillin, vancomycin, and methicillin); and carbapenem antibiotics (ertapenem, doripenem, imipenem/cilastatin, and meropenem).

Meanwhile, in another preferred mode of such administration, the composition of the present invention is administered after (or simultaneously with) a treatment using vancomycin, metronidazole, linezolid, ramoplanin, or fidaxomicin, for example.

EXAMPLES

Hereinafter, the present invention is described more specifically on the basis of Examples. However, the present invention is not limited to Examples below.

Note that mice used in Examples were prepared or produced as follows. In the following description, mice may be referred to with "SPF" or "GF" attached in front thereof. These "SPF" and "GF" indicate that the mice were maintained in the absence of specific pathogenic bacteria (specific pathogen-free, SPF), and that the mice were maintained under Germ-Free (GF) conditions, respectively.

Mice

C57BL/6, Balb/c, and IQI mice maintained under SPF or GF conditions were purchased from Sankyo Labo Service Corporation, Inc. (Japan), JAPAN SLC, INC. (Japan), CLEA Japan, Inc. (Japan), or The Jackson Laboratory (USA). GF mice and gnotobiotic mice were bred and

maintained within the gnotobiotic facility of The University of Tokyo, Yakult Central Institute for Microbiological Research, or Sankyo Labo Service Corporation, Inc. *Myd88^{-/-}*, *Rip2^{-/-}*, and *Card9^{-/-}* mice were produced as described in Non-Patent Documents 1 to 3, and backcrossed for 8 generations or more, so that a C57BL/6 genetic background was achieved. *Foxp3^{eGFP}* mice were purchased from the Jackson Laboratory.

Il10^{venus} Mice

To form a bicistronic locus encoding both Il10 and Venus under control of an Il10 promoter, a targeting construct was first created. Specifically, a cassette (IRES-Venus-SV40 polyA signal cassette, refer to Non-Patent Document 4) which was made of an internal ribosome entry site (IRES), a yellow fluorescent protein (Venus), and a SV40 polyA signal (SV40 polyA) and which was arranged next to a neomycin-resistant gene (neo), was inserted between a stop codon and a polyA signal (Exon 5) of a Il10 gene. Next, the obtained targeting construct was used to cause homologous recombination with the Il10 gene region in the genome of mice. Thus, Il10^{venus} mice having an Il10^{venus} alleles were produced (refer to FIG. 1). Note that in FIG. 1 "tk" represents a gene coding thymidine kinase, "neo" represents the neomycin-resistant gene, and "BamH1" represents a cleavage site by the restriction enzyme BamH1.

Genomic DNAs were extracted from the Il10^{venus} mice, treated with BamH1, and Southern blotted by use of a probe shown in FIG. 1. FIG. 2 shows the obtained results. Wild-type and Il10^{venus} alleles were detected as bands having sizes of 19 kb and 5.5 kb, respectively. Hence, as is apparent from the results shown in FIG. 2, it was found that the homologous recombination shown in FIG. 1 occurred in the genome of the Il10^{venus} mice.

Further, CD4⁺ Venus⁻ cells or CD4⁺ Venus⁺ cells in the colonic lamina propria of the Il10^{venus} mice were sorted by use of a FACS Aria. Then, real-time RT-PCR was carried out on an ABI 7300 system by a method to be described later, to determine the amount of IL-10 mRNA expressed. FIGS. 3 and 4 show the obtained results. As is apparent from the results shown in FIGS. 3 and 4, it was found that, since the development of the IL-10 mRNA was detected only in the CD4⁺ Venus⁺ cells, the expression of IL-10 mRNA in the Il10^{venus} mice was correctly reflected in the expression of Venus. Note that the germ-free states of such Il10^{venus} mice were established in Central Institute for Experimental Animals (Kawasaki, Japan). The Il10^{venus} mice in the germ-free states were maintained in vinyl isolators in Sankyo Labo Service Corporation, Inc. (Tokyo, Japan), and used in the following Examples.

Meanwhile, experiments and analyses in Examples were carried out as follows.

Method for Colonization of Mice with Bacteria and Analysis Thereof

According to the description in Non-Patent Documents 5 and 6, mice in which SFB or *Clostridium* were colonized were produced. Cecal contents or feces of the obtained gnotobiotic mice were dissolved in sterile water or an anaerobic dilution solution. The dissolved cecal contents or feces as they were or after a chloroform treatment were orally administered to GF mice. Three strains of the *Lactobacillus* and 16 strains of the *Bacteroides* were cultured separately from each other in a BL or EG agar medium in an anaerobic manner. The cultured bacteria were harvested, suspended in an anaerobic TS broth, and orally administered forcibly to GF mice. The state of the colonization of the bacteria in the mice was assessed by microscopic observation conducted on a smear preparation of fecal pellets.

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Cell Separation and Flow Cytometry

In order to isolate lymphocytes from the colonic lamina propria and the small intestinal lamina propria, the small intestine and the colon were collected, and cut open longitudinally. Then, fecal content and the like there inside were washed to remove. Subsequently, the small intestine and the colon were shaken in HBSS containing 5 mM of EDTA at 37° C. for 20 minutes. After removal of epithelium and fat tissue, the intestinal tissues were cut into small pieces. To the small pieces, RPMI 1640 (4% fetal bovine serum (FBS), 1 mg/ml of collagenase D, 0.5 mg/ml of dispase, and 40 µg/ml of DNaseI (all of which were manufactured by Roche Diagnostics K.K.)) were added, and the mixture was shaken in a water bath kept at 37° C. for 1 hour. The digested tissues were washed with HBSS containing 5 mM of EDTA, and resuspended in 5 ml of 40% PERCOLL® (GE Healthcare). The suspension was overlaid on 2.5 ml of 80% PERCOLL® in a 15-ml Falcon tube. Then, centrifugation was carried out at room temperature and at 2000 rpm for 20 minutes to conduct cell separation by PERCOLL® density gradient centrifugation. Cells at the interface were collected, and used as lamina propria lymphocytes. The collected cells were suspend in a staining buffer (PBS, 2% FBS, 2 mM EDTA, and 0.09% Na₂S₂O₃), and stained by use of an anti-CD4 antibody (RM4-5, BD Biosciences) labeled with PE or PE-Cy7. After the staining of CD4, Foxp3 in the cells were stained by use of CYTOFIX/CYTOPERM® Kit Plus with GOLGISTOP® (BD Biosciences) or Foxp3 Staining Buffer Set (eBioscience), as well as an anti-Foxp3 antibody (FJK-16s, eBioscience) labeled with Alexa647. Flow cytometry was performed by use of a FACScant II, and the data were analyzed by FLOWJO® software (TreeStar Inc.). The sorting of the cells were performed by use of a FACSARIA™.

Real-Time RT-PCR

From an RNA prepared by using RNeasy Mini Kit (Qiagen), a cDNA was synthesized by use of a MMV reverse transcriptase (Promega KK). The obtained cDNA was analyzed by real-time RT-PCR using Power SYBR Green PCR Master Mix (Applied Biosystems) and ABI 7300 real time PCR system (Applied Biosystems), or real-time RT-PCR using SYBR Premix Ex Taq (TAKARA) and Light Cycler 480. For each sample, a value obtained was normalized for the amount of GAPDH. A primer set was designed by using Primer Express Version 3.0 (Applied Biosystems), and those exhibiting a 90% or higher sequence identity at an initial evaluation were selected. The primer set used was as follows:

Foxp3 (SEQ ID NO: 1)
5' -GGCAATAGTTCCTTCCAGAGTT-3'
(SEQ ID NO: 2)
5' -GGGTCGCATATTGTGGTACTTG-3'
CTLA4 (SEQ ID NO: 3)
5' -CCTTTTGTAGCCCTGCTCACTCT-3'
(SEQ ID NO: 4)
5' -GGGTCACCTGTATGGCTTCAG-3'
GATR (SEQ ID NO: 5)
5' -TCAGTGCAAGATCTGCAAGCA-3'
(SEQ ID NO: 6)
5' -ACACCGGAAGCCAAACACA-3'

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IL-10 (SEQ ID NO: 7)
5' -GATTTTAATAAGCTCCAAGACCAAGGT-3'
(SEQ ID NO: 8)
5' -CTTCTATGCAGTTGATGAAGATGTCAA-3'
GAPDH (SEQ ID NO: 9)
5' -CCTCGTCCCGTAGACAAAATG-3'
(SEQ ID NO: 10)
5' -TCTCCACTTTGCCACTGCAA-3'
Mmp2 (SEQ ID NO: 11)
5' -GGACATTGTCTTTGATGGCA-3'
(SEQ ID NO: 12)
5' -CTTGTCACGTGGTGTCTACTG-3'
Mmp9 (SEQ ID NO: 13)
5' -TCTCTGGACGTCAAATGTGG-3'
(SEQ ID NO: 14)
5' -GCTGAACAGCAGAGCCTTC-3'
Mmp13 (SEQ ID NO: 15)
5' -AGGTCTGGATCACTCCAAGG-3'
(SEQ ID NO: 16)
5' -TCGCCTGGACCATAAAGAA-3'
Ido1 (SEQ ID NO: 17)
5' -AGAGGATGCGTGACTTTGTG-3'
(SEQ ID NO: 18)
5' -ATACAGCAGACCTTCTGGCA-3'

Preparation and Culturing of Large Intestinal Epithelial Cells (IECs)

First, the colon was collected, cut open longitudinally, and rinsed with PBS. Subsequently, the colon was treated with 1 mM dithiothreitol (DTT) at 37° C. for 30 minutes on a shaker, and then vortexed for one minute to disrupt the epithelial integrity. The released IECs were collected, and suspended in 5 ml of 20% PERCOLL®. The suspension was overlaid on 2.5 ml of 80% PERCOLL® in a 15-ml Falcon tube. Then, the tube was centrifuged at 25° C. and 780 g for 20 minutes to conduct cell separation by PERCOLL® density gradient centrifugation. Cells at the interface were collected, and used as colonic IECs (purity: 90% or higher, viability: 95%). The obtained IECs thus collected were suspended in RPMI containing 10% FBS, and 1×10⁵ cells of the IECs were cultured in a 24-well plate for 24 hours. Thereafter, the culture supernatant was collected, and measured for active TGF-β1 level by ELISA (Promega).

Meanwhile, for culturing T cells in vitro, 1.5×10⁵ MACS-purified splenic CD4⁺ T cells were cultured in each well of a round-bottomed 96-well plate, together with a 50% conditioned medium in which IECs isolated from GF mice or *Clostridium*-colonized mice were cultured, and with 25 ng/ml of hIL-2 (Peprotech), in the presence or absence of 25 µg/ml of an anti-TGF-β antibody (R&D). Note that 10 µg/ml of an anti-CD3 antibody and an anti-CD28 antibody (BD Bioscience) were bound to the round-bottomed plate. After a 5-day culture, the CD4⁺ T cells were collected, and subjected to a real-time PCR.

Colitis Experimental Model

A fecal suspension of *Clostridium*-colonized mice was orally administered to C57BL/6 mice (2-week old), and grown in a conventional environment for six weeks.

For preparing a DSS-induced colitis model, 2% (wt/vol) DSS (reagent grade, DSS salt, molecular weight=36 to 50 kD, manufactured by MP Biomedicals), together with drinking water, was given to the mice for six days.

Meanwhile, for preparing an oxazolone-induced colitis model, the mice were presensitized by transdermally applying, onto the mice, 150 μ l of a 3% oxazolone (4-ethoxymethylene-2-phenyl-2-oxazolin-5-one, Sigma-Aldrich)/100% ethanol solution. Five days after that, 150 μ l of a 1% oxazolone/50% ethanol solution was intrarectally administered again to the presensitized mice under a light anesthesia. Note that the intrarectal administration was conducted by using a 3.5F catheter.

Each mouse was analyzed daily for body weight, occult blood, bleeding visible with the naked eyes (gross blood), and the hardness of stool. Moreover, the body weight loss percentage, intestinal bleeding (no bleeding, occult blood (hemocult+), or bleeding visible with the naked eyes), and the hardness of stool (normal stool, loose stool, or diarrhea) were evaluated numerically, and the disease activity index (DAI) was calculated in accordance with the description in "S. Wirtz, C. Neufert, B. Weigmann, M. F. Neurath, Nat Protoc 2, 541 (2007)."

OVA Specific IgE Reaction

BALB/c SPF mice were inoculated with a fecal suspension of *Clostridium*-colonized mice (2-week old), and grown in a conventional environment. Then, 1 μ g of OVA (grade V, Sigma) and 2 mg of alum (Thermo Scientific), 0.2 ml in total, were intraperitoneally injected to the mice (at their ages of 4 weeks and 6 weeks). Sera were collected every week from the mice at the root of their tail, and OVA-specific IgE was measured by ELISA (Chondrex). Then, at their ages of 8 weeks, splenic cells were collected, inoculated in a 96-well plate at 1×10^6 cells per well, and stimulated with OVA (100 μ g/ml) for three days. Thereafter, the culture supernatant was collected, and measured for IL-4 and IL-10 levels by ELISA (R&D).

Statistical Analysis

The difference between control and experimental groups was evaluated by the Student's t-test.

Example 1

First, it was investigated whether or not accumulation of regulatory T cells (Treg cells) in the colonic lamina propria was dependent on commensal bacteria. Specifically, lymphocytes were isolated from peripheral lymph nodes (pLN) of Balb/c mice bred in the absence of specific pathogenic bacteria (SPF) or from lamina propria of the colon or the small intestine (SI) of the mice. The CD4 and Foxp3 were stained by antibodies. Then, the ratio of Foxp3⁺ cells in CD4⁺ lymphocytes was analyzed by flow cytometry. FIG. 5 shows the obtained results. As is apparent from the results shown in FIG. 5, it was found that Foxp3⁺ Treg cells were present at a high frequency in the lamina propria of the gastrointestinal tracts, especially in the colonic lamina propria, of the mice kept under the environment free from specific pathogenic microorganisms (SPF). In addition, it was also found that the number of the Foxp3⁺ Treg cells in the colonic lamina propria gradually increased up to three months after their birth, whereas the number of the Foxp3⁺ Treg cells in the peripheral lymph nodes was basically constant from the time of two weeks after their birth.

Example 2

Next, it was investigated whether or not the temporal accumulation of the Treg cells in the colon as found in

Example 1 had a relationship with the colonization of intestinal commensal microbiota. Specifically, the expression of CD4 and the expression of Foxp3 in lymphocytes isolated from the small intestine, the colon, and the peripheral lymph nodes of mice bred under a germ-free (GF) or SPF environment (8 weeks old: Balb/c mice, IQI mice, and C57BL/6 mice) were analyzed. Similar results were obtained in three or more independent experiments. FIGS. 6 and 7 show the obtained results. Note that, in FIG. 7, each white circle represents the absolute number of CD4⁺ Foxp3⁺ cells in an individual mouse, and the error bars represent standard deviations (SDs).

In addition, lamina propria lymphocytes were collected from SPF mice and GF mice (Balb/c mice or C57BL/6 mice). CD4 and Foxp3 were stained with antibodies. Then, the lamina propria lymphocytes were analyzed by FACS. FIG. 8 shows the obtained results. Note that in FIG. 8 each white circle represents the absolute number of CD4⁺ Foxp3⁺ cells in an individual mouse, ** indicates that "P<0.001", and * indicates that "P<0.01."

Further, lymphocytes were isolated from the lamina propria of the colon, the lamina propria of the small intestine (SI), Peyer's patches (PPs), and mesenteric lymph nodes (MLNs) of mice (SPF C57BL/6 mice) to which antibiotics were orally administered with water for eight weeks. CD4 and Foxp3 were stained with antibodies. Then, the lymphocytes were analyzed by FACS. Similar results were obtained in two or more independent experiments. FIG. 9 shows the obtained results (the ratio of the Foxp3⁺ cells in the CD4⁺ cells of an individual mouse). Note that the following antibiotics were used in combination in accordance with the description in the following document:

ampicillin (A; 500 mg/L, Sigma)
vancomycin (V; 500 mg/L, NACALAI TESQUE, INC.)
metronidazole (M; 1 g/L, NACALAI TESQUE, INC.)
neomycin (N; 1 g/L, NACALAI TESQUE, INC.)
Rakoff-Nahoum, J. Paglino, F. Eslami-Varzaneh, S. Edberg, R. Medzhitov, Cell 118, 229 (Jul. 23, 2004)

Fagarasan et al., Science 298, 1424 (Nov. 15, 2002)
In FIG. 9, each white circle represents the absolute number of the CD4⁺ Foxp3⁺ cells in an individual mouse, each horizontal bar represents the average value of the absolute numbers, * indicates that "P<0.01," and "AVMN" represents the kinds of the administered antibiotics by using the first letters of the antibiotics.

As is apparent from the results shown in FIGS. 6 to 9, the frequencies and the absolute numbers of Foxp3⁺ CD4⁺ cells in the small intestine and the peripheral lymph nodes of the GF mice were equal to or greater than those of the SPF mice (refer to FIGS. 6 to 8). In addition, the numbers of the Treg cells in the small intestinal lamina propria, Peyer's patches, and mesenteric lymph nodes of the SPF mice to which the antibiotics were orally administered for eight weeks were equal to or greater than those of the SPF mice (refer to FIG. 9). Meanwhile, the number of the Foxp3⁺ CD4⁺ cells in the colonic lamina propria of the GF mice was decreased significantly in comparison with that of the SPF mice (refer to FIGS. 6 and 7). This decrease was commonly observed among mice of different genetic backgrounds (Balb/c, IQI, and C57BL/6), as well as among mice bred in different animal facilities (refer to FIG. 7 for the data regarding the different genetic backgrounds, the data regarding the mice bred in the different animal facilities are not shown in the drawings). In addition, it was also shown that the number of Treg cells in the colonic lamina propria of the SPF C57BL/6 mice to which the antibiotics were administered was decreased significantly (refer to FIG. 9).

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Example 3

Next, it was directly checked whether or not the decrease in the number of the Treg cells in the colonic lamina propria of the GF mice shown in Example 2 was attributed to the absence of microbiota. Specifically, a fecal suspension of B6 SPF mice purchased from The Jackson Laboratory was orally administered to GF-IQI mice (conventionalization). Three weeks after the administration, lymphocytes were isolated from the colonic lamina propria, and the expression of Foxp3 in CD4⁺ lymphocytes was analyzed. FIGS. 10 and 11 show the obtained results. Note that each white circle in FIG. 11 represents the absolute number of CD4⁺ Foxp3⁺ cells in an individual mouse, the error bars represent standard deviations (SD), * indicates that “P<0.01” in Student’s t-test, and ** indicates that “P<0.001.” As is apparent from the results shown in FIGS. 10 and 11, the number of Treg cells in the small intestinal lamina propria did not change. However, the number of the Treg cells in the colonic lamina propria increased significantly. Hence, it was shown that host-microbial interaction played an important role in the accumulation of Foxp3⁺ Treg cells in the colonic lamina propria, while the accumulation of the Treg cells in the small intestinal lamina propria had a different mechanism.

Example 4

Next, the relationship between the gut-associated lymphoid tissues of mice and the number of Foxp3⁺ cells in the colonic lamina propria of the mice was investigated in accordance with the method described in M. N. Kweon et al., J Immunol 174, 4365 (Apr. 1, 2005). Specifically, 100 µg of an extracellular domain recombinant protein (a fusion protein (LTβR-Ig) between a lymphotoxin β receptor (LTβR) and a Fc region of human IgG1, refer to Honda et al., J Exp Med 193, 621 (Mar. 5, 2001)) was injected intraperitoneally into pregnant C57BL/6 mice 14 days after conception. The LTβR-Ig was again injected intraperitoneally into fetuses obtained from such mice, so that mice from which isolated lymphoid follicles (ILFs), Peyer’s patches (PPs), and colonic-patches (CPs) were completely removed were produced. Then, the ratios of Foxp3⁺ cells in CD4⁺ cells in the colonic lamina propria of the mice treated with the LTβR-Ig, and mice treated with rat IgG (control) were analyzed by FACS. FIG. 12 shows the obtained results. Note that in FIG. 12 each white circle represents the ratio of Foxp3⁺ cells in an individual mouse, and the error bars represent standard deviations. As is apparent from the results shown in FIG. 12, it was found that the ratio of the Foxp3⁺ cells in the colonic lamina propria of the mice deficient in isolated lymphoid follicles, Peyer’s patches, and the colonic-patches (the mice treated with the LTβR-Ig) rather increased. Accordingly, it was suggested that the decrease in the number of the Treg cells in the colonic lamina propria of the GF mice and the mice treated with the antibiotics was caused because the transmission of specific signals which promotes the accumulation of Treg cells in the colonic lamina propria and which is caused by the intestinal microbes did not occur, rather than simply because of a secondary effect of disorganized gut-associated lymphoid tissues.

Example 5

To investigate whether or not a specific intestinal flora induced the accumulation of colonic Treg cells, vancomycin as an antibiotic against Gram-positive bacteria or polymyxin

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B as an antibiotic against Gram-negative bacteria was administered to SPF mice (from 4 weeks of age) for four weeks, and analyzed for the ratio of Foxp3⁺ cells in the CD4⁺ cell group ([%] Foxp3⁺ in CD4). FIG. 30 shows the obtained results. Note that, in FIG. 30, “SPF” indicates the result of SPF mice (control), “poly B” indicates the result of the SPF mice to which polymyxin B was administered, and “Vanco.” indicates the result of the SPF mice to which vancomycin was administered. Meanwhile, * indicates that “P<0.01.”

As is apparent from the results shown in FIG. 30, the number of Treg cells in the colon of the mice to which vancomycin was administered was markedly decreased in comparison with that of the control. In contrast, no influence was observed on the number of Treg cells of the mice to which polymyxin B was administered. Those facts suggested that Gram-positive commensal bacteria played a major role in accumulation of Treg cells.

Example 6

A recent report has suggested that spore-forming bacteria play an important role in intestinal T cells response (see V. Gaboriau-Routhiau et al., Immunity 31, 677 (Oct. 16, 2009)). In this respect, fecal microorganisms (spore-forming fraction) resistant to 3% chloroform were orally administered to GF mice, which were then analyzed for the ratio of Foxp3⁺ cells in the CD4⁺ cell group ([%] Foxp3⁺ in CD4). FIG. 31 shows the obtained results. Note that, in FIG. 31, “GF” indicates the result of GF mice, and “+chloro” indicates the result of the GF mice to which the chloroform-treated feces were administered. Meanwhile, ** indicates that “P<0.001.”

As is apparent from the results shown in FIG. 31, three weeks after the administration of the chloroform-treated feces, the number of Treg cells in the administered mice was markedly increased to the same level as those of the SPF mice and the GF mice to which the untreated feces was forcibly administered (see FIGS. 7 and 11).

Accordingly, considering the results shown in Example 5 in combination, it was revealed that the specific components of the indigenous microbiota were highly likely to belong to the Gram-positive group, and that the spore-forming fraction played an important role in the induction of Treg cells.

Example 7

Next, the species of the intestinal microbiota which induced the accumulation of Treg cells in the colon as suggested in Examples 4 to 6 were identified. Specifically, segmented filamentous bacteria (SFB), 16 strains of the *Bacteroides* spp. (Bactero. (6 strains of *B. vulgatus*, 7 of the *B. acidifaciens* group 1, and 3 of the *B. acidifaciens* group 2)), 3 strains of the *Lactobacillus* (Lacto. (*L. acidophilus*, *L. fermentum*, and *L. murinum*)), and 46 strains of *Clostridium* spp. (Clost., refer to “Itoh, K., and Mitsuoka, T. Characterization of clostridia isolated from faeces of limited flora mice and their effect on caecal size when associated with germ-free mice. Lab. Animals 19: 111-118 (1985))”), or microbiota collected from mice (SPF) bred under a conventional environment was orally administered to GF-Balb/c mice or GF-IQI mice. The mice were maintained in vinyl isolators for three weeks. Then, CD4 cells were isolated from the colon and the small intestine of these mice. The numbers of Treg cells in the colon and the small intestine were analyzed by flow cytometry.

FIG. 13 shows FACS dot-plots obtained when a gate was set on CD4⁺ cells of the Balb/c mice. FIG. 14 shows the ratio of Foxp3⁺ cells in CD4⁺ cells of each mouse.

Note that, the bacteria belonging to the genus *Clostridium* are classified by sequencing of 16S rRNA gene, as follows. Specifically, the 16S rRNA genes of the bacteria were amplified by PCR using 16S rRNA gene-specific primer pairs: 5'-AGAGTTTGTATCMTGGCTCAG-3' (SEQ ID NO: 19) and 5'-ATTACCGCGGCKGCTG-3' (SEQ ID NO: 20) (see T. Aebischer et al., Vaccination prevents *Helicobacter pylori*-induced alterations of the gastric flora in mice. FEMS Immunol. Med. Microbiol. 46, 221-229(2006)). The 1.5-kb PCR product was then introduced into pCR-Blunt Vector. The inserts were sequenced and aligned using the ClustalW software program. The resulting sequences of 16S rRNA genes derived from strain 1-41 of 46 strains of *Clostridium* spp. were shown in SEQ ID NO: 21-61. Phylogenetic tree which was constructed by the neighbor-joining method with the resulting sequences of the 41 strains of *Clostridium* and those of known bacteria obtained from Genbank database using Mega software was shown in FIG. 49.

As is apparent from the results shown in FIGS. 13 and 14, no effect on the number of the Treg cells in the colon was observed in the GF mice in which the segmented filamentous bacteria (SFB) were colonized (refer to FIG. 14). Moreover, mice in which the cocktail of three strains of *Lactobacillus* was colonized gave similar results (refer to FIG. 14). On the other hand, it was shown that the accumulation of Foxp3⁺ cells in the colonic lamina propria was strongly induced in the mice in which 46 strains of *Clostridium* spp. were colonized. Importantly, such accumulation was promoted irrespective of the genetic backgrounds of the mice, and led to the increase in number similar to that in the SPF mice although intestinal microbiota of only a single genus were colonized. It was also shown that the colonization of the *Clostridium* did not change the number of Treg cells in the small intestinal lamina propria (refer to FIG. 14). Note that, when the 16 strains of *Bacteroides* spp. were colonized, the number of Treg cells in the colon was increased significantly. However, the extent of the increase varied depending on the genetic background of the mice in which the bacteria were colonized (refer to FIGS. 13 and 14).

Example 8

Next, CD4 expression, Foxp3 expression, and Helios expression in LP lymphocytes of the thymuses and the colons of SPF mice, GF mice, *Lactobacillus*-colonized mice, and *Clostridium*-colonized mice were analyzed by flow cytometry.

FIGS. 32 and 33 show the obtained results. Note that, in FIGS. 32 and 33, "GF" or "Germ Free" indicates the results of the GF mice, "SPF" indicates the results of the SPF mice, "Lacto." indicates the results of the *Lactobacillus*-colonized mice, and "Clost." indicates the results of the *Clostridium*-colonized mice. In FIG. 32, the vertical axis represents the ratio of Helios⁺ cells in the Foxp3⁺ cell group ([%] Helios⁺ in Foxp3⁺), and ** indicates that "P<0.001."

As is apparent from the results shown in FIGS. 32 and 33, most Foxp3⁺ cells found in the SPF mice or the *Clostridium*-colonized mice did not express Helios. Note that Helios is a transcription factor known to be expressed in thymic-derived natural Treg cells (see A. M. Thornton et al., J Immunol 184, 3433 (Apr. 1, 2010)). Accordingly, it was suggested that most of the Treg cells in the SPF mice and the

Clostridium-colonized mice were Treg cells induced in peripheral portions, i.e., so-called iTreg cells.

Example 9

Next, it was investigated whether or not the colonization of the *Clostridium* or the like had an influence on other T cells. Specifically, SFB, 16 strains of *Bacteroides* spp. (Bactero.), 46 strains of *Clostridium* spp. (Clost.), or microbiota collected from mice bred under a conventional environment (SPF) was colonized in GF IQL mice. Three weeks later, lymphocytes in the colonic lamina propria were isolated from these mice, and stimulated with PMA (50 ng/ml) and ionomycin (1 µg/ml) for four hours in the presence of GOLGISTOP® (BD Bioscience). After the stimulation was given, intracellular cytokines were stained by using an anti-IL-17 PE antibody (TC11-18H10) and an anti-IFN-γ FITC antibody (BD Bioscience) in accordance with the manual of a CYTOFIX/CYTOPERM® kit (BD Bioscience). Then, the ratio of IFN-γ⁺ cells or IL-17⁺ cells in CD4⁺ leucocytes was analyzed by flow cytometry. FIGS. 15 and 16 show the obtained results. Note that, in FIGS. 15 and 16, each white circle represents the absolute number of CD4⁺ IFN-γ⁺ cells or the absolute number of CD4⁺ IL-17⁺ cells in each individual mouse, and the error bars represent standard deviations (SD). As is apparent from the results shown in FIGS. 15 and 16, the colonization of the *Clostridium* did not have any influence on Th1 cells (CD4⁺ IFN-γ⁺ cells) in the colon, and caused only a slight increase of Th17 cells (CD4⁺ IL-17⁺ cells). Accordingly, it was suggested that the genus *Clostridium* was a genus of bacteria which specifically induced Treg cells.

Example 10

It has been reported that 46 strains of *Clostridium* spp. exert an influence on the accumulation of CD8⁺ intestinal tract intraepithelial lymphocytes (IELs) in the colon. Accordingly, it is conceivable that *Clostridium* regulates the immune system in various aspects, and that *Clostridium* exhibits a marked ability to induce and maintain Treg cells especially in the colon, as described above. In addition, a kind of cytokines, transforming growth factor-β (TGF-β), is known to play an important role in regulation of Treg cell generation.

In this respect, it was examined whether or not the colonization of *Clostridium* provided a colonic environment rich in TGF-β. Specifically, first, the whole colons of GF mice, *Clostridium*-colonized mice, and *Lactobacillus*-colonized mice were cultured for 24 hours, and the culture supernatants thereof were measured for the concentration of active TGF-β (TGF-β1) by ELISA (the number of mice analyzed was four per group). FIG. 34 shows the obtained results. Note that, in FIG. 34, "GF" indicates the result of the GF mice, "Clost." indicates the result of the *Clostridium*-colonized mice, and "Lacto." indicates the result of *Lactobacillus*-colonized mice. Meanwhile, * indicates that "P<0.02," and ** indicates that "P<0.001."

As is apparent from the results shown in FIG. 34, the amount of TGF-β produced in the colons of the *Clostridium*-colonized mice was significantly larger than those of the GF mice and the *Lactobacillus*-colonized mice.

Next, intestinal epithelial cells (IECs) of GF mice and *Clostridium*-colonized mice were cultured for 24 hours, and the culture supernatants thereof were measured for the concentration of active TGF-β (TGF-β1) by ELISA (the number of mice analyzed was four per group). FIG. 35

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shows the obtained results. Note that, in FIG. 35, "GF" indicates the result of the GF mice, and "Clost." indicates the result of the *Clostridium*-colonized mice. Meanwhile, ** indicates that " $P < 0.001$."

As is apparent from the results shown in FIG. 35, TGF- β was detected in the culture supernatant of the IECs isolated from the *Clostridium*-colonized mice, whereas no TGF- β was detected in the culture supernatant of the IECs isolated from the GF mice.

Next, as described above, splenic CD4⁺ T cells were cultured for five days together with a 50% conditioned medium in which IECs isolated from the GF mice or the *Clostridium*-colonized mice were cultured, and with the anti-CD3 antibody, in the presence or absence of an anti-TGF- β antibody. Then, the T cells were collected, and analyzed for expression of Foxp3 by real-time RT-PCR. FIG. 36 shows the obtained results. Note that, in FIG. 36, "Medium" indicates the result of a medium in which no cells were cultured, "GF" indicates the result of the conditioned medium in which the IECs of the GF mice were cultured, "Clost." indicates the result of the conditioned medium in which the IECs of the *Clostridium*-colonized mice were cultured, and "Clost.+ α TGF β " indicates the result of the conditioned medium to which the anti-TGF- β antibody was added and in which the IECs of the *Clostridium*-colonized mice were cultured. Meanwhile, ** indicates that " $P < 0.001$."

As is apparent from the results shown in FIG. 36, when the culture supernatant of the IECs derived from the *Clostridium*-colonized mice was added to the splenic CD4⁺ T cells, the differentiation into Foxp3-expressing cells was accelerated. Meanwhile, the differentiation into the Treg cells was inhibited by the anti-TGF- β antibody.

Moreover, the expression of MMP2, MMP9, and MMP13, which are thought to contribute to the activation of latent TGF- β was investigated. The expression of indoleamine 2,3-dioxygenase (IDO), which is thought to be involved in the induction of Treg cells, was also investigated. Specifically, 46 bacterial strains of the genus *Clostridium* (Clost.), or three bacterial strains of the genus *Lactobacillus* (Lacto.) were orally administered to C57BL/6 germ-free mice. Three weeks after the administration, IECs were collected, and analyzed for relative mRNA expression levels of MMP2, MMP9, MMP13, and IDO genes by real-time RT-PCR (the number of mice analyzed was three per group). FIGS. 37 to 40 show the obtained results. Note that, in FIGS. 37 to 40, "GF #1 to 3" indicate the results of GF mice, "Clost. #1 to 3" indicate the results of the *Clostridium*-colonized mice, and "Lacto. #1 to 3" indicate the results of the *Lactobacillus*-colonized mice.

For the relationship between the activation of latent TGF- β and the above-described MMP, see D'Angelo et al., J. Biol. Chem. 276, 11347-11353, 2001; Heidinger et al., Biol. Chem. 387, 69-78, 2006; Yu et al., Genes Dev. 14, 163-176, 2000. For the relationship between IDO and the induction of Treg cells, see G. Matteoli et al., Gut 59, 595 (May, 2010).

As is apparent from the results shown in FIGS. 37 to 39, in agreement with the production of TGF- β described above, transcription products of the genes encoding MMP2, MMP9, and MMP13 were expressed at higher levels in the IECs derived from the *Clostridium*-colonized mice than those in the GF mice and in the *Lactobacillus*-colonized mice.

Moreover, as is apparent from the results shown in FIG. 40, IDO was expressed only in the *Clostridium*-colonized mice.

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Accordingly, it was revealed that the *Clostridium* activated the IECs, and led to the production of TGF- β and other Treg cell-inducing molecules in the colon.

Example 11

Next, it was investigated whether or not the Treg cell accumulation induced by the colonization of the *Clostridium* was dependent on signal transmission by pathogen-associated molecular pattern recognition receptors. Specifically, the numbers of Treg cells in the colonic lamina propria of each of SPF mice of *Myd88*^{-/-} (deficient in Myd88 (signaling adaptor for Toll-like receptor)), *Rip2*^{-/-} (deficient in Rip2 (NOD receptor adaptor)), and *Card9*^{-/-} (deficient in Card9 (essential signal transmission factor for Dectin-1 signal transmission)) were examined. In addition, *Clostridium* spp. were caused to be colonized in the *Myd88*^{-/-} GF mice, and the change in the number of Treg cells was investigated. FIGS. 17 and 18 show the obtained results. As is apparent from the results shown in FIGS. 17 and 18, the number of Treg cells of each kind of the SPF mice deficient in the associated factors of the pathogen-associated molecular pattern recognition receptors did not change relative to that of wild-type mice of the same litter, which served as a control. In addition, it was found that also when *Clostridium* spp. were colonized in GF mice deficient in Myd88, the accumulation of Treg cells in the colonic lamina propria was induced. Accordingly, it has been suggested that the mechanism of inducing the accumulation of Treg cells in the colonic lamina propria relies not on activation of recognition pathway for major pathogen-associated molecular patterns as is caused by most of bacterium, but on specific commensal bacterial species.

Example 12

Intestinal tract Foxp3⁺ Treg cells are known to exert some immunosuppressive functions through IL-10 production (refer to Non-Patent Document 9). Meanwhile, animals having CD4⁺ Foxp3⁺ cells from which IL-10 is specifically removed are known to develop inflammatory bowel disease (refer to Non-Patent Document 18). In this respect, first, the expression of IL-10 in lymphocytes of various tissues was examined. Specifically, lymphocytes were isolated from various tissues of SPF *Il10*^{venus} mice, and the expression of CD4 and the expression of Venus were analyzed by flow cytometry. FIG. 19 shows the obtained results. Note that each numeric value in FIG. 19 represents the ratio of cells within the corresponding one of regions divided into four.

Moreover, lymphocytes in the colonic lamina propria were isolated from *Il10*^{venus} mice, and the expression of T cell receptor β chain (TCR β) on the surfaces of the cells was detected by FACS. FIG. 20 shows the obtained results (FACS dot-plots obtained when a gate was set on CD4⁺ cells). Note that each numeric value in FIG. 20 represents the ratio of cells within the corresponding one of regions divided into four.

Furthermore, lymphocytes in the colonic lamina propria were isolated from *Il10*^{venus} mice. The lymphocytes were stimulated with PMA (50 ng/ml) and ionomycin (1 μ g/ml) for four hours in the presence of GOLGISTOP® (BD Bioscience). Then, after the stimulation was given, intracellular cytokines were stained by using an anti-IL-17 PE antibody, an anti-IL-4 APC antibody (11B11), and an anti-IFN- γ FITC antibody (BD Bioscience) in accordance with the manual of a CYTOFIX/CYTOPERM® kit (BD Biosci-

ence). FIG. 21 shows the obtained results (FACS dot-plots obtained when a gate was set on CD4⁺ cells).

Note that each numeric value in FIG. 21 represents the ratio of cells within the corresponding one of regions divided into four.

In addition, Foxp3⁺ CD4⁺ cells and Foxp3⁻ CD4⁺ cells were isolated from the spleen (Spl) of Foxp3^{eGFP} reporter mice, and Venus⁺ cells were isolated from the colonic lamina propria and the small intestine (SI) lamina propria of Il10^{venus} mice. Then, the obtained cells were analyzed in terms of the expression of predetermined genes. The gene expression was analyzed by real-time RT-PCR using a Power SYBR Green PCR Master Mix (Applied Biosystems) and an ABI 7300 real time PCR system (Applied Biosystems). Here, the value for each cell was normalized for the amount of GAPDH. FIG. 22 shows the obtained results. Note that in FIG. 22 the error bars represent standard deviations.

As is apparent from the results shown in FIGS. 19 to 22, almost no Venus⁺ cells (IL-10-producing cells) were detected in the cervical lymph nodes (peripheral lymph nodes), thymus, peripheral blood, lung, and liver of mice kept under the SPF conditions. Meanwhile, in the spleen, Peyer's patches, and mesenteric lymph nodes thereof, Venus⁺ cells were slightly detected (refer to FIG. 19). On the other hand, many Venus⁺ cells were found in the lymphocytes in the small intestine lamina propria and colonic lamina propria. In addition, most of the Venus⁺ cells in the intestines were positive for CD4, and also positive for T cell receptor β chain (TCR β) (refer to FIGS. 19 and 20). Moreover, it was found that the Venus⁺ CD4⁺ T cells expressed Foxp3 and other Treg cell-associated factors such as a cytotoxic T-Lymphocyte antigen (CTLA-4) and a glucocorticoid-induced TNFR-associated protein (GITR) although the Venus⁺ CD4⁺ T cells showed none of the phenotypes of Th2 (IL-4-producing) and Th17 (IL-17-producing) (refer to FIGS. 21 and 22). In addition, it was shown that the expression level of CTLA-4 in the intestinal Venus⁺ cells was higher than that in the splenic GFP⁺ Treg cells isolated from the Foxp3^{eGFP} reporter mice (refer to FIG. 22).

Example 13

Venus⁺ cells can be classified into at least two subsets, namely, Venus⁺ Foxp3⁺ double positive (DP) Treg cells and Venus⁺ Foxp3⁻ Treg cells on the basis of intracellular Foxp3 expression. Cells of the latter subset correspond to type 1 regulatory T cells (Tr1) (refer to Non-Patent Documents 8 and 9). In this respect, the Venus⁺ cells (IL-10-producing cells) observed in Example 8 were investigated in terms of the expression of Foxp3. Specifically, the expression of CD4, Foxp3, and Venus in the lamina propria of the colon and the lamina propria of the small intestine of Il10^{venus} mice kept under GF or SPF conditions was analyzed by FACS, and the numbers of Venus⁺ cells in the intestinal tract lamina propria were compared between SPF and GF Il10^{venus} mice. FIG. 23 shows the obtained results (dot-plots obtained when a gate was set on CD4⁺ cells).

In addition, the intracellular expression of Venus and Foxp3 in CD4 cells in various tissues of SPF Il10^{venus} mice was analyzed by flow cytometry. FIG. 24 shows the obtained results (dot-plots obtained when a gate was set on CD4⁺ cells). Note that each numeric value in FIG. 24 represents the ratio of cells within the corresponding one of regions divided into four.

Moreover, in order to investigate whether or not the presence of commensal bacteria had any influence on the

expression of IL-10 in regulatory cells in the gastrointestinal tracts, germ-free (GF) Il10^{venus} mice were prepared. Then, predetermined species of bacteria were caused to be colonized in the obtained GF Il10^{venus} mice. Three weeks after the species of bacteria were colonized, a CD4⁺ cell group (V⁺F⁻, Venus⁺ Foxp3⁻ cells; V⁺F⁺, Venus⁺ Foxp3⁺ cells; and V⁻F⁺, Venus⁻ Foxp3⁺ cells) in which Foxp3 and/or Venus were expressed in the colon and the small intestine was analyzed by flow cytometry. FIG. 25 shows dot-plots obtained when a gate was set on colonic CD4⁺ cells, and FIGS. 26 and 27 show the ratios in the CD4⁺ cell group of each mouse. Note that each numeric value in FIG. 25 represents the ratio of cells within the corresponding one of regions divided into four. Meanwhile, the error bars in FIGS. 26 and 27 represent standard deviations, * indicates that "P<0.02," and ** indicates that "P<0.001."

Moreover, in order to check whether or not the presence of commensal bacteria had any influence on the expression of IL-10 in regulatory cells in the gastrointestinal tracts, antibiotics were orally given with water to five or six Il10^{venus} mice per group for 10 weeks. The following antibiotics were used in combination.

ampicillin (A; 500 mg/L Sigma)

vancomycin (V; 500 mg/L NACALAI TESQUE, INC.)

metronidazole (M; 1 g/L NACALAI TESQUE, INC.)

neomycin (N; 1 g/L NACALAI TESQUE, INC.)

Then, CD4 and Foxp3 of lymphocytes in the lamina propria of the colon, the lamina propria of the small intestine (SI), mesenteric lymph nodes (MLN), and Peyer's patches (PPs) were stained with antibodies, and analyzed by FACS. The results were obtained from two or more independent experiments which gave similar results. FIG. 28 shows the obtained results (the ratio of Venus⁺ cells in CD4⁺ cells in each sample). Note that each white circle in FIG. 28 represents an individual sample, each horizontal bar represents an average value, * indicates that "P<0.02," and "AVMN" represents the kinds of the administered antibiotics by using the first letters of the antibiotics.

As is apparent from the results shown in FIGS. 23 and 24, it was shown that the small intestinal lamina propria was rich in Venus⁺ Foxp3⁻ cells, namely, Tr1-like cells, and that the Venus⁺ Foxp3⁺ DP Treg cells were present at a high frequency in the colon of the SPF mice (refer to FIGS. 23 and 24). In contrast, although sufficient numbers of Foxp3⁺ cells were observed also in other tissues, the expression of Venus was not observed in almost all of the cells (refer to FIG. 24).

In addition, as is apparent from the results shown in FIGS. 23 and 25 to 28, it was shown that all regulatory T cell fractions of Venus⁺ Foxp3⁻, Venus⁺ Foxp3⁺, and Venus⁻ Foxp3⁺ in the colon significantly decreased under the GF conditions (FIGS. 23 and 26 to 27). Moreover, similar decrease in Venus⁺ cells was observed also in the SPF Il10^{venus} mice treated with the antibiotics (refer to FIG. 28).

Moreover, as is apparent from the results shown in FIGS. 25 to 27, the colonization of *Clostridium* spp. strongly induced all regulatory T cell fractions of Venus⁺ Foxp3⁻, Venus⁺ Foxp3⁺, and Venus⁻ Foxp3⁺ in the colon, and the degrees of the induction thereof were equal to those in the SPF mice (refer to FIGS. 25 and 27). In addition, it was found that the colonization of the three strains of *Lactobacillus* or the colonization of SFB had an extremely small influence on the number of Venus⁺ and/or Foxp3⁺ cells in the colon (refer to FIGS. 25 and 27). Moreover, the colonization of 16 strains of *Bacteroides* spp. also induced Venus⁺ cells, but the influence of the colonization was specific to Venus⁺ Foxp3⁻ Tr1-like cells (refer to FIGS. 25 and 27). On the other hand, it was found that none of the bacterial species

tested exerted any significant influence on the number of IL-10-producing cells in the small intestinal lamina propria (refer to FIG. 26).

Hence, it was shown that the genus *Clostridium* colonized in the colon or a physiologically active substance derived from the bacteria provided a signal for inducing the accumulation of IL-10⁺ regulatory T cells in the colonic lamina propria or the expression of IL-10 in T cells. Meanwhile, it was shown that the number of Venus⁺ cells in the small intestine was not significantly influenced by the situation where no commensal bacteria were present or commensal bacteria were decreased (refer to FIGS. 23 and 26 to 28), and that IL-10⁺ regulatory cells (Tr1-like cells) accumulated in the small intestinal lamina propria independently of commensal bacteria.

Example 14

It was investigated whether or not Venus⁺ cells induced by the genus *Clostridium* had an immunosuppressive function similar to that of Venus⁺ cells in the colon of SPF mice. Specifically, CD4⁺ CD25⁻ cells (effector T cells, T_{eff} cells) isolated from the spleen were seeded in a flat-bottomed 96-well plate at 2×10⁴/well, and cultured for three days together with 2×10⁴ splenic CD11c⁺ cells (antigen-presenting cells) subjected to 30 Gy radiation irradiation treatment, 0.5 μg/ml of an anti-CD3 antibody, and a lot of Treg cells. In addition, for the last six hours, the CD4⁺ CD25⁻ cells were cultured, with [³H]-thymidine (1 μCi/well) was added thereto. Note that, Treg cells used in Example 14 were CD4⁺ GFP⁺ T cells isolated from the spleen of Foxp3^{eGFP} reporter mice, or CD4⁺ Venus⁺ T cells in the colonic lamina propria of GF II10^{Venus} mice in which *Clostridium* spp. were colonized or SPF II10^{Venus} mice. Then, proliferation of the cells was determined based on the uptake amount of [³H]-thymidine, and represented by a count per minute (cpm) value.

As is apparent from the results shown in FIG. 29, Venus⁺ CD4⁺ cells of the mice in which the genus *Clostridium* was colonized suppressed in vitro proliferation of CD25⁻ CD4⁺ activated T cells. The suppression activity was slightly inferior to that of GFP⁺ cells isolated from the Foxp3^{eGFP} reporter mice, but equal to that of Venus⁺ cells isolated from the SPF II10^{Venus} mice. Accordingly, it has been shown that the genus *Clostridium* induces IL-10-expressing T cells having sufficient immunosuppressive activities, and thereby plays a critical role in maintaining immune homeostasis in the colon.

Example 15

Next, the influence, on the local immune response, of the colonization of a large number of *Clostridium* and the resultant proliferation of Treg cells was investigated.

Dextran Sulfate Sodium (DSS)-Induced Colitis Model

First, the DSS-induced colitis model was prepared as described above, and the influence, on the model mice, of the inoculation of the *Clostridium* and the proliferation of Treg cells was investigated. Specifically, control mice and *Clostridium*-inoculated mice were treated with 2% DSS, then observed and measured for six days for the body weight loss, the hardness of stool, and bleeding, and then were evaluated numerically. In addition, on day 6, the colons were collected, dissected, and analyzed histologically by HE staining. FIGS. 41 to 43 show the obtained results. Note that, in FIGS. 41 to 43, “SPF+Clost.” or “SPF+Clost. #1 to 3” indicate the results of C57BL/6 mice inoculated with a fecal

suspension of *Clostridium*-colonized mice, and grown in a conventional environment for six weeks, and “SPF” or “SPF #1 to 3” indicate the results of C57BL/6 mice (control mice) grown in a conventional environment for six weeks without being inoculated with the fecal suspension. In addition, in FIG. 41, the vertical axis “Disease score” represents the disease activity index (DAI) described above, and the horizontal axis “post 2% DSS(d)” represents the days elapsed after the initial administration of 2% DSS to the mice. Moreover, in FIG. 41, * indicates that “P<0.02,” and ** indicates that “P<0.001.” Meanwhile, Treg cells induced by regulatory dendritic cells are known to play a preventive role in a DSS-induced colitis model (see S. Manicassamy et al., Science 329, 849 (Aug. 13, 2010)).

As is apparent from the results shown in FIGS. 41 to 43, the symptoms of the colitis such as body weight loss and rectal bleeding were significantly suppressed in the mice having a large number of *Clostridium* (hereinafter also referred to as “*Clostridium*-abundant mice”) in comparison with the control mice (see FIG. 41). All the features typical for colonic inflammation, such as shortening of the colon, edema, and hemorrhage, were observed markedly in the control mice in comparison with the *Clostridium*-abundant mice (see FIG. 42). Moreover, histological features such as mucosal erosion, edema, cellular infiltration, and crypt loss were less severe in the DSS-treated *Clostridium*-abundant mice than in the control mice (see FIG. 43).

Oxazolone-Induced Colitis Model

Next, the oxazolone-induced colitis model was prepared as described above, and the influence, on the model mice, of the inoculation of *Clostridium* and the proliferation of Treg cells was investigated. Specifically, control mice and *Clostridium*-inoculated mice were sensitized with oxazolone, and subsequently the inside of the rectums thereof were treated with a 1% oxazolone/50% ethanol solution. Then, the body weight loss was observed and measured. In addition, the colons were dissected, and analyzed histologically by HE staining. FIGS. 44 and 45 show the obtained results. Note that, in FIGS. 44 and 45, “SPF+Clost.” indicates the results of C57BL/6 mice (*Clostridium*-abundant mice) inoculated with a fecal suspension of *Clostridium*-colonized mice, and grown in a conventional environment for six weeks, and “SPF” indicates the results of C57BL/6 mice (control mice) grown in a conventional environment for six weeks without being inoculated with the fecal suspension. In addition, in FIG. 44, the vertical axis “Weight (% of initial)” represents the body weight after the administration of 1% oxazolone where the body weight before the administration was taken as 100%, and the horizontal axis “post 1% oxazolone (d)” represents the days elapsed after the administration of 1% oxazolone to the mice. Meanwhile, it is known that Th2-type T cells are involved in colitis induced by oxazolone. (see M. Boirivant, I. J. Fuss, A. Chu, W. Strober, J Exp Med 188, 1929 (Nov. 16, 1998)).

As is apparent from the results shown in FIGS. 44 and 45, the colitis proceeded along with persistent body weight loss in the control mice. Meanwhile, the body weight loss of the *Clostridium*-abundant mice was reduced (see FIG. 44). In addition, it was also revealed that portions having histological diseases such as mucosal erosion, edema, cellular infiltration, and hemorrhage were reduced in the colon of the *Clostridium*-abundant mice (see FIG. 45).

Example 16

Next, the influence, on the systemic immune response (systemic IgE production), of the colonization of a large

number of *Clostridium* and the resultant proliferation of Treg cells was investigated. Specifically, as described above, control mice and *Clostridium*-inoculated mice were immunized by administering alum-adsorbed ovalbumin (OVA) twice at a 2-week interval. Then, sera were collected from these mice, and the OVA-specific IgE level thereof was investigated by ELISA. In addition, splenic cells were collected from the mice in each group, and IL-4 and IL-10 production by in-vitro OVA restimulation was investigated. FIGS. 46 to 48 show the obtained results. Note that, in FIGS. 46 to 48, "SPF+Clost." indicates the results of BALB/c SPF mice (*Clostridium*-abundant mice) inoculated with a fecal suspension of *Clostridium*-colonized mice, and grown in a conventional environment, "SPF" indicates the results of BALB/c SPF mice (control mice) grown in a conventional environment without being inoculated with the fecal suspension, and ** indicates that "P<0.001." Meanwhile, in FIG. 46, the vertical axis "OVA-specific IgE (ng/ml)" represents the concentration of OVA-specific IgE in the sera. Moreover, in FIG. 46, the horizontal axis represents the days elapsed after the initial administration of the alum-adsorbed ovalbumin to the *Clostridium*-abundant mice or the control mice (4-week old), and "OVA+Alum" indicates the timing of the administration of the alum-adsorbed ovalbumin. In addition, in FIGS. 47 and 48, "OVA" on the horizontal axis indicates the results in the case where the in-vitro OVA restimulation was performed, and "-" indicates the results in the case where no in-vitro OVA restimulation was performed. Moreover, in FIGS. 47 and 48, the vertical axes "IL-4 (pg/ml)" and "IL-10 (pg/ml)" show the IL-4 concentration and the IL-10 concentration in culture supernatants of splenic cells, respectively.

As is apparent from the results shown in FIGS. 46 to 48, the IgE level was significantly lower in the *Clostridium*-abundant mice than in the control mice (see FIG. 46). Moreover, the IL-4 production by the OVA restimulation was reduced (see FIG. 47) and the IL-10 production thereby was increased (see FIG. 48) in the splenic cells of the *Clostridium*-abundant mice sensitized with OVA and alum, in comparison with those of the control mice.

Accordingly, in consideration of the results shown in Example 15 in combination, it has been revealed that the induction of Treg cells by *Clostridium* in the colon plays an important role in local and systemic immune responses.

Example 17

Next, GF Balb/c were colonized with three strains of *Clostridium* belonging to cluster IV (strains 22, 23 and 32 listed in FIG. 49). Three weeks later, colonic Foxp3⁺ Treg cells were analyzed by FACS. FIG. 50 shows the obtained results. As is apparent from the results shown in FIG. 50, gnotobiotic mice colonized with three strains of *Clostridium* showed an intermediate pattern of Treg induction between GF mice and mice inoculated with all 46 strains.

Example 18

Next, it was investigated whether or not a spore-forming (for example, a chloroform resistant) fraction of a fecal sample obtained from humans had the effect of inducing proliferation or accumulation of regulatory T cells similar to the spore-forming fraction of the fecal sample obtained from mice.

Specifically, human stool from a healthy volunteer (Japanese, male, 29 years old) was suspended with phosphate-buffered saline (PBS), mixed with chloroform (final con-

centration 3%), and then incubated in a shaking water bath for 60 min. After evaporation of chloroform by bubbling with N₂ gas, the aliquots containing chloroform-resistant (for example, spore-forming) fraction of human intestinal bacteria were orally inoculated into germ-free (GF) mice (IQI, 8 weeks old). The treated mice were kept in a vinyl isolator for 3 weeks. The colon was collected and opened longitudinally, washed to remove fecal content, and shaken in Hanks' balanced salt solution (HBSS) containing 5 mM EDTA for 20 min at 37° C. After removing epithelial cells and fat tissue, the colon was cut into small pieces and incubated with RPMI1640 containing 4% fetal bovine serum, 1 mg/ml collagenase D, 0.5 mg/ml dispase and 40 µg/ml DNase I (all manufactured by Roche Diagnostics) for 1 hour at 37° C. in a shaking water bath. The digested tissue was washed with HBSS containing 5 mM EDTA, resuspended in 5 ml of 40% PERCOLL® (manufactured by GE Healthcare) and overlaid on 2.5 ml of 80% PERCOLL® in a 15-ml Falcon tube. PERCOLL® gradient separation was performed by centrifugation at 780 g for 20 min at 25° C. The interface cells were collected and suspended in staining buffer containing PBS, 2% FBS, 2 mM EDTA and 0.09% NaN₃ and stained for surface CD4 with Phycoerythrin-labeled anti-CD4 Ab (RM4-5, manufactured by BD Biosciences). Intracellular staining of Foxp3 was performed using the Alexa647-labeled anti-Foxp3 Ab (FJK-16s, manufactured by eBioscience) and Foxp3 Staining Buffer Set (manufactured by eBioscience). The percentage of Foxp3 positive cells within the CD4 positive lymphocyte population was analyzed by flow cytometry. FIGS. 51 and 52 show the obtained results.

In figures, representative histograms (FIG. 51) and combined data (FIG. 52) for Foxp3 expression by CD4 positive lymphocytes from GF mice (GF) or GF mice gavaged with chloroform-treated human stool (GF+Chloro.) are shown. In addition, numbers in FIG. 51 indicate the percentages of cells in the gate. Each circle in FIG. 52 represents a separate animal, error bars indicate the SD, and ** indicates that "P<0.001."

As is apparent from the results shown in FIGS. 51 and 52, it was found that also when the spore-forming (for example, the chloroform resistant) fraction of human intestinal bacteria was colonized in GF mice, the accumulation of Foxp3⁺ regulatory (Treg) cells in the colonic lamina propria of the mice was induced.

Next, it was investigated what species of bacteria grew by gavaging with chloroform-treated human stool.

Specifically, using a QIAamp DNA Stool mini kit (manufactured by QIAGEN), bacterial genomic DNA was isolated from the human stool from a healthy volunteer as described above (human stool) or fecal pellets from GF mice gavaged with chloroform-treated human stool (GF+Chloro.). Quantitative PCR analysis was carried out using a LightCycler 480 (manufactured by Roche). Relative quantity was calculated by the ΔCt method and normalized to the amount of total bacteria, dilution, and weight of the sample. The following primer sets were used:

```
total bacteria (SEQ ID NO: 62)
5'-GGTGAATACGTTCCCG-3' and
5'-TACGGCTACCTTGTTACGACTT-3' (SEQ ID NO: 63)
```


43

-continued

Clostridium cluster XIVa
(Clostridium coccoides subgroup)

5'-AAATGACGGTACCTGACTAA-3' and (SEQ ID NO: 64)

5'-CTTTGAGTTTCATTCTTGCGAA-3' (SEQ ID NO: 65)

Clostridium cluster IV (Clostridium leptum)

5'-GCACAAGCAGTGGAGT-3' and (SEQ ID NO: 66)

5'-CTTCCTCCGTTTGTCAA-3' (SEQ ID NO: 24)

Bacteroides

5'-GAGAGGAAGGTCCCCAC-3' and (SEQ ID NO: 67)

5'-CGCTACTTGGCTGGTTAG-3' . (SEQ ID NO: 68)

FIG. 53 shows the obtained results.

As is apparent from the results shown in FIG. 53, mice gavaged with chloroform-treated human stool exhibited high amounts of spore-forming bacteria, such as *Clostridium* clusters XIVa and IV, and a severe decrease of non-spore-

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forming bacteria, such as *Bacteroides*, compared with the human stool before chloroform treatment.

INDUSTRIAL APPLICABILITY

As has been described above, the present invention makes it possible to provide an excellent composition for inducing proliferation or accumulation of regulatory T cells (Treg cells) by utilizing bacteria belonging to the genus *Clostridium* or a physiologically active substance or the like derived from the bacteria. Since the composition of the present invention has immunosuppressive effects, the composition can be used, for example, to prevent or treat autoimmune diseases or allergic diseases, as well as to suppress immunological rejection in organ transplantation or the like. In addition, healthy individuals can easily and routinely ingest the composition as a food or beverage, such as a health food, to improve their immune functions.

SEQUENCE LISTING

SEQ ID NO:1 to 20, 62 to 69

<223> Artificially synthesized primer sequence

SEQ ID NO:21 to 61

<223> 16S rRNA coding gene sequence of each *Clostridium* strain

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 69

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<211> LENGTH: 23

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<220> FEATURE:

<223> OTHER INFORMATION: Artificially synthesized primer sequence

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23

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<212> TYPE: DNA

<213> ORGANISM: Artificial

<220> FEATURE:

<223> OTHER INFORMATION: Artificially synthesized primer sequence

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22

<210> SEQ ID NO 3

<211> LENGTH: 23

<212> TYPE: DNA

<213> ORGANISM: Artificial

<220> FEATURE:

<223> OTHER INFORMATION: Artificially synthesized primer sequence

<400> SEQUENCE: 3

cctttttagt cctgtctac tct

23

<210> SEQ ID NO 4

<211> LENGTH: 21

<212> TYPE: DNA

<213> ORGANISM: Artificial

<220> FEATURE:

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<210> SEQ ID NO 6
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<400> SEQUENCE: 6
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<210> SEQ ID NO 7
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<400> SEQUENCE: 7
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<210> SEQ ID NO 8
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<210> SEQ ID NO 14
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<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: Artificially synthesized primer sequence

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<210> SEQ ID NO 15
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<212> TYPE: DNA
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<400> SEQUENCE: 15

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<210> SEQ ID NO 16
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: Artificially synthesized primer sequence

<400> SEQUENCE: 16

tcgcctggac cataaagaa 19

<210> SEQ ID NO 17
<211> LENGTH: 20
<212> TYPE: DNA
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<223> OTHER INFORMATION: Artificially synthesized primer sequence

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20

<210> SEQ ID NO 18

<211> LENGTH: 20

<212> TYPE: DNA

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<220> FEATURE:

<223> OTHER INFORMATION: Artificially synthesized primer sequence

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20

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<223> OTHER INFORMATION: Artificially synthesized primer sequence

<400> SEQUENCE: 19

agagtttgat cmtggctcag

20

<210> SEQ ID NO 20

<211> LENGTH: 16

<212> TYPE: DNA

<213> ORGANISM: Artificial

<220> FEATURE:

<223> OTHER INFORMATION: Artificially synthesized primer sequence

<400> SEQUENCE: 20

attaccgcgg ckgtg

16

<210> SEQ ID NO 21

<211> LENGTH: 1460

<212> TYPE: DNA

<213> ORGANISM: Clostridium leptum

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1) .. (1460)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 1

<400> SEQUENCE: 21

agagtttgat cctggctcag gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac

60

gagaaccatt ggatcgagga ttcgtccaag tgaagggtggg gaaagtggcg gacgggtgag

120

taacgcgtga gcaatctgcc ttggagtggg gaataacggc tggaaacagc cgctaatacc

180

gcatgataca gctgggaggc atctccctgg ctgtcaaaga tttatcgctc tgagatgagc

240

tcgcgtctga ttagctagtt ggcggggtaa cggcccacca aggcgacgat cagtagccgg

300

actgagaggt tggccggcca cattgggact gagacacggc ccagactcct acgggaggca

360

gcagtgggga atattgggca atggggcga gctgaccca gcaacgccgc gtgaaggaag

420

aaggctttcg ggttgtaaac ttcttttgtc agggacgaag caagtgacgg tacctgacac

480

ggctactacg gtcagcagcg cgtatacgta ggtgccagcg tatccggaat tacctgggtt

540

aaggcgtgtt agccggactg cagtcagatg tgaatcacgg gctcaacttg tgctgcattg

600

gaactgtagt tctgagtact gagagcagac ggaattctag gtagcggtag atgcgtagat

660

ataggaggac acagtgcgag gcgtctgctg acagcaactg acgctgaggg gggaagcgtg

720

ggggagccaa caggattaga tacctggtag ttcacgcctg gtaaacgat ggatactagg

780

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tgtgggggga ctgacccct cgtggcgcc agttaacacc aataaagtat cccacctggg	840
agtacgatcg caaggttgaa actcaaagga attgacgggg cccgcacaag cggtaggagta	900
tgtggtttaa ttcgaagcaa cgcgaagaac cttaccaggg cttgacatcc cgaggaccgg	960
actagagata gtcttttctc ttcggagacc tcggtgacag gtggtgcatg gttgtcgtca	1020
gctcgtgtcg taagatgttg ggtaagtcc cgcaacgagc gcaaccctta ttgttagttg	1080
ctacgcaaga gcactctagc gagactgccg ttgacaaaac ggaggaaggt ggggacgacg	1140
tcaaatacatc atgcccctta tgcctgggc cacacacgta ctacaatggt ggtcaacaga	1200
gggaagcaat accgcgaggt ggagcaaatc cctaaaagcc atcccagttc ggatcgcagg	1260
ctgcaacccg cctcgtgtaa gttggaatcg ctagtaatcg cggatcagca tgcgcggtg	1320
aatacgttcc cgggccttgt acacacgcc cgtcacacca tgagagtcgg gaacaccga	1380
agtcctagc ctaaccgcaa gggggggcgc ggcgaaggt gggttcgata attgggggta	1440
agtcgtaaca aggtagccgt	1460

<210> SEQ ID NO 22

<211> LENGTH: 1485

<212> TYPE: DNA

<213> ORGANISM: Clostridium leptum

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1) .. (1485)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 2

<400> SEQUENCE: 22

agagtttgat cctggctcag gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac	60
ggagcaccctc tgaaggagtt ttcggacaac ggaagggact gcttagtggc ggacgggtga	120
gtaacgcgtg aggaacctgc cttggagtgg ggaataacag ctggaacag ctgctaatac	180
cgcataatat atctgggcgc catggctctg gatatacaag atttatcgct ctgagatgga	240
ctcgcgtctg attagctagt tggcggggta acggcccacc aaggcgacga tcagtagccg	300
gactgagagg ttggccggcc acattgggac tgagacacgg cccagactcc tacgggaggc	360
agcagtgggg aatattgggc aatgggcgca agcctgaccc agcaacgccg cgtgaaggaa	420
gaaggcttcc ggggtgtaaa cttcttttgt cagggacgaa gcaagtgcg gtacctgacg	480
aataagccac ggctaactac gtgccagcag ccgcggtaat acgtaggtgg caagcgttat	540
ccggatttac tgggtgtaaa gggcgtgtag gcgggactgc aagtcagatg tgaaccac	600
gggctcaacc tgtgggcctg catttgaaac tgtagtctt gagtactgga gaggcagacg	660
gaattctagt tgtagcgtga aatgcgtaga tatagaagaa cacagttagc gagccgtct	720
gcaactgacg ctgagcgcga aagcgtgggg agcaaacagg attagatacc ctggtagtcc	780
acgctgtaaa cgatggatta ctaggtgtgg ggggactgac cccctccgtg ccgcagttaa	840
cacaataagt atcccacctg gggagtagca tcgcaaggtt gaaactcaaa aggaattgac	900
gggggcccgc acaagcgtg gagtatgtgg tttaaattcg aagcaacgcg aagaacctta	960
ccagggcttg acatcccgtg gaccgtccta gagataggat ttcccttcg gggacactgg	1020
agacaggtgg tgcattgttg tcgtcagctc gtgtcgtgag atgttgggtt aagtcgccga	1080
acgagcgcaa cccttattgt tagttgctac gcaagagcac tctagcgaga ctgccgttga	1140
caaacgggag gaaggtgggg acgacgtcaa atcatcatgc cccttatgtc ctgggccaca	1200
cacgtactac aatggtggtc aacagaggga agcaaacgcg cgaggtggag caaatcccta	1260

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aaagccatcc cagttcggat cgcaggtgc aaccgcctg cgtgaagttg gaatcgctag 1320
taatcgcgga tcagaatgcc gcggtgaata cgttcccgga ccttgtagac accgcccgtc 1380
acaccatgag agtcggggaac acccgaagtc cgtagcctaa ccgcaagggg ggcgcgccg 1440
aaggtggggtt cgataattgg ggtgaagtcg taacaaggta gccgt 1485

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<210> SEQ ID NO 23
<211> LENGTH: 1491
<212> TYPE: DNA
<213> ORGANISM: Clostridium leptum
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1)..(1491)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
strain 3

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<400> SEQUENCE: 23
agagtttgat cctggctcag gacgaacgct ggcgcggtgc ttaacacatg caagtcgaac 60
ggagcacctt cgaaagagac ttcggtcaat ggaaaagaat gcttagtggc ggacgggtga 120
gtaacgcgtg aggaacctgc ctttcagtgg gggacaacag ttggaaacga ctgctaatac 180
cgcataacgt acgggtatcg catggtatct gtaccaaaga tttatcgctg agagatggcc 240
tcgctgtcga ttagctagtt ggtagggtaa cggcctacca aggcgacgat cagtagccgg 300
actgagaggt tggccggcca cattgggact gagatacggc ccagactcct acgggaggca 360
gcagtgggga atattgggca atgggcgaaa gcctgaccca gcaacgcgcg gtgaagggaag 420
aaggctttcg ggttgtaaac ttcttttgac ggggaagagc agaagacggt acctgtcgaa 480
taagccacgg ctaactacgt gccagcagcc gcggtaatag gtaggtggca agcgtgtgcc 540
ggatttactg ggtgtaaagg gcgtgtagcc gggctgacaa gtcagatgtg aaatccgggg 600
gtcacaaccc cgaactgcat ttgaaactgt tggctttgag tatcggagag gcaggcggaa 660
ttcctagtgt agcggtgaaa tgcgtagata ttagggggaa caccagtggc gaagcggcct 720
gtcggacgac aactgacggt gaggcgcgaa agcgtgggga gcaaacagga ttagataccc 780
tggtagtcca cgctgtaaac gatggatact aggtgtgcgg ggactgacct ctgcgtgccg 840
cagctaacgc aataagtatc ccacctgggg agtacgatcg caaggttgaa actcaaagga 900
attgacgggg gcccgacaaa gcggtggatt atgtggttta attcgatgca acgcgaagaa 960
ccttaccagg gcttgacatc ctactaacga agtagagata cattaggtac ccttcggggg 1020
aagtagagac aggtgggtgca tggttgtcgt cagctcgtgt cgtgagatgt tgggttaagt 1080
cccgcaacga gcgcaacctt tattgttagt tgctacgcaa gagcactcta gcgagactgc 1140
cgttgacaaa acggaggaag gtggggacga cgtcaaatca tcatgcccct tatgtcctgg 1200
gtacacacg taatacaatg gcggtcaaca gagggatgca aaaccgcgag gtggagcgaa 1260
cccctaaaag ccgtcccagt tcagatcgca gtctgcaacc cgactgcgtg aagtcggaat 1320
cgctagtaat cgcggatcag catgccgcgg tgaatacgtt cccgggcctt gtacacaccg 1380
cccgtcacac catgagagtc ggaacacccc gaagtccgta gcctaaccgc aaggagggcg 1440
cggccgaagg tgggttcgat aattgggggtg aagtcgtaac aaggtagccg t 1491

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<210> SEQ ID NO 24
<211> LENGTH: 1491
<212> TYPE: DNA
<213> ORGANISM: Clostridium coccoides
<220> FEATURE:
<221> NAME/KEY: rRNA

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<222> LOCATION: (1) .. (1491)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 4

<400> SEQUENCE: 24

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agagtttgat cctggctcag gatgaacgct ggcggcgtgc ctaacacatg caagtcgaac    60
gggtgtacgg ggaggaaggc ttcggccgga aaacctgtgc atgagtggcg gacgggtgag    120
taacgcgtgg gcaacctggc ctgtacaggg ggataaact tagaaatagg tgctaatacc    180
gcataacggg ggaagccgca tggcttttcc ctgaaaactc cggtggtaca ggatgggccc    240
gcgtctgatt agccagtggg cagggtaacg gcctaccaa gcgacgatca gtagccggcc    300
tgagaggggc gacggccaca ctgggactga gacacggccc agactcctac gggaggcagc    360
agtgggggat attgcacaat ggggggaaac cctgatgcag cgacgccgcg tgagtgaaga    420
agtatttcgg tatgtaaagc tctatcagca gggaagaaaa tgacggtacc tgactaagaa    480
gccccggcta actacgtgcc agcagccgcg gtaatacgtg gggggcaagc gttatccgga    540
tttactgggt gtaaaggagg cgtagacggc agcgcaagtc tgagtgaat cccatggctt    600
aaccatggaa ctgctttgga aactgtgcag ctggagtgcg ggagagtaag cggaattcct    660
agtgtagcgt gaaatgcgta gattatagga ggaacaccag tggcgaagcg ggctaactga    720
actgtaactg acgttgaggc tcgaaagcgt ggggagcaaa caggattaga taccctggta    780
gtccacgccc taaacgatga ttactagggt ttgggggacc aaggtcttcg gtgccggcgc    840
aaacgcatta agtaatccac ctggggagta cgttcgcaag aatgaaactc aaaggaattg    900
acggggaccc gcacaagcgg tggagcatgt ggtttaattc gaagcaacgc gaagaacctt    960
acctggtctt gacatcccg aacgagtgta gcaaagtcac ttcccttcg gggcattgga    1020
gacaggtggt gcattggtgt cgtcagctcg tgtcgtgaga tgttgggtta agtccgcaa    1080
cgagcgcaac ccctatttcc agtagccagc aggtagagct gggcactctg gagagactgc    1140
cgggataaac cgggaggaag gcgggggatga cgtcaaatca tcatgcccc ttagatcagg    1200
gtacacacag tgctacaatg gcgtaaacaa aggggaagcga gacggtgacg ttgagcaaat    1260
ccccaaaata acgtcccagt tcggattgta gtctgcaact cgactacatg aagctggaat    1320
cgctagtaat cgcaatcag aatgtcgcg tgaatacgtt cccgggtctt gtacacaccg    1380
cccgtcacac catgggagtc ggaatgccc gaagtcagtg acctaaccga aaggaaggag    1440
ctgccgaagg tggagccggt aactgggggtg aagtcgtaac aaggtagccg t          1491

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<210> SEQ ID NO 25

<211> LENGTH: 1467

<212> TYPE: DNA

<213> ORGANISM: Clostridium leptum

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1) .. (1467)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 5

<400> SEQUENCE: 25

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agagtttgat cctggctcag gacgaacgct ggcggcacgc ctaacacatg caagtcgaac    60
ggagtgaaga tgctcgcatc tgaacttagt ggcggaaggc tgagtaaacac gtgagcaacc    120
tgcccttcag agggggatta cgtttggaaa cgaacgctaa taccgataa aatatcggag    180
tcgcatggca ctgatataa aggagcaatc cgctgaaaga tgggctcgcg tccgattagg    240
cagttggcgg ggtatcggcc caccaaaccg acaatcggtg gccggactga gaggttgaac    300

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ggccacattg ggactgagac gcgggcccaga ctctctacggg aggcagcagt gggggatatt	360
gcacaatggg ggaacccttg atgcagcgat gccgcgtgaa tgaagacggc cttcgggttg	420
taaagttctg tcgcagggga cgaaaatgac ggtaccctgc aagaaagctc cggctaacta	480
cgtgccagca gccgcggtaa tacgtaggga gcaagcgttg tccggaatta ctgggtgtaa	540
agggagcgta ggcgggagga taagtgaat gtgaaatcta tgggctcaac ccatagtcgc	600
gttcaaaactg ttcttcttga gtgaagtaga ggcaggcgga attcctagtg tagcggtgaa	660
atgcgtagat attaggagga caccagtggc gaaggcgggc tgctgggctt tactgacgct	720
gaggctcgaa agcgtgggta gcaaacagga ttagataccc tggtagtcca cgcggtaaac	780
gatgattact aggtgtgggt ggactgacct catccgtgcc ggagttaaca caataagtaa	840
tccacctggg gagtacggcc gcaaggttga aactcaaagg aattgacggg ggcccgcaca	900
agcagtggag tatgtggttt aattcgacgc aacgcgaaga accttaccag gtcttgacat	960
cgagtgcagg acatagagat atgtctttcc ttcgggacac gaagacaggt ggtgcatggt	1020
tgtcgtcagc tcgtgtcgtg agatgttggg ttaagtcccg caacgagcgc aacccttacc	1080
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ggatgacgtc aaatcatcat gcccttatg acctgggcga cacacgtact acaatggcgg	1200
tcaacagagg gaggcaaagc cgcgaggcag agcaaacccc taaaagccgt ctgagttcgg	1260
attgcaggct gcaactcgcc tgcataaagt cggaattgct agtaatcgcg gatcagcatg	1320
ccgcggtgaa tacgttcccg ggccttgtac acaccgcccg tcacaccatg agagccggtg	1380
acaccogaag tcaatagtct aaccgcaagg aggacattgc cgaaggtagg attggtatt	1440
ggggtgaagt cgtaacaagg tagccgt	1467

<210> SEQ ID NO 26

<211> LENGTH: 1474

<212> TYPE: DNA

<213> ORGANISM: Clostridium coccoides

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1) .. (1474)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 6

<400> SEQUENCE: 26

agagtttgat catggctcag gatgaacgct gccggcgtgc ctaacacatg caagtcgaac	60
gggtgtacgg gaaggaaggc ttccggccgga aaacctgtgc atgagtggcg gacgggtgag	120
taacgcgtgg gcaacctggc ctgtacaggg ggataaact tagaaatagg tgctaatacc	180
gcataacggg ggaagccgca tggcttttcc ctgaaaactc cgggtgtaca ggatgggccc	240
gcgtctgatt agccagttag cagggtaacg gcttaccaaa gcgacgatca gtagccggcc	300
tgagaggggc gacggccaca ctgggactga gacacggccc agactcctac gggaggcagc	360
agtgggggat attgcacaat ggggggaacc ctgatgcagc gacgcccggt ggggtgaagaa	420
gcgcctcgcc gcgtaaaagg ctgtcagcag ggaagaaaat gacggtacct gaagaagaag	480
ccccggctaa ctacgtgccg gcagccggcg taatacgtag gggggcaagc gttatccgga	540
tttactgggt gtaaaagggg cgcagacggc gatgcaagcc aggagtgaag gcccgggggc	600
caaccccggg actgctcttg ggaactgcgt ggctggagtg cagagggcag cggaattcct	660
ggtgaaatgc gtagatatca gaagacacgg tcgcaggcgg cctgctgact gactgacgt	720
tgagccgaag cgtggggagc aaacaggatt agataccgtg gtagtcacgc cgtaaacgat	780

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gattactagg tgtcggggag cagagactgc ccggtgccgc agccaacgca ttaagtaatc	840
cacctgggga gtacgttcgc aagaatgaaa ctcaaaggaa ttgacgggga cccgcacaag	900
cggtggagca tgtggtttaa ttcgaagcaa cgcgaagaac cttacctggt cttgacatcc	960
cgatgacgag tgagcaaagt cactttccct tcggggcatt ggagacaggt ggtgcatggt	1020
tgtcgtcagc tcgtgtcgtg agatgttggg ttaagtcccg caacgagcgc aaccctatt	1080
tccagtagcc agcaggtaga gctgggcaact ctggagagac tgcccgggat aaccgggagg	1140
aaggcgggga tgacgtcaaa tcatcatgcc cttatgatc agggctacac acgtgctaca	1200
atggcgtaaa caaagggaag cgagacgggtg acgttaagca aatcccaaaa ataacgtccc	1260
agttcggatt gtagtctgca actcgactac atgaagctgg aatcgctagt aatcgcgaa	1320
cagaatgtcg cgggtgaatac gttcccgggt cttgtacaca ccgcccgtca caccatggga	1380
gtcgaaaatg cccgaagtca gtgacctaac cgaaagggaag gagctgccga aggtggagcc	1440
ggtaactggg gtgaagtcgt aacaaggtag ccgt	1474

<210> SEQ ID NO 27

<211> LENGTH: 1484

<212> TYPE: DNA

<213> ORGANISM: Clostridium leptum

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1)..(1484)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 7

<400> SEQUENCE: 27

agagtttgat cctggctcag gatgaacgct ggcggcgtgc ttaacacatg caagtogaac	60
gagaatccag tgaaggagtt ttcggacaac ggatctggag gaaagtggcg gacgggtgag	120
taacgcgtga gcaatctgcc ttggagtggg gaataacggt tggaacagc cgctaatacc	180
gcatgatgcg tctgggagcg atctctctgg acgcaaaaga tttatcgctc tgagatgagc	240
tcgcgtctga ttagcttggt ggcggggtaa agggccacca aggcgacgat cagtagccgg	300
actgagaggt tggccggcca cattgggact gagacacggc ccagactcct acgggaggca	360
gcagtgggga atattgggca atgggcgcaa gcctgaccca gcaacgccgc gtgaagggaag	420
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ataagccacg gctaactacg tgccagcagc cgcggaata cgtaggtggc aagcgttatc	540
cggatttatt ggggtgaaag ggcgtgtagg cgggaaagca agtcagatgt gaaaactcag	600
ggctcaacc tgagcctgca tttgaaactg tttttcttga gtgctggaga ggcaatcgga	660
attccgtgtg tagcggtaga atgcgtagat atacggagga caccagtggc gagcggattg	720
ctggacagta ctgacgtga agcgcgaaaag cgtgggagca aacagataga tacctggtag	780
tcacgcgtaa acgatggata ctagggtgtg ggggactgac cccctccgtg ccgcagctaa	840
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ggggcccgca caagcgttg agtatgtggt ttaattcgaa gcaacgcgaa gaaccttacc	960
agggcttgac atcctgctaa cgaaccagag atggattagg tgcccttcgg ggaaagcaga	1020
gacaggtggt gcattggtgt cgtcagctcg tgtcgtgaga tgttgggtta agtcccgcaa	1080
cgagcgcaac ccttattggt agttgctacg caagagcact ctagegagac tgccgttgac	1140
aaaacggagg aaggtgggga cgacgtcaaa tcatcatgcc ccttacgtcc tggggcacac	1200
acgtactaca atggcggcca acaaagagag gcaagaccgc gaggtggagc aaatctcaaa	1260

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aagccgtccc agttcggatc gcaggtctgca acccgctgc gtgaagttgg aatcgctagt 1320
aatcgcggat cagcatgccg cgggtgaatac gttcccgggc cttgtacaca ccgccgtca 1380
caccatgaga gtcgggaaca ccgaagtcc gtagcctaac cgcaaggggg gcgcggccga 1440
agggtgggttc gataattggg gtgaagtcgt aacaaggtag ccgt 1484

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<210> SEQ ID NO 28
<211> LENGTH: 1483
<212> TYPE: DNA
<213> ORGANISM: Clostridium leptum
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1) .. (1483)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
strain 8

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<400> SEQUENCE: 28

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agagtttgat cctggctcag gacgaacgct ggcggcgtgc ttaacacatg caagtcgaac 60
ggagaccccc tgaaggagtt ttcggacaac ggatgggaat gcttagtggc ggactggtga 120
gtaacgcgtg aggaacctgc cttccagagg gggacaacag ttggaaacga ctgctaatac 180
cgcatgatgc gttggagccg catgactccg acgtcaaaga tttatcgctg gaagatggcc 240
tcgcgtctga ttagctagtt ggtgaggtaa cggcccacca aggcgacgat cagtagccgg 300
actgagaggt tggccggcca cattgggact gagatacggc ccagactcct acgggaggca 360
gcagtgggga atattgggca atggacgcaa gtctgaccca gcaacgccgc gtgaagggaag 420
aaggctttcg ggtgtgaaac ttcttttaag ggggaagagc agaagacggt accccttgaa 480
taagccacgg ctaactacgt gccagcagcc gcggtaatat gtaggtggca agcgttgtcc 540
ggatttactg ggtgtaaagg gcgtgcagcc ggagagacaa gtcagatgtg aaatccacgg 600
gtcacaacccg tgaactgcat ttgaaactgt ttcccttgag tgctggagag gtaatcggga 660
ttccctgtgt agcgggtgaat gcgtagatat agagaccaca gtgccgacgc cgaatactga 720
cgatactgac ggtgagcgcg aaagcgtggg gagcaaacag gattagatac cctggtagtc 780
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acaataagta tcgcacctgg ggagtacgat cgcaagggtg aaactcaaag gaattgacgg 900
gggcccgcac aagcggtgga ttatgtggtt taattcgaag caacgcgaag aaccttacca 960
gggcttgaca tcctgctaac gaagtagaga tacattaggt gcccttcggg gaaagcagag 1020
acaggtggtg catggtgtgc gtcagctcgt gtcgtgagat gttgggttaa gtcccgaac 1080
gagcgcaacc cctattgtta gttgtacgc aagagcactc tagcgagact gccgttgaca 1140
aaacggagga aggcggggac gacgtcaaat catcatgccc cttatgtcct gggctacaca 1200
cgtaatacaa tggcggttaa caaagggatg caaagccgcg aggcagagcg aacccccaaa 1260
agccgtccca gttcggatcg caggctgcaa cccgcctgcg tgaagtcgga atcgctagta 1320
atcgcggatc agcatgccg ggtgaatacg ttcccgggcc ttgtacacac cgcccgtcac 1380
accatgagag tcgggaacac ccgaagtcg tagcctaacc gcaaggaggg gcgcggccgaa 1440
gggtgggttcg ataattgggg tgaagtcgta acaaggtagc cgt 1483

```

```

<210> SEQ ID NO 29
<211> LENGTH: 1480
<212> TYPE: DNA
<213> ORGANISM: Clostridium coccoides
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1) .. (1480)

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-continued

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 9

<400> SEQUENCE: 29

```

ggagtttgat cctggctcag gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac      60
gggctcatat tgaaacctag tgatgtatga gttagtggcg gacgggtgag taacgcgtgg      120
agaacctgcc gtatactggg ggataacact tagaaatagg tgctaatacc gcataagcgc      180
acagcttcgc atgaagcagt gtgaaaaact ccggtgggat acgatggatc cgcgtctgat      240
tagctgggtg cgggggtaac agcccaccaa ggcgacgac agtagccggc ctgagagggt      300
gaacggccac attgggactg agacacggcc caaactccta cgggaggcag cagtggggaa      360
tattgcacaa tgggggaaac cctgatgcag cgacgcccg tgagtgaaga agtatttcgg      420
tatgtaaagc tctatcagca gggaagaaat actgacctta cggtcagcag acggtacctg      480
actaagaagc cccgggctaa ctacgtgcc a gcagccgcg taatacgtag gggcaagcgt      540
tatccggatt tactgggtgt aaagggggcg cagacggcga tgcaagccag gagtgaagc      600
cggggcccaa cccgggact gctcttgac tgcgtggctg gagtgcagag ggcagcgaat      660
tcctgtgtag cgtgaatgcg tagattcaga ggacacgtgc gagcgctgc tgactgcact      720
gacgtgagcc cgaagcgtgg ggagcaaca ggattagata cctggtagtc cagccgtaa      780
acgatgatta ctagggtgct gggagcagag actgcccgtt gccgcagcca acgcattaag      840
taatccacct ggggagtacg ttcgcaagaa tgaaactcaa aggaattgac ggggacccgc      900
acaagcggtg gagcatgtgg tttaattcga agcaacgcga agaaccttac caggccttga      960
catccccctg gatggccctg aacgggggtc gcccttcggg gcaggggaga cagggtgtgc     1020
atggttgctg tcagctcgtg tctgtagatg ttgggttaag tcccgcaacg agcgcaaccc     1080
ctgccgcgag tagccagcat tttagatggg gactctgcgg ggactgccgg ggacaacccg     1140
gaggaaggcg gggatgacgt caaatcatca tgccccctat ggctgggct acacacgtgc     1200
tacaatggcg ccgacagagg gaggcgaagc ggcgacgcgg agcgaacccc aaaaacggcg     1260
tcccagttcg gattgtagtc tgcaacccga ctacatgaag ccggaatcgc tagtaatcgc     1320
ggatcagaat gcccggtga atacgttccc ggggtcttga cacaccgcc gtcacaccat     1380
gggagccggg aatgcccgaa gtctgtgacc gaaccgtaa ggggaggggc agccgaaggc     1440
aggcccggtg actgggtga agtcgtaaca aggtagccgt                               1480

```

<210> SEQ ID NO 30

<211> LENGTH: 1489

<212> TYPE: DNA

<213> ORGANISM: Clostridium leptum

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1) .. (1489)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 10

<400> SEQUENCE: 30

```

agagtttgat cctggctcag gatgaacgct ggcggcgtgc ttaacacatg caagtcgagc      60
gaagcacttt tttagaactc ttcggaggga agagagggtg acttagcggc ggacgggtga      120
gtaacgcgtg ggcaacctgc cttacacagg gggataacaa ttagaaatga ttgctaatac      180
cgcataagac cacggtactg catggtacag tggtaaaaac tgagggtggtg taagatgggc      240
ccgcgtctga ttaggtagtt ggtggggtag aagcctacca agccgacgat cagtagccga      300
cctgagaggg cgaccggcca cattgggact gagacacggc ccaaactcct acgggaggca      360

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-continued

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gcagtgggga atattgcaca atgggggaaa ccctgatgca gcgacgcgc gtgagtgagg 420
aagtatttcg gtatgtaaag ctctatcagc agggaagaaa atgacgggtac ctgactaaga 480
agcccccgcc taactacgtg ccagcagccg cggtaatacg tagggggcaa gcgttatccg 540
gatttactgg gtgtaaaggg agcgtagacg gacttgcaag tctgatgtga aaatccgggg 600
cccaaccgag gactgcattg aaactgtatt ttttgagggg gtccgaggag gcaagtggaa 660
tcctgggtag cggtgaaatg gcgtagaatt caggaggagaa caccagtggc ggaaggcgaa 720
ttactggacg ataactgacg gtgaggcgcg aagcgtggga gcaaacaaga attagatacc 780
ctggtagtca cgctgtaacg atcgatacta ggtgtgctgg gactgacccc tgcgtgccgg 840
agttaacaca ataagtatcg cactggggag tacgatcgca aggttgaaac tcaaaggaat 900
tgacgggggc ccgcacaagc ggtggattat gtggtttaat tcgaagcaac gcgaagaacc 960
ttaccagggc ttgacatcct gctaacgaag tagagataca ttaggtgccc ttcggggaaa 1020
gcagagacag gtggtgcatg gttgtcgtca gctcgtgtcg tgagatgttg ggttaagtcc 1080
cgcaacgagc gcaaccctta ttgttagttg ctacgcaaga gcaactctagc gagactgccg 1140
ttgacaaaac ggaggaaggc ggggacgacg tcaaatcatc atgcccctta tgtcctgggc 1200
tacacacgta atacaatggc ggttaacaaa gggatgcaaa gccgcgaggc agagcgaacc 1260
ccaaaagacc gtcccagttc ggatcgacg ctgcaaccgg cctgcgtgaa gtcggaatcg 1320
ctagtaatcg cggatcagca tgcgcgggtg aatacgttcc cgggccttgt acacaccgcc 1380
cgtcacacca tgagagtcgg gaacacccga agtccgtagc ctaaccgcaa ggaggcgcg 1440
gccgaagggt ggttcgataa ttggggtgaa gtcgtaacaa ggtagccgt 1489

```

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<210> SEQ ID NO 31
<211> LENGTH: 1490
<212> TYPE: DNA
<213> ORGANISM: Clostridium leptum
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1)..(1490)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
strain 11

```

```

<400> SEQUENCE: 31
agagtttgat cctggctcag gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac 60
gagaatccag tgaaggagtt ttcggacaac ggatctggag gaaagtggcg gacgggtgag 120
taacgcgtga gcaatctgcc ttggagtggg gaataacggt tggaacacgc cgctaatacc 180
gcatgatgag tctgggaggc atctctctgg acgcaaaaga tttatcgctc tgagatgagc 240
tcgcgtctga ttagcttggt ggcggggtaa aggccacca aggcgacgat cagtagccgg 300
actgagaggt tggccgcca cattgggact gagacacggc ccagactcct acgggaggca 360
gcagtgggga atattgggca atgggcgcaa gcctgaccca gcaacgccgc gtgaagggaag 420
aaggctttcg ggttgtaaac ttcttttctg agggacgaag aaagtgacgg tacctcagga 480
ataagccacg gctaactacg tgccagcagc cgcggtataa cgtagggtgc aagcgttatc 540
cggatttatt ggggtgaaag ggcgtgtagg cgggaaagca agtcagatgt gaaaactcag 600
ggctcaaccc tgagcctgca tttgaaactg tttttcttga gtgctggaga ggcaatcgga 660
attccgtggt gtacgggtga aatgcgtaga ttataccgga ggaaccacca gtggcggaag 720
gcggattgct ggaacagtaa ctgacgctga ggcgccgaaa gcgtggggag caaacaggat 780
agataccctg gtagtccacg ccgtaaacga tggatactaa gtgtggggga ctgaccctt 840

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cgtgcccagc taagcaataa gtttcccacc tggggagtag gatcgaggt gaaactcaaa 900
ggaattgacg ggggcccgcc caagcgggtg gagtaggggt taattggagc aacgggaaga 960
accttaccag ggcttgacat cctgtaacga accagaagag ggattaggtg ccttcgggga 1020
aagcagagac aggtggtgca tggttgtcgt cagctcgtgt cgtgagatgt gggtaaagt 1080
ccgcaacgag cgcaaccctt attgttagtt gctacgcaag agcactctag cgagactgcc 1140
gttgacaaaa cggaggaagg tggggacgac gtcaaatcat catgccctt acgtcctggg 1200
ccacacacgt actacaatgg cggccaacaa agagaggcaa gaccgagagg tggagaaaat 1260
ctcaaaaagc cgtcccagtt cggatcgag gctgcaaccc gcctgcgtga agttggaatc 1320
gctagtaatc gcggtacgac atgcccggt gaatacgttc ccgggccttg tacacaccgc 1380
ccgtcacacc atgagagtcg ggaacacccg aagtcgtag cctaaccgca aggggggcgc 1440
ggccgaaggt gggttcgata attgggggtga agtcgtaaca aggtagccgt 1490

```

<210> SEQ ID NO 32

<211> LENGTH: 1489

<212> TYPE: DNA

<213> ORGANISM: Clostridium coccoides

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1) .. (1489)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 12

<400> SEQUENCE: 32

```

agagtttgat catggctcag gatgaacgct ggccggcgtg ttaacacatg caagtcgaac 60
gggctcatat tgaaacctag tgatgtatga gttagtggcg gacgggtgag taacgcgtgg 120
agaacctgcc gtatactggg ggataaacct tagaaatagg tgctaatacc gcataagcgc 180
acagcttcgc atgaagcagt gtgaaaaact ccggtgggtat acgatggatc cgcgtctgat 240
tagctgggtg gcggggtaac agcccaccaa ggcgacgac agtagccggc ctgagagggg 300
gaacggccac attgggactg agacacggcc caaactccta cgggaggcag cagtggggaa 360
tattgcacaa tgggggaaac cctgatgcag cgacgccgcg tgagtgaaga agtatttcgg 420
tatgtaaagc tctatcagca ggaagaaaat actgacctta cggtcagcag acggtacctg 480
actaagaagc cccggctaac tacgtgccag cagccggcgt aatacgtagg gggcaagcgt 540
tatccgatt tactgggtgt aaaggagcgc tagacggcag cgcaagtctg aagtgaatc 600
ccatggctta accatggaac tgctttggaa actgtgcagc tggagtgcag gagaggtaa 660
cgggaattcct agtgtagcgg tgaatgcgta gatattagag gacaccagtg gcgatgcggc 720
ttactggact gtactgacgt tgagctcgaa agcgtgggga gcaccagaat tagaatactg 780
tagtcacgcc gtaaccgatg atactaggtg tgggggacca aggtctcgtg ccggcggcaa 840
acgcattaag taatccacct ggggagtagc ttcgcaagaa tgaactcaa aggaattgac 900
ggggaccgcc acaagcgggt gagcatgtgg tttaattcga agcaacgcga agaacttac 960
ctggtcttga catcccgatg acgagtgagc aaagtcactt tcccttcggg gcattggaga 1020
cagggtggtg atggttgtcg tcagctcgtg tcgtgagatg ttgggttaag tcccgcaacg 1080
agcgcaaccc ctatttcag tagccagcag gtagagctgg gcactctgga gagactgccc 1140
gggataaccg ggaggaagcg ggggatgacg tcaaatcatc atgcccctta tgatcaggcg 1200
tacacacgtg ctacaatggc gtaaacaaaag ggaagcgaga cggtgacgtt aagcaaatcc 1260
caaaaataac gtcccagttc ggattgtagt ctgcaactcg actacatgaa gctggaatcg 1320

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-continued

```

ctagtaatcg cgaatcagaa tgcgcggtg aatacgttcc cggtcttgt acacaccgcc 1380
cgtcacacca tgggagtcgg aatgcccga agtcagtgac ctaaccgaaa ggaaggagct 1440
gccgaaggtg gagccggtaa ctgggggtgaa gtcgtaacaa ggtagccgt 1489

```

```

<210> SEQ ID NO 33
<211> LENGTH: 1456
<212> TYPE: DNA
<213> ORGANISM: Clostridium coccoides
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1) .. (1456)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
strain 13

```

```

<400> SEQUENCE: 33

```

```

agagtttgat catggctcag gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac 60
gaagcacttg agaacgattc ttcggatgag gacttttgtg actgagtggc ggacgggtga 120
gtaacgcgtg ggtaacctgc cctatacagg gggataacag ttagaaatga ctgctaatac 180
cgcataagcg cactaaaacc gcatggttcg gtgtgaaaaa ctgaggtggt ataggatgga 240
cccgctctg attagcttgt tggtagggta acggctcacc aaggcgacga tcagtagccg 300
gcctgagagg gcgaccggcc acattgggac tgagacacgg cccaaactcc tacgggagggc 360
agcagtgggg gatattgcac aatgggggga accctgatgc agcgacgccg cgtgggtgaa 420
gaagcgctc ggcgctaaa gccctgtcag cagggaagaa aatgacggta cctgaagaag 480
aagccccgcg taactacgtg ccagcagccg cggtaatatg taggggcaag cgttattccg 540
ggatttactg ggtgtaaagg gggcgacgac ggcgatgcaa gccaggagtg aagccccggg 600
ccccccggg actgctcttg gactgcgtgc tggagtgcag aaggggcagc gatcctgtgt 660
accgtgaatt gcgtagatat cagagacacg ttgcgagcgc tgctgactgc actgacgtga 720
gcgaagctgg agcacagata gatactgtag tcagcgtaac gatgatacta gtgtcgggag 780
cagagactgc ccgttgccgc agcccaacgc attagtattc cacttgggga gtacgtttcg 840
cagaatgaac ttcaaggaaa tgacggggac ccgcacaagg cgggtggagca tgtggtttaa 900
ttcgaagcaa cgcaagaac cttaccaggc cttgacatcc cccctggatg gcccgtaacg 960
gggtcagcct ttcggggcag gggagacagg tggtgcatgg ttgtcgtcag ctcgtgtcgt 1020
gagatgttgg gttaagtccc gcaacgagcg caacccctgc ccgagtagc cagcatttta 1080
gatggggact ctgcggggac tgccggggac aacccggagg aaggcgggga tgacgtcaaa 1140
tcacatgcc ccttatggcc tgggtacac acgtgctaca atggcgcca cagaggagg 1200
cgaagcggcg acgcggagcg aacccccaaa acggcgtccc agttcgatt gtagtctgca 1260
acccgactac atgaagccgg aatcgctagt aatcgcgat cagaatgccg cggatgaatac 1320
gttcccggt cttgtacaca ccgccgtca cccatggga gccgggaatg cccgaagtct 1380
gtgaccgaac ccgtaagggg aggggcagcc gaaggcaggc tcggtgactg gggatgaagtc 1440
gtaacaaggt agccgt 1456

```

```

<210> SEQ ID NO 34
<211> LENGTH: 1475
<212> TYPE: DNA
<213> ORGANISM: Clostridium leptum
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1) .. (1475)

```

-continued

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 14

<400> SEQUENCE: 34

```

agagtttgat cctggctcag gacgaacgct ggccggcgtgc ttaacacatg caagtcgaac      60
ggagcacccc tgaaggagtt ttcggacaac ggatgggaat gcttagtggc ggactggtga      120
gtaacgcgtg aggaacctgc ctccagagg gggacaacag ttgaaacga ctgctaatac      180
cgcatgatgc gttggagcgc catgactccg acgtcaaaga tttatcgtg gaagatggcc      240
tcgcgtctga ttagctagtt ggtgaggtaa cggcccacca aggcgacgat cagtagccgg      300
actgagaggt tggccggcca cattgggact gagatacggc ccagactcct acgggaggca      360
gcagtgggga atattgggca atggacgcaa gtctgaccca gcaacgccgc gtgaaggaa      420
aaggctttcg ggtgtaaac ttcttttaag ggggaagagc agaagacggt accccttgaa      480
taagccacgg ctaactacgt gccagcagcc gcggtatac gtaggtggca agcgttgtcc      540
ggatttactg ggtgtaaagg gcgtgcagcc ggagagacaa gtcagatgtg aaatccacgg      600
gtcacaaccc tgaactgcac ttgaaactgt ttcccttgag tgctggagag gtaatcggaa      660
tttcttggtg tagcggtgaa tgcgtagata taaggaagga cacagtggcg agcggattac      720
tggacgatac tgacgtgagc gcgaaagcgt gggggagcaa cagaaattag atactgtagt      780
gcagctgtaa cgatcgatac tagttgcggg actgaccctc tgctgctgag ttacacaata      840
agtatcgcac ctgggagtag gatcgcaagg ttggaactca aaggaattga cggggcccgc      900
acaagcgttg gattatgtgg ttttaattcg agcaacgcga agaaccttac cagggtctga      960
catcctgcta acgaagtaga gatacattag gtgcccttcg gggaaagtag agacaggtgg      1020
tgcatggttg tcgtcagctc gtgtcgtgag atgttgggtt aagtcccga acgagcgcaa      1080
cccctattgt tagttgtcac gcaagagcac tctagcgaga ctgccgttga caaacggag      1140
gaaggcgggg acgacgtcaa atcatcatgc cccttatgtc ctgggttaca cacgtaatac      1200
aatggcgggt aacaaaggga tgcaaagccg cgaggcagag cgaaccccaa aaagccgtcc      1260
cagttcggat cgcaggctgc aaccgcctc cgtgaagtcg gaatcgtag taatcgcgga      1320
tcagcatgcc gcggtgaata cgttcccggt ccttgtagac accgccgtc acaccatgag      1380
agtcgggaac acccgaagtc cgtagcctaa ccgcaaggag ggcgcggccg aaggtgggtt      1440
cgataattgg ggtgaagtcg taacaaggta gccgt                                     1475

```

<210> SEQ ID NO 35

<211> LENGTH: 1480

<212> TYPE: DNA

<213> ORGANISM: Clostridium coccoides

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1) .. (1480)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 15

<400> SEQUENCE: 35

```

agagtttgat cctggctcag gatgaacgct ggccggcgtgc ttaacacatg caagtcgaac      60
gggctcatat tgaaacctag tgatgtatga gttagtggcg gacgggtgag taacgcgtgg      120
agaacctgcc gtatactggg ggataacact tagaaatagg tgctaatacc gcataagcgc      180
acagcttcgc atgaagcagt gtgaaaaact ccggtggtat acgatggatc cgcgtctgat      240
tagctggttg gcggggaac agcccaccaa ggcgacgac agtagccggc ctgagagggt      300
gaacggccac attgggactg agacacggcc caaactccta cgggaggcag cagtggggaa      360

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tattgcacaa tgggggaaac cctgatgcag cgacgccgag tgagtgaaga agtatttcgg	420
tatgtaaagc tctatcagca gggaagaaat actgacctta cggtcagcag acggtacctg	480
actaagaagc cccggctaac tacgtgccag cagccgcggt aatacgtagg ggcaagcgtt	540
atccggattt actgggtgta aaggggagcgt agacggcagc gcaagtctga agtgaaatcc	600
catggcttaa cccatggaac tgctttggaa actgtgcagc tggagtgcag gagaggtaa	660
cggaattcct agtgtagcgt gaaatgcgta gattattagg aggacaacag tgcgagcgct	720
actgacgtga ggctcgaagc gtgggagcaa acaggattag atacctggta gtcacgcgta	780
aacgatgatt actaggggtg tgggggacca aggtcttcgg tgccgggcga aacgcattaa	840
gtaatccacc tggggagtac gtctcgcaaga atgaaactca aaggaattga cggggacccg	900
cacaagcggg ggagcatgtg gtttaattcg aagcaacgcg aagaacctta cctgggtctg	960
acatcccgat gacgagttag caaagtcact ttcccttcgg ggcattggag acagggtggtg	1020
catggtgtgc gtcagctcgt gtcgtgagat gttgggttaa gtcccgcaac gagcgcaacc	1080
cctattttcca gtagccagca ggtagagctg ggcactctgg agagactgcc cgggataacc	1140
gggaggaagg cggggatgac gtcaaatcat catgcccctt atgatcaggg ctacacacgt	1200
gtacaatgg cgtaaacaaa ggaagcgcag acgggtgacgt taagcaaadc ccaaaaaata	1260
cgccccagtt cggattgtag tctgcaactc gactacatga agctggaate gctagtaate	1320
gcgaatcaga atgtcgcggt gaatacgttc ccgggtcttg tacacaccgc ccgtcacacc	1380
atgggagtcg gaaatgcccg aagtcagtga cctaaccgaa aggaaggagc tgccgaagg	1440
ggagccggta actggggtga agtcgtaaca aggtagccgt	1480

<210> SEQ ID NO 36
 <211> LENGTH: 1486
 <212> TYPE: DNA
 <213> ORGANISM: Clostridium papyrosolvens
 <220> FEATURE:
 <221> NAME/KEY: rRNA
 <222> LOCATION: (1)..(1486)
 <223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 16

<400> SEQUENCE: 36

agagtttgat cctggctcag gataaacgct ggcggcgcac ataagacatg caagtcgaa	60
ggacttaact cattotttta gattgagagc ggttagtggt ggactggtga gtaaacgta	120
agcaacctgc ctatcagagg ggaataacag tgagaaatca ttgctaatac cgcataatgct	180
cacagtatca catgatacag tgaggaaagg agcaatccgc tgatagatgg gcttgcgcct	240
gattagttag ttggtggggt aacggcctac caagacgacg atcagtagcc ggactgagag	300
gttgaaacgc cacattggga ctgagatacg gccagactc ctacgggagg cagcagtcgg	360
gaatattgag caatggagga aactctgacg cagtgcgcgc gcgtatagga agaaggtttt	420
cggattgtaa actattgtcg ttagggaaga taaaagactg tacctaagga ggaagccccg	480
gctaactatg tgccagcagc cgcggtaata catagggggc aagcgttatc cggaattatt	540
gggtgtaaag ggtgcgtaga cggaagaaca agttggttgt gaaatccctc ggctcaactg	600
aggaactgca accaaaaacta ttctccttga gtgtcggaga ggaaagtgga attcctagt	660
tagcgggtgaa atgcgtagat attaggagga acaccagtgg cgaaggcgac tttctggacg	720
ataactgacg ttgaggcagc aaagtgtggg gagcaaacag gattagatac cctggtagtc	780
cacactgtaa acgatggata ctagggttag ggtgtattaa gcaactctgtg ccgccgctaa	840

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cgcattaagt atcccacctg gggagtagc cgcgaagggt gaaactcaaa ggaattgacg    900
ggggcccgca caagcagtggt agtatgtggt ttaattcgaa gcaacgcgaa gaaccttacc    960
agggccttgac atataccgga atatactaga gatagtatag tccttcggga ctggtataca   1020
ggtggtgcat ggtgtgctgc agctcgtgtc gtgagatggt gggttaagtc ccgcaacgag   1080
cgcaaccctc atcgttagtt gctagcaggt aatgctgaga actctagcga gactgccggt   1140
gataaatcgg aggaaggtgg ggatgacgtc aaatcatcat gccctttatg tcctgggcta   1200
cacacgtact acaatggcgc taacagaggg aagcaatata gtgatatgga gcaaaaccct   1260
aaaagcggtc tcagttcgga ttgaaggctg aaattcgctc tcatgaagcc ggaattgcta   1320
gtaatggcag gtcagcatac tgccgtgaat acgttcccg gacctgtaca caccgcccgt   1380
cacaccatga gagttgaaa taccgaagc ctgtgagcta actgtaaaga ggcagcagtc   1440
gaaggtagag ccaatgattg ggggaagtc gtaacaaggt agccgt                    1486

```

<210> SEQ ID NO 37

<211> LENGTH: 1493

<212> TYPE: DNA

<213> ORGANISM: Clostridium leptum

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1) .. (1493)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 17

<400> SEQUENCE: 37

```

agagtttgat catggctcag gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac    60
gagaaccaac ggattgagga ttcgtccaaa tgaagttggg gaaagtggcg gacgggtgag   120
taacgcgtga gcaatctgcc ttggagtggg gaataacggt tggaacagc cgctaatacc   180
gcatgatgcg tctgggaggc atctctctgg acgccaaaga tttatcgctc tgagatgagc   240
tcgcgtctga ttagctagtt ggcggggcaa cggcccacca aggcgacgat cagtagccgg   300
actgagaggt tggcgggcca cattgggact gagacacggc ccagactcct acgggaggca   360
gcagtgggga atattgggca atgggcgcaa gcctgaccca gcaacgcgcg gtgaaggaaag   420
aaggctttcg ggttgtaaac tctttttaag ggggacgaac aaatgacggt accccttgaa   480
taagccacgg ctaactacgt gccagcagcc gcggtaatat gtaggtggca agcgttatcc   540
ggatttattg ggtgtaaagg gcgtgtaggc gggaatgcaa gtcagatgtg aaaactatgg   600
gctcaaccca tagcctgcat ttgaaactgt atttcttgag tgctggagag gcaatcgga   660
ttccgtgtgt agcggtgaaa tgcgtagata tacggaggaa caccagtggc gaagcggatt   720
gctggacagt aactgacgct gaggcgcgaa agcgtgggga gcaaacaggg attagatacc   780
ctggtagtca cgccgtaaac gatggatact aggtgtgggg ggactgaccc cctccgtgcc   840
gcagctaacy caataagtat ccacactggg gagtacgac gcaagggttg aaactcaaag   900
gaattgacgg gggccgcac aagcggtgga gtatgtggtt taattcgaag caacgcgaag   960
aaccttacca gggcttgaca tcctgctaac gaaccagaga tggatcaggt gcccttcggg  1020
gaaagcagag acaggtgggt catggtgtgc gtcagctcgt gtcgtgagat gttgggttaa  1080
gtcccgcaac gagcgcaacc cctattgtta gttgctacgc aagagcactc tagcgagact  1140
gccgttgaca aaacggagga aggtggggac gacgtcaaat catcatgcc cttacgtcct  1200
gggccacaca cgtactacaa tggcggccaa caaagagagg caagaccgcg aggtggagca  1260
aatctcaaaa agccgtccca gttcggatcg caggctgcaa ccgcctgcg tgaagttgga  1320

```

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atcgctagta atcgcggtac agcatgccgc ggtgaatacg ttcccgggcc ttgtacacac 1380
cgcccgctcac accatgagag tcgggaacac ccgaagtcgc tagcctgacc gcaagggggg 1440
cgcgcccgaa ggtgggttcg ataattgggg tgaagtagta acaaggtagc cgt 1493

```

```

<210> SEQ ID NO 38
<211> LENGTH: 1493
<212> TYPE: DNA
<213> ORGANISM: Clostridium leptum
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1) .. (1493)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
strain 18

```

```

<400> SEQUENCE: 38

```

```

agagtttgat cctggctcag gacgaacgct ggccggcgcc ctaacacatg caagtcgaac 60
ggagcttata ttccagaagt ttccggatgg acgagagata agcttagtgg cggacgggtg 120
agtaaacacg gagcaacctg cctttcagag ggggataaca gttggaaacg actgctaata 180
ccgcataacg ctgcgatggg gcaccccgat gcagccaaag gagcaatccg ctgaaagatg 240
ggctcgcgcc cgattagcta gttggtgggg caacggccca ccaaggcgac gatcggtagc 300
cggactgaga ggttgatcgg ccacattggg actgagacac ggcccagact cctacgggag 360
gcagcagtgg gggatattgc acaatggagg aaactctgat gcagcgacgc cgcgtgaggg 420
aagacggtct tcggattgta aacctctgtc ttgggggaag aaaatgacgg tacccaaaga 480
ggaagctccg gctaactacg tgccagcagc gcgggtaata cgtaaggagc gagcgttgtc 540
cggaattact ggggtgaaag ggagcgtacg cgggcgagaa agttgaatgt taaatctacc 600
ggcttaactg gtatgtcgtg tcaaaacttc ttgtcttgag tgaagtagag gcaggcgga 660
ttcctagtgt agcggtgaaa tgcgtagata taggaggaca ccagtgggag aagccgctg 720
ctgggcttta actgacgctg aggctcgaaa gcgtggggag caaaccagga ttagataccc 780
tggtagtcaa cgctgtaaac gatgattact aggtgtgggg gggactgacc ccctcgtgc 840
cgcagttaac acaataagta tccacctggg gagtacggcc gcaaagtttg aaaactcaaa 900
aggaatgacg ggggcccgcg caaagcagtg gagtatgtgg tttaatttcg aagcaacgcg 960
aagaacctta ccaggctctg acatcgtgcg catagcctag agataggtga agcccttcg 1020
ggcgcacaga cagggtgtgc atggtgtgcg tcagctcgtg tcgtgagatg ttgggttaag 1080
tcccgcacag agcgcaaccc ttattattag ttgtacgca agagcaactc aatgagactg 1140
ccgttgacaa aacggaggaa ggtggggatg acgtcaaac atcatgccc ttatgacctg 1200
ggctacacac gtactacaat ggcactgaaa cagagggaag cgacatcgcg aggtgaagcg 1260
aatcccaaaa aagtgtccca gttcggttg caggctgcaa ctgcctgca tgaagtcgga 1320
attgctagta atcgcggtac agcatgccgc ggtgaatacg ttcccgggcc ttgtacacac 1380
cgcccgctcac accatgggag tcggtaacac ccgaagccag tagcctaacc gcaaggaggg 1440
cgctgtcgaa ggtgggattg atgactgggg tgaagtcgta acaaggtagc cgt 1493

```

```

<210> SEQ ID NO 39
<211> LENGTH: 1483
<212> TYPE: DNA
<213> ORGANISM: Clostridium leptum
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1) .. (1483)

```

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<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 19

<400> SEQUENCE: 39

```

agagtttgat cctggctcag gacgaacgct ggccggcacgc ctaacacatg caagtcgaac      60
ggagtgaaga tgcttgcatc tgaacttagt ggccggacggg tgagtaaacac gtgagcaacc      120
tgcccttcag agggggataa cgtttggaaa cgaacgctaa taccgcataa aatatcggag      180
tcgcatggca ctgatatcaa aggagcaatc cgctgaaaga tgggctcgcg tccgattagc      240
cagttggcgg ggtaacggcc caccaaaccg acaatcggtg gccggactga gaggttgaa      300
ggccacattg ggactgagac acggcccaga ctctacggg aggcagcagt gggggatatt      360
gcacaatggg ggaaacctcg atgcagcgat gccgcgtgaa tgaagacggc cttcgggttg      420
taaagttctg tcgcagggga cgaatatgac ggtaccctgc aagaaagctc cggctaacta      480
cgtgccagca gccgcggtaa tacgtaggga gcaagcgttg tccggaatta ctgggtgtaa      540
agggagcgta ggcgggagga taaagttgaa tgtgaaatct atgggctcaa cccatagctg      600
cgttcaaaac tgtttctctt gagtgaagta gaggcaggcg gaattcctag tgtagcggtg      660
aaatcgctag atattaggag gaacaccagt ggcgaaagcg gcctgctggg cttttactga      720
cgctgaggct cgaagcgtg ggtagcaaac agaattagat taccctgtta ttcacggcgg      780
taaacgatga ttactaggtt tgggttgacc tgaccccatc tcgtgccgga agtaacacca      840
taaagtaatc cacctggggg agtacggccg ccaggttgaa acttcaaaag gaattgacgg      900
gggcccgcac aagcagtgga ggtatgtggt ttaatttcga cgcaaacgcg aagaacctta      960
ccaggtctct gacatcgagt gacggacata gagatatgtc tttcctttcg ggacacgaag     1020
acaggtggtg catggttgtc gtcagctcgt gtcgtgagat gttgggttaa gtcccgaac     1080
gagcgcaacc cttaccatta gttgctacgc aagagcactc tgatgggact gccgttgaca     1140
aaacggagga aggtggggat gacgtcaaat catcatgccc cttatgacct gggcgacaca     1200
cgtactacaa tggcgggtcaa cagagggagg caaagccgcg aggcagagca aaccctaaa     1260
agccgtctca gttcggattg caggctgcaa ctgcctgca tgaagtcgga attgctagta     1320
atcgcggtac agcatgccg ggtgaatacg ttccggggcc ttgtacacac cgcctctcac     1380
accatgagag ccggtaacac ccgaagtcaa tagtctaacc gcaaggagga cattgccgaa     1440
ggtgggattg gtaattgggg tgaagtcgta acaaggtagc cgt                                1483

```

<210> SEQ ID NO 40

<211> LENGTH: 1511

<212> TYPE: DNA

<213> ORGANISM: Clostridium coccoides

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1) .. (1511)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 20

<400> SEQUENCE: 40

```

agagtttgat cctggctcag gatgaacgct ggccggcgtgc ttaacacatg caagtcgaac      60
gggctcatat tgaaacctag tgatgtatga gttagtggcg gacgggtgag taacgcgtgg      120
agaacctgcc gtatactggg ggataaact tagaaatagg tgctaatacc gcataagcgc      180
acagcttcgc atgaacacgt gtgaaaaact ccggtggtat acgatggatc cgcgtctgat      240
tagctggttg gcggggtaac agcccaccaa ggcgacgac agtagccggc ctgagagggg      300
gaacggccac attgggactg agacacggcc caaactccta cgggaggcag cagtggggaa      360

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tattgcacaa tgggggaaac cctgatgcag cgacgccgcg tgagtgaaga agtatttcgg	420
tatgtaaagc tctatcagca gggaagaaat actgacctta cggtcagcag acggtacctg	480
actaagaagc cccggctaac tacgtgccag cagccgcggt aatacgtagg ggcaagcgtt	540
atccggattt actgggtgta aagggagcgt agacggcagc gcaagtctga gtgaaatccc	600
atggcttaac catggaactg ctttggaac tgtgcagctg gagtgcagga gaggtaaagc	660
ggaattccta gtgtagcggg tgaatgcgt agatatagga ggaacaacag tggcgggaagg	720
cggctactgg gactgtaact gacgttgagg ctcgaaagcg tggggagcaa acaggattag	780
ataccctggt agtcacgccg taaacgatga ttactaggtg ttgggggacc ataggctctc	840
ggtgccggcg caaacgcaat taagtaatcc acctggggga gtacgttcgc aagaatgaaa	900
ctcaaaggaa ttgacgggga cccgcacaaa gcggtggagc atgtggttta attcgaaagc	960
aaacgcgaag aaacottacc tggctctgac atcccgatga cgagtgcagc aagtcacttt	1020
cccttcgggg caattggaga caggtgggtg atgggttgct gtcagctcgt gtcgtgagat	1080
gttgggttaa gtcgcccaac gagcgcaacc cctatttcca gtagccagca ggtagagctg	1140
ggcactctgg agagactgcc cgggataacc gggaggaagg cggggatgac gtcaaatcat	1200
catgcccctt atgatcaggg ctacacacgt gctacaatgg cgtaaacaaa gggaagcgag	1260
acggtgacgt taagcaaatc ccaaaaataa cgtcccagtt cggattgtag tctgcaactc	1320
gattacatga agctggaatc gctagtaatc gcgaatcaga atgtcgcggt gaatacgttc	1380
ccgggtcttg tacacaccgc ccgtcacacc atgggagtcg gaaatgcccg aagtcagtga	1440
cctaaccgaa aggaaggagc tgcgaaggt ggagccggt actggggtga agtagataac	1500
aaggtagccg t	1511

<210> SEQ ID NO 41

<211> LENGTH: 1495

<212> TYPE: DNA

<213> ORGANISM: Clostridium leptum

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1) .. (1495)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 21

<400> SEQUENCE: 41

agagtttgat cctgcgctca ggacgaacgc tggcggcgcg cctaacacat gcaagtcgaa	60
cgggactatt ttgggagaag ttttcggatg gatctcgga tagtttagtg gcggacgggt	120
gagtaacgcg tgggcaacct gccttacaca gggggataac aattagaaat gattgctaata	180
accgcataag accacggtac tgcattggtac agtggtaaaa actgaggtgg tgtaagatgg	240
gcccgcgtct gattaggtag ttggtgggggt agaagcctac caagccgacg atcagtagcc	300
gacctgagag ggcgaccggc cacattggga ctgagacacg gcccaaaactc ctacgggagg	360
cagcagtggtg gaattattgca caatggggga aacctgatg cagcgacgcc gcgtgagtga	420
ggaagtattt cggtatgtaa agctctatca gcagggaaga aaatgacggt acctgactaa	480
gaagccccgg ctaactacgt gccagcagcc gcggaatac gtagggggca agcgttatcc	540
ggatttactg ggtgtaaagg gagcgtagac ggacttgcaa gtctgatgtg aaaatccggg	600
ggcccaaccc cggaactgca ttggaaactg tatatctaga gtgtcgga ggaagtgga	660
atttcctggt gtagcgtgta aatgcgtaga tatcagagga acaccagtgg cgaaggcgct	720
tgctgacga tgactgacgt tgaagctcga aaagcgtggg tagcaaacag aattagatac	780

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cctggtaagt caaccocgta aacgatgatt actagggtttt ggttggaactg accccatccg      840
tgccggagta acaccaataa gttatccaac ctgggaagta cggccggcag gttgaaactc      900
aaaaggaaat gacgggggcc cgcacaagca gttgaagtat gtgggttaat tcgacgcaaa      960
cgcaagaac cttaccaggt cttgacatcg agtgacggac atagagatat gtctttcctt     1020
cgggacacga agacaggtgg tgcattggtg tcgtcagctc gtgtcgtgag atgttgggtt     1080
aagtcgccga acgagcgcaa cccttaccat tagttgtac gcaagagcac tctgatggga      1140
ctgccgttga caaaacggag gaaggtgggg atgacgtcaa atcatcatgc cccttatgac      1200
ctgggcgaca cacgtactac aatggcggtc aacagaggga ggcaaagccg cgaggcagag      1260
caaaccctta aaagccgtct cagttcggat tgcaggctgc aactcgctg catgaagtcg      1320
gaattgctag taatcgcgga tcagcatgcc gcggtgaata cgttccggg ccttgtacac      1380
accgcccgtc acaccatgag agccggtaac acccgaagtc aatagtctaa ccgcaaggag      1440
gacattgccg aaggtgggat tggttaattg ggtgaagtcg taacaaggta gccgt      1495

```

<210> SEQ ID NO 42

<211> LENGTH: 1491

<212> TYPE: DNA

<213> ORGANISM: Clostridium leptum

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1)..(1491)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 22

<400> SEQUENCE: 42

```

agagtttgat cctggctcag gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac      60
gagaatccag tgaaggagtt ttcggacaac ggatctggag gaaagtggcg gacgggtgag      120
taacgcgtga gcaatctgcc ttggagtggg gaataacggt tggaaacagc cgctaatacc      180
gcatgatcgc tctgggaggc atctctctcg acgccaaga tttatcgctc tgagatgagc      240
tcgcgtctga ttagcttgtt ggcggggtaa agggccacca aggcgacgat cagtacccgg      300
actgagaggt tggccggcca cattgggact gagacacggc ccagactcct acgggaggca      360
gcagtgggga atattgggca atgggcgcaa gcctgacca gcaacgccgc gtgaagggaag      420
aaggctttcg ggtgtgaaac ttctttctcg agggacgaag aaagtgcagg tacctcagga      480
ataagccacg gctaactacg tgccagcagc cgcggttaata cgtagggtggc aagcgttatc      540
cggatttatt ggggtgaaag ggcgtgtagg cgggaaagca agtcagatgt gaaaactcag      600
ggctcaaccc tgagcctgca ttgaaactg tttttcttga gtgtcgaga ggcaatcgga      660
attccgtgtg tagcggtgaa atgcgtagat atacggagga caccagtggc gaagcggatt      720
gctggacagt aactgacgct gaggcgcgaa gcgtggggag caaacaggat tagataccct      780
ggtagtccac gccgtaaacg atggatacta ggtgtggggg gactgacccc ctccgtgccg      840
cagctaacgc aataagtatc ccacctgggg agtacgatcg caaggttgaa actcaaagga      900
attgacgggg gcccgacaaa gcggtggagt atgtggttta attcgaagca acgcgaagaa      960
ccttaccagg gcttgacatc ctgctaacga accagagatg gattaggtgc ccttcgggga     1020
aagcagagac aggtggtgca tggttgctgt cagctcgtgt cgtgagatgt tgggttaagt     1080
cccgaacga gcgcaacct tattgttagt tgctacgcaa gagcactcta gcgagactgc     1140
cgttgacaaa acggaggaag gtggggacga cgtcaaatca tcatgccct tacgtcctgg     1200
gccacacacg tactacaatg gcggccaaca aagagaggca agaccgcgag gtggagcaaa     1260

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tctcaaaaag cgtgccagt tcggatcgca ggctgcaacc cgctgctgtg aagttggaat 1320
cgctagtaat cgcggatcag catgccgcgg tgaatacgtt cccgggcctt gtacacaccg 1380
cccgtcacac catgagagtc gggaacaccc gaagtccgta gcctaaccgc aaggggggcg 1440
cggccgaagg tgggttcgat aattgggggtg aagtcgtaac aaggtagccg t 1491

```

```

<210> SEQ ID NO 43
<211> LENGTH: 1495
<212> TYPE: DNA
<213> ORGANISM: Clostridium leptum
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1) .. (1495)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
strain 23

```

```

<400> SEQUENCE: 43

```

```

agagtttgat cctgtgcctc aggatgaacg ctggcggcgt gcttaacaca tgcaagtcga 60
acgagaacca acggattgag gattcgtcca aatgaagttg gggaaagtgg cggacgggtg 120
agtaacgcgt gagcaatctg ccttgagtg ggaataacg gttggaaca gccgctaata 180
ccgcatgatg cgtctgggag gcattctctt ggacgcaaaa gatttatcgc tctgagatga 240
gctcgcgtct gattagctag ttggcggggc aacggccac caaggcgacg atcagtagcc 300
ggactgagag gttggcgggc cacattggga ctgagacacg gccagactc ctacgggagg 360
cagcagtggtg gaattattgg caatgggcgc aagcctgacc cagcaacgcc gcgtgaagga 420
agaaggcttt cgggttgtaa acttctttta agggggacga acaaatgacg gtacccttg 480
aataagccac ggctaactac gtgccagcag ccgcggtaat acgtaggtag caagcgttat 540
ccggatttat tgggtgtaaa gggcgtgtag gcgggaatgc aagtcagatg tgaatactat 600
gggtcaacc catagcctgc atttgaaact gtatttcttg agtgctggag aggcgaatcg 660
aattccgtgt gtacgcgtga aatgcgtaga tatacggagg aacaccagtg gcgaaggcgg 720
attgctggac agtaactgac gctgaggggc gaaagcgtgg ggagcaaaca ggattagata 780
ccctggtagt ccacgcccga aacgatggat actaagtggt gggggactga cccctccgt 840
gccgcagcta acgcaataag tatcccacct ggggagtacg atcgcaaggt tgaaactcaa 900
aggaattgac gggggccgcg acaagcgggt gagtatgtgt tttaattcga agcaacgcga 960
agaaccttac cagggttgta catcctgcta acgaaccaga gatggatcag gtgcccttcg 1020
gggaaagcag agacaggtgg tgcattggtt tcgtcagctc gtgtcgtgag atgttgggtt 1080
aagtcgccga acgagcgcaa cccctattgt tagttgttac gcaagagcac tctagcgaga 1140
ctgccgttga caaacggag gaaggtgggg acgacgtcaa atcatcatgc cccttacgtc 1200
ctgggccaca cacgtactac aatggcggcc aacaaagaga ggcaagaccg cgagggtggag 1260
caaatctcaa aaagccgtcc cagttcggat cgcaggctgc aaccgcctcg cgtgaagttg 1320
gaatcgctag taatcgcgga tcagcatgcc gcggtgaata cgttccggg ccttgtagac 1380
accgcccgtc acaccatgag agtcgggaac acccgaagtc cgtagcctga ccgcaagggg 1440
ggcgcggccc aaggtgggtt cgataattgg ggtgaagtcg taacaaggta gccgt 1495

```

```

<210> SEQ ID NO 44
<211> LENGTH: 1440
<212> TYPE: DNA
<213> ORGANISM: Clostridium leptum
<220> FEATURE:
<221> NAME/KEY: rRNA

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<222> LOCATION: (1) .. (1440)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
strain 24

<400> SEQUENCE: 44
agagtttgat cctggctcag gacgaacgct ggcggcacgc ctaacacatg caagtcgaac    60
ggagtgaaga tgctcgcatc tgaacttagt ggcggaacgg tgagtaacac gtgagcaacc    120
tgcccttcag agggggataa cgtttgaaa cgaacgctaa taccgcataa aatatcgag    180
tcgcatggca ctgatatcaa aggagtaatc cgctgaaaga tgggctcgcg tccgattagg    240
cagttggcgg ggtaacggcc caccaaaccg acaatcggtg gccggactga gaggttgaa    300
ggccacattg ggactgagac acggcccaga ctctacggg aggcagcagt gggggatatt    360
gcacaatggg ggaaaccctg atgcagcgat gccgcgtgaa tgaagacggc cttcgggttg    420
taaagttctg tcgcagggga cgaaaatgac ggtaccctgc aagaagctc cgctaacta    480
ctgtccagca gcccggttaa tacgtaggga gcaagcgttg tccggaatta ctgggtgtaa    540
agggagcgta ggccgggagga taagttgaat gtgaaatcta tgggctcaac ccatagttgc    600
gttcaaaact gttcttcttg agtgaagtag aggcaggcgg aattcctagt gtagcgtgta    660
aatgcgtaga tattagagga acaccagtgg cgaagcggcc tgctgggctt ttactgacgc    720
tgagctcgaa agcgtgggta gcaacaggat tagataccct ggtagtccac gcggtaaacg    780
atgattacta gtgtgggttg actgacccat ccatgccgga gttaacacaa tagtaatcca    840
cctggggagt acgcgcagtg aactcaaagg attgacgggg ccgcacaaag cagtgagtat    900
gtggtttatt cgacgcacgc gagactacag tcttgacatc gatgacggac tagagatatg    960
tctttctcgg acacgaagac aggtggtgca tgggtgtcgt cagctcgtgt cgtgagatgt   1020
tgggttaagt ccgcacaacga gcgcaaccct taccattagt tgttacgcaa gagcactcta   1080
atgggactgc cgttgacaaa acggaggaag gtggggatga cgtcaaatca tcatgccctt   1140
tatgacctgg gcgacacacg tactacaatg gcggtcaaca gagggaggca aagccgcgag   1200
gcagagcaaaa cccctaaaag ccgtctcagt tcggattgca ggctgcaact cgctgcatg   1260
aagtcggaat tgctagtaat cgcggatcag catgccgcgg tgaatacgtt cccgggcctt   1320
gtacacaccg ccgctcacac catgagagcc ggtaacaccc gaagtcaata gtctaaccgc   1380
aaggaggaca ttgccgaagg tgggatggta attggggtga agtagtaaca aggtagccgt   1440

<210> SEQ ID NO 45
<211> LENGTH: 1495
<212> TYPE: DNA
<213> ORGANISM: Clostridium leptum
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1) .. (1495)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
strain 25

<400> SEQUENCE: 45
agagtttgat cctggctcag gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac    60
gagaaccatt ggatcgagga ttctgtccaag tgaagggtgg gaaagtggcg gacgggtgag   120
taacgcgtga gcaatctgcc ttggagtggg gaataacggc tggaaacagc cgctaatacc   180
gcatgatata gctgggaggg atctccctgg ctgtcaaaga tttatcgctc tgagatgagc   240
tcgcgtctga ttagctagtt ggccggggtaa cggcccacca aggcgacgat cagtagccgg   300
actgagaggt tggccggcca cattgggact gagacacggc ccagactcct acgggaggca   360

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gcagtgggga atattgggca atgggcgcaa gcctgaccca gcaacgccgc gtgaagggaag 420
aaggcttttc ggttgtaaac ttcttttgc agggacgaag caagtgcagg tacctgacga 480
ataagccacg gctaactacg tgccagcagc cgcggaata cgtagtggca agcgttatcc 540
ggattttatt ggggtgtaaag ggcgtgtagg cgggaatgca agtcagatgt gaaaactatg 600
gggtcacaac catagcctgc atttgaaact gtatttcttg agtgctggag aggcaatcga 660
attcctgtgt tagcgggtga aatgcgtaga tatacggagg aacaccagtg gcgaagcgga 720
ttgctggaca agtaactgac gctgaggcgc gaaagcgtgg ggagcaaca ggattagata 780
ccctggtagt ccacgcgta aacgatggat actaggtgtg gggggactga cccctccgt 840
gccgcagcta acgcaataag tatcccacct ggggagtagc atcgcaaggt tgaaactcaa 900
aggaattgac gggggccgc acaagcgtg gagtatgtg tttaattcga cgcaacgcga 960
agaaccttac caggcctga catcctacta acgaaccaga gatggattag gtgcccttcg 1020
gggaaagtag agacaggtg tgcatggtt tcgtcagctc gtgctgtgag atgttgggtt 1080
aagtcgccga acgagcgcaa cccctattgt tagttgtac gcaagagcac tctagcgaga 1140
ctgccgttga caaaccgag gaaggtggg acgacgtcaa atcatcatgc cccttacgtc 1200
ctgggccaca cacgtactac aatggcgcc aacaagaga ggcaagccg cgaggtggag 1260
caaatctcaa aaagccgtc cagttcggat cgcaggctgc aaccgcctg cgtgaagttg 1320
gaatcgctag taatcgcgga tcagcatgcc gcggtgaata cgttcccggt ccttgtacac 1380
accgccgtc acaccatgag agtcgggaac acccgaaagtc cgtagcctaa ccgcaagggg 1440
ggcgcgccg aaggtgggtt cgataattgg ggtgaagtcg taacaaggta gccgt 1495

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<210> SEQ ID NO 46

<211> LENGTH: 1495

<212> TYPE: DNA

<213> ORGANISM: Clostridium leptum

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1) .. (1495)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 26

<400> SEQUENCE: 46

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agagtttgat catggctcag gacgaacgct ggcggcgtgc ttaacacatg caagtcgaac 60
ggagcacccc tgaaggagtt ttcggacaac ggatgggaat gcttagtggc ggactggtga 120
gtaacgcgtg aggaacctgc cttccagagg gggacaacag ttggaaacga ctgctaatac 180
cgcctgatgc gttggagccg catgactccg acgtcaaaga ttatcgtcga gaagatggcc 240
tcgcgtctga ttagctagtt ggtgaggtaa cggcccacca aggcgacgat cagtagccgg 300
actgagaggt tggccggcca cattgggact gagatacggc ccagactcct acgggaggca 360
gcagtgggga atattgggca atggacgcaa gtctgaccca gcaacgccgc gtgaagggaag 420
aaggcttttc ggttgtaaac ttcttttaag ggggaagagc agaagacggt accccttgaa 480
taagccacgg ctaactacgt gccagcagcc gcggaatac gtaggtggca agcggtgtcc 540
ggatttactg ggtgtaaagg gcgtgcagcc ggagagacaa gtcagatgtg aaatccacgg 600
gctcaacccc tgaactgcat ttgaaactgt ttcccttgag tgctggagag ggtaatcgga 660
attcctttgt gtagcgtgta aatgcgtaga tataagaaga acaccagtgg cgaagcgga 720
ttactggacg ataactgacg gtgaggcgcg aaagcgtggg ggagcaacag attaaatacc 780
ctggtagtc acgctgttaa cgatcgatac taggtgtgcc gggactgacc ccctgcgtgc 840

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ccggaggttaa ccacaataag tatcgcacct ggggagtagc atcgcaaggt gaacttcaaa	900
ggaattgacg ggggcccgcc ccaagccgtg gattatgttg ttaattcgaa gcaacgcgaa	960
gaacctaccc agggcttgac atcctgctaa cgaagtagag atacattagg tgccctttcg	1020
gggaaagcag agacaggtgg tgcattggtg tcgtcagctc gtgtcgtgag atgttgggtt	1080
aagtcccgcg acgagcgcaa ccctattgt tagttgtac gcaagagcac tctagcgaga	1140
ctgccgttga caaaacggag gaaggcgggg acgacgtcaa atcatcatgc cccttatgtc	1200
ctgggctaca cacgtaatac aatggcgggt aacaaaggga tgcaaagccg cgaggcagag	1260
cgaaccccaa aaagccgtcc cagttcggat cgcaggctgc aaccgcctg cgtgaagtcg	1320
gaatcgctag taatcgcgga tcagcatgcc gcggtgaata cgttccggg ccttgtagac	1380
accgcccgtc acaccatgag agtcgggaac acccgaagtc cgtagcctaa ccgcaaggag	1440
ggcgcgccg aaggtgggtt cgataattgg ggtgaagtcg taacaaggta gccgt	1495

<210> SEQ ID NO 47

<211> LENGTH: 1509

<212> TYPE: DNA

<213> ORGANISM: Clostridium leptum

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1)..(1509)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 27

<400> SEQUENCE: 47

agagtttgat cctggctcag gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac	60
ggagtagccc tgaaggagtt ttoggacaac tgatgggact acttagtggc ggacgggtga	120
gtaacgcgtg agtaacctgc cttggagtgg ggaataacag ctggaacag ctgctaatac	180
cgcataatat gtctgtgtcg catggcactg gacatcaaa atttatcgct ctgagatgga	240
ctcgcgtctg attagctagt tggcggggta acggcccacc aaggcgacga tcagtagccg	300
gactgagagg ttggccggcc acattgggac tgagacacgg ccagactcc tacgggaggc	360
agcagtgggg aatattgggc aatgggcgca agcctgaccc agcaacgccg cgtgaaggaa	420
gaaggcttcc ggggtgtaaa cttcttttaa gggggaagag cagaagacgg tacccttga	480
ataagccacg gctaactacg tgccagcagc cgcggtaata cgtagggtgg aagcgttgtc	540
cggatttact ggggtgaaag ggcgtgcagc cggagagaca agtcagatgt gaaatccacg	600
ggctcaaccc gtgaactgca ttgaaactg tttcctggag ttccggaggt atggaattct	660
tgtagcgggt gaaatgctgt agatatggga gaaccaccag tgcgaggggg cttccgggac	720
tgtagctgac tgtagaggtc tcaaagctgg gggagcaccg aggaatgaga taccgtgata	780
gtccacacgg gtaacggatg attactaggt gttgggggga ccaggtctt ttccgtgccc	840
ggcgcaaacc ctttaggaat tccacctggg gaattacgtt tggcaagaaa ggaacttcaa	900
agaaattgaa cgggggaccc ccccaaccgg tggaggcatg gtgttttatt tcggaggaa	960
gggaagaacc ttaccttgt tctgacctc cggatgacga agtgagcaaa gtcaacttcc	1020
cttcggggcc atggaggaca ggtggtggca tgggtggtcg tcagctcgtg tcgtgagatg	1080
ttgggttaag tcccgaacg agcgaaccc ctatttccag tagccagcag gtagagctgg	1140
gcactctgga gagactgccc gggataaccg ggaggaaggc ggggatgacg tcaaatcatc	1200
atgccctta tgatcagggc tacacacgtg ctacaatggc gtaaacaaa ggaagcgaga	1260
cggtagcgtt aagcaaatcc caaaaataac gtcccagttc ggattgtagt ctgcaactcg	1320

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actacatgaa gctggaatcg ctagtaatcg cgaatcagaa tgctcgcggtg aatacgttcc	1380
cgggtcttgt acacaccgcc cgtcacacca tgggagtcgg aaatgccga agtcagtgac	1440
ctaaccgaaa ggaaggagct gccgaagggtg gagccggtaa ctggggtgaa gtcgtaacaa	1500
ggtagccgt	1509

<210> SEQ ID NO 48
 <211> LENGTH: 1583
 <212> TYPE: DNA
 <213> ORGANISM: Clostridium leptum
 <220> FEATURE:
 <221> NAME/KEY: rRNA
 <222> LOCATION: (1) .. (1583)
 <223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 28

<400> SEQUENCE: 48

agagtttgat cctggctcag gacgaacgct ggccggcgcg ctaacacatg caagtcgaac	60
ggagcttata ttccagaagt ttccggatgg acgagagata agcttagtgg cggacgggtg	120
agtaaacagt gagcaacctg cctttcagag ggggataaca gttggaaacg actgctaata	180
ccgcataacg ctgcgatggg gcatcccgat gcagccaaag gagcaatccg ctgaaagatg	240
ggctcgcgcg cgattagcta gttggtgggg caacggccca ccaaggcgac gatcggtagc	300
cggactgaga ggttgatcgg ccacattggg actgagacac ggcccagact cctacgggag	360
gcagcagtg gggatattgc acaatggagg aaactctgat gcagcgacgc cgcgtgaggg	420
aagacggtct tcgattgta aacctctgtc ttgggggaag aaaatgacgg tacccaaaga	480
ggaagctccg gctaactacg tgccagcagc cgcggtaata cgtaggggag cgagcgttgt	540
ccggaattac tgggtgtaaa gggagcgtag cgggcgagaa agttgaatgt taaatctacc	600
ggcttaactg gtactgcgt ccaaaacttc ttggtcttga gtgaaagtaa gaggccaggg	660
cggaaattct tagtgtaagc ggggtgaaat gcgttagata ttagggagga accaccaggt	720
gggcgaaggg cggcttgctg ggctttaact ggacggctgg aggcttgga aaggcgtggg	780
gagagcaaac acagggaatt aagtataccc tggatatgt cacacgcttg taaagagtat	840
gattaactta ggggtgtggg gggaaactga cctttctgtg tgcgcgcagg ttaacacaca	900
tttagagtat atccaacttg gggagagtac ggccggcaaa gtttgaaact tcaaaaggga	960
aattgagacc gggggggccc gccaccaagc acagtggaga gtatggtggg tttaatttcg	1020
agaagcaacc ggcggaagag aaactttacc agtccttgac atcgggtggcg gcataagccc	1080
tagagattag gtgaagccct tcggggggccc caccagacag gtggtgcatg gttgtcgtca	1140
gctcgtgtcg tgagatgttg ggttaagtcc ccgcaaacga gcgcaaccct tattattagt	1200
ttgctacgca agagcactct aatgagactg ccgttgacaa aacggaggga ggtggggatg	1260
acgtcaaacc atcatgcccc ttatgacctg ggctacacac gtactacaat ggcaactgaaa	1320
cagaggggaag cgacatcgcg aggtgaagcg aatcccaaaa aagtgtccca gttcggattg	1380
caggctgcaa ctgcctgca tgaagtcgga attgctagta atcgcggatc agcatgccgc	1440
ggtgaatacg ttccggggcc ttgtacacac cgcctgtcac accatgggag tcggtaacac	1500
ccgaagccag tagcctaacc gcaaggaggg cgtgtgtgaa ggtgggattg atgactgggg	1560
tgaagtcgta acaaggtagc cgt	1583

<210> SEQ ID NO 49
 <211> LENGTH: 1519
 <212> TYPE: DNA

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<213> ORGANISM: Clostridium coccoides
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1)..(1519)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
strain 29

<400> SEQUENCE: 49
agagtttgat catggctcag gatgaacgct ggcggcgtgc ctaacacatg caagtcgaac    60
gggtgtacgg gaaggaaggc ttcggccgga aaacctgtgc atgagtggcg gacgggtgag    120
taacgcgtgg gcaacctggc ctgtacaggg ggataacact tagaaatagg tgctaatacc    180
gcataacggg ggaagccgca tggtttttcc ctgaaaactc cgggtgtaca ggatgggccc    240
gcgtctgatt agccagtggc cagggtaacg gcctaccaa gcgacgatca gtagccggcc    300
tgagaggcg gacggccaca ctgggactga gacacggccc agactcctac gggaggcagc    360
agtgggggat attgcacaat ggggggaacc ctgatgcagc gacgccgct gggatgaagaa    420
gcgcctcggc gcgtaaagcc ctgtcagcag ggaagaaaa tgacggtacc tgaagaagaa    480
gccccgggct aactacgtgc cagcagccgg cggttaattac gtaggggggc aggcgttatc    540
cggatttact ggggtgtaaa ggggggcgca aacggcgatg gcaggccagg aatggaaagc    600
ccggggggccc aaccccgga cttgctcttg ggaaactggc ttggctggga gtggcaggag    660
gggcaggcgg aaattcctgg tggtagcggg ggaaatggc taaaaatcaa gaagaaaaac    720
cgggtgggaa aggcggcctg gtgggactgc gaactgacgt tgaaggcccg aaagcgtggg    780
gaacaaacag gatagattcc ctggtagttc cacgccgtaa acgatgatta ctagggtgctg    840
gggagcagag actgcccggt gccgcagcca acgcattaag taatccacct ggggagtacg    900
ttcgcaagaa tgaaactcaa aggaattgac ggggacccgc acaagcggtg gagcatgtgg    960
tttaattcga agcaacgcga agaaccttac ctggtcttga catcccgatg acgagtgagc   1020
aaagtcactt tcccttcggg gcattggaga caggtgggtc atggttgctg tcagctcgtg   1080
tcgtgagatg ttgggttaag tcccgcaacg agcgcaaccc ctatttccag tagccagcag   1140
gtagagctgg gcactctgga gagactgccc gggataaacc ggaggaaggc ggggatgacg   1200
tcaaatacat atgcccctta tgatcagggc tacacacgtg ctacaatggc gtaaacaaaag   1260
ggaagcgaga cggtgacgtt aagcaaatcc caaaaataac gtcccagttc ggattgtagt   1320
ctgcaactcg actacatgaa gctggaatcg ctagtaatcg cgaatcagaa tgtcgcggtg   1380
aatacgttcc cgggtcttgt acacaccgcc cgtcacacca tgggagtcgg aaatgcccgga   1440
agtcagtgac ctaaccgaaa ggaaggagct gccgaaggtg gagccggtaa ctggggtgaa   1500
gtcgtaacaa ggtagccgt                                     1519

<210> SEQ ID NO 50
<211> LENGTH: 1497
<212> TYPE: DNA
<213> ORGANISM: Clostridium coccoides
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1)..(1497)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
strain 30

<400> SEQUENCE: 50
agagtttgat cctggctcag gatgaacgct ggcggcgtgc ctaacacatg caagtcgaac    60
ggggtatata agcggaggtt tacggatgga aggttatata cttagtggcg gacgggtgag    120
taacgcgtgg gcaacctgcc ccgtgccggg ggataaccgc tggaacacag cgctaatacc    180

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gcataagcgc atacagccgc atgggtgtat gcggaaagct ccggcggcac gggatgggcc	240
cgcgcccgat tagccagttg gcggggtaac ggcccaccaa agcgacgac ggtagccggc	300
ctgagagggc ggacggccac attgggactg agacacggcc caaactccta cgggaggcag	360
cagtggggaa tattgcacaa tgggggaaac cctgatgcag caacgccgcg tgggtgaagg	420
agcgtttcgg cgcgtaaagc cctgtcagcg gggaagaaga aagacggtag ccgaccaaga	480
agccccggct aactacgtgc cagcagccgc ggtaatacgt agggggcgag cgttatccgg	540
aattactggg tgtaaaggga gcgtagacgg cgaggtgaag ctgaagtga agcccgcggc	600
ccaaccggcg aactgctttg ggaactgttt tgcctggagta tgggaggggt aagcggaatt	660
cctggtgtag cggtgaaatg cgtagatata aggaggaaca ccggtggcga agcggttta	720
ctggaccata actgacgttg aggctcgaaa gcgtggggag cgaacaggat tagataccct	780
ggtagtccac gcgtaaacga tgattaccag gtgtcgggtg tcgaaggacg gcccggtgcc	840
gcagcgaacg cagtaagtaa tccacctggg gagtacgttc gcaagaatga aactcaaagg	900
aattgacggg gaccgcgaca agcgggtggag catgtggttt aattcgaagc aacgcgaaga	960
accttaccgg gccttgacat cccctggaca gcatatgtaa tgtatgtttc ctccgggacc	1020
agggagacag gtggtgcatg gttgtcgtca gctcgtgtcg tgagatgttg ggtcaagtcc	1080
cgcaacgagc gcaaccctcg ccccgagtag ccagcattta agatgggcac tctgggggga	1140
ctgccgggga taaccgggag gaaggcgggg atgacgtcaa atcatcatgc cccttatggc	1200
cggggctaca cacgtgtac aatggcgtaa acagagggag gcgagacagc gatgttaagc	1260
gaaccccaaa aataacgtcc cagttcggat tgcagcctgc aactcggctg catgaagctg	1320
gaatcgctag taatcgcgga tcagaatgcc gcggtgaata cggtcccggg tcttgtagac	1380
accgcccgtc acaccatggg agtcgggaac gcccgaaagc ggtgaccgaa cccgaaaggg	1440
gaggagccgt cgaaggcggg cctggtgact ggggtgaagt cgtaacaagg tagccgt	1497

<210> SEQ ID NO 51

<211> LENGTH: 1475

<212> TYPE: DNA

<213> ORGANISM: Clostridium leptum

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1)..(1475)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 31

<400> SEQUENCE: 51

gagtttgatc ctggctcagg ataaacgtcg gcggcgacaa taagacatgc aagtccaacg	60
aacttaatac cttgcttgca aggtaagcgg ttagtggcgg actggtgagt aacacgtaag	120
aaatctgcct atcagagggg aataacagtg agaaatcact gctaataccg catatgccat	180
agttatcgca tgataatagt gggaaagaag caattcgctg atagatgagc ttgcggctga	240
ttagctagtt ggtggggtaa cgccctacca aggcgacgat cagtagccgg cctgagaggg	300
tgaacggcca cattgggact gagacacggc ccaaactcct acgggaggca gcagtgggga	360
atattgcaca atgggggaaa cctgatgca gcgacgccgc gtgagtgaag aagtatttcg	420
gtatgtaaag ctctatcagc agggaagaaa atgacggtag ctgactaaga aagccccggc	480
taactacgtg ccagcagcgg cggtaatagc tagggggcaa gcgttatccg gatttactgg	540
tgtaaaggga gcgtagacgg cagcgcaagt ctgagtgaag tcccatggct taccatgaa	600
actgctttgg aaactgtgca gctggagtgc aggagaggta agcggaatcc tagttagcgc	660

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gttgaaatgc gtagattatc agaaggaaca cgggtggccg aggcggcctg ctgggccttt 720
actgacgctg aggcctgaag cgtgggtagc aaacaggatt agataccctg gtagtccacg 780
cggtaaacga tgattactag gtgtgggtgg actgacccca tccgtgccgg agttaacaca 840
ataagtaatc cacctgggga gtacggccgc aaggttgaaa ctcaaaggaa ttgacggggg 900
cccgcacaa gtagtgagta tgtgggttaa ttcgacgcaa cgcaagaac cttaccaggt 960
cttgacatcg agtgacggac atagagatat gtcttccctt cgggacacga agacaggtag 1020
tgcatggttg tcgtcagctc gtgtcgtgag atgttgggtt aagtcccgca acgagcgcaa 1080
cccttaccat tagttgctac gcaagagcac tctaattggga ctgccgttga caaaacggag 1140
gaaggtgggg atgacgtcaa atcatcatgc cccttatgac ctgggcgaca cacgtactac 1200
aatggcggtc aacagaggga ggcaaagccg cgaggcagag caaaccccta aaagcgtct 1260
cagttcggat tgcaggctgc aactcgctg catgaagtcg gaattgctag taatcgcgga 1320
tcagcatgcc gcggtgaata cgttcccggg ccttgtagac accgcccgtc acaccatgag 1380
agccggtaac acccgaagtc aatagtctaa ccgcaaggag gacattgccg aaggtgggat 1440
tggttaattg ggtgaagtcg taacaaggta gccgt 1475

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<210> SEQ ID NO 52

<211> LENGTH: 1491

<212> TYPE: DNA

<213> ORGANISM: Clostridium leptum

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1) .. (1491)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 32

<400> SEQUENCE: 52

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agagtttgat cctggctcag gatgaacgct ggcggcgtgc ttaacacatg caagtcgaac 60
gagaatcagt ggattgagga ttcgtccaaa tgaaactgag gaaagtggcg gacgggtgag 120
taacgcgtga gcaatctgcc ttggagtggg gaataacggc tggaacagc cgctaatacc 180
gcatgataca gttgggaggc atctctctga ctgtcaaaga tttatcgctc tgagatgagc 240
tcgcgtctga ttagctagtt ggcggggtaa cggcccacca aggcgacgat cagtagccgg 300
actgagaggt tggccggcca cattgggact gagacacggc ccagactcct acgggaggca 360
gcagtgggga atattgggca atgggcgcaa gcctgaccca gcaacgccgc gtgaaggaa 420
aaggctttcg ggttgtaaac tcttttctg ggggacgaac aaatgacggt accccaggaa 480
taagccacgg ctaactacgt gccagcagcc gcggttaatac gtaggtggca agcgttatcc 540
ggatttattg ggtgtaaagg gcgtgtaggc gggaatgcaa gtcagatgtg aaaactatgg 600
gctcaaccca tagcctgcat ttgaaactgt atttcttgag tgctggagag gcaatcgaa 660
ttccgtgtgt agcggtgaaa tgcgtagata tacggaggaa caccagtggc gaagcggatt 720
gctggacagt aactgacgct gaggcgcgaa agcgtgggga gcaaacagga ttagataccc 780
tggtagtcca cgccgtaacg atggatacta gtgtgggggg actgaccccc tccgtgccgc 840
agctaacgca ataagtatcc ccacctgggg agtacgatcg caaggttgaa actcaaagga 900
attgacgggg gcccgacaaa gcggtggagt atgtggttta attcgaagca acgcgaagaa 960
ccttaccagg gcttgacatc ctgctaacga accagagatg gattaggtgc ccttcgggga 1020
aagcagagac aggtggtgca tggttgtcgt cagctcgtgt cgtgagatgt tgggttaagt 1080
cccgcaacga gcgcaacccc tattgttagt tgctacgcaa gagcactcta gcgagactgc 1140

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cgttgacaaa acggaggaag gtggggacga cgtcaaatca tcatgcccct tacgtcctgg 1200
gccacacacg tactacaatg gcggttaaca aagagaggca agaccgcgag gtggagcaaa 1260
tctcaaaaag ccgteccagt tcggatcgca ggctgcaacc cgctgctgtg aagttggaat 1320
cgctagtaat cgcggtatcg catgccgcgg tgaatacgtt cccgggcctt gtacacaccg 1380
cccgtcacac catgagagtc gggaacaccc gaagtccgta gcctaaccgc aaggggggcg 1440
cggccgaagg tgggttcgat aattgggggtg aagtcgtaac aaggtagccg t 1491

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<210> SEQ ID NO 53
<211> LENGTH: 1495
<212> TYPE: DNA
<213> ORGANISM: Clostridium coccoides
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1)..(1495)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
strain 33

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<400> SEQUENCE: 53

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agagtttgat cctggctcag gatgaacgct ggcggcgtgc ctaacacatg caagtcgaac 60
gggtgtacgg ggaggaaggc ttcggccgga aaacctgtgc atgagtggcg gacgggtgag 120
taacgcgtgg gcaacctggc ctgtacaggg ggataaact tagaaatagg tgctaatacc 180
gcataacggg ggaagccgca tggcttttcc ctgaaaactc cggtggtaca ggatgggccc 240
gcgtctgatt agccagttag cagggtaacg gcctacaaa gcgacgatca gtagccggcc 300
tgagaggggc gacggccaca ctgggactga gacacggccc agactcctac gggaggcagc 360
agtgggggat attgcacaat ggggggaacc ctgatgcagc gacgccgcgt gggatgaaga 420
gcgcctcggc gcgtaaagcc ctgtcagcag ggaagaaaat gacggtacct gaagaagaag 480
ccccggctaa ctacgtgcc aacgcgcgg taatacgtag ggggcaagcg ttatccggat 540
ttactgggtg taaagggggc gcagacggcg atgcaagcca ggagtgaag cccggggccc 600
aaccgccgga ctgctcttgg aactgcgtgg ctggagtgc gagggggcag gcggaattcc 660
tgggtgtacg gtgaaatgcg tagatatcag aggaacaccg gtggcgaaag cggcctgctg 720
gactgcaact gacgttagg cccgaaagcg gtgggagcaa acaggattag ataccctggt 780
agtccacgcc gtaaacgatg attactaggt gtcggggagc agagactgcc cgggtgccga 840
gcccacgcga ttaagtatcc acctggggag tacgttcgca agaataaac tcaaaggaat 900
tgacggggac ccgcacaagc ggtggagcat gtggtttaat tcgaagcaac gcgaagaacc 960
ttaccaggcc ttgacatccc cctggatggc ccgtaacggg gccagccctt tttgggcagg 1020
ggagacaggt ggtgcatggt tgcgtcagc tcgtgtcgtg agatgttggg ttaagtcccg 1080
caacgagcgc aaccctgcc cgcagtagcc agcatcttag atggggactc tcgggggact 1140
gccggggaca acccggagga aggcggggat gacgtcaaat catcatgcc cttatggcct 1200
gggctacaca cgtgctacaa tggcgccgac agaggaggga gaagcggcga cgcggagcga 1260
acccccaaaa cggcgctcca gttcggattg tagtctgcaa cccgactaca tgaagccgga 1320
atcgctagta atcgcggtac agaatgccgc ggtgaatacg ttcccgggtc ttgtacacac 1380
cgcccgctac accatgggag cccgggaatgc ccgaagtctg tgaccgaacc cgtaagggga 1440
ggggcagccg aaggcaggcc cggtgactgg ggtgaagtcg taacaaggta gccgt 1495

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<210> SEQ ID NO 54
<211> LENGTH: 1493

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<212> TYPE: DNA
<213> ORGANISM: Clostridium leptum
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1)..(1493)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
        strain 34

<400> SEQUENCE: 54
agagtttgat cctggctcag gacgaacgct ggcggcgtgc ttaacacatg caagtcgaac      60
ggagcacccc tgaaggagtt ttcggacaac ggatgggaat gcttagtgtc ggactgggtga     120
gtaacgcgtg agaacctgc cttccagagg gggacaacag ttggaaacga ctgctaatac      180
cgcatgatgc gttggagcgc catgactccg acgtcaaaga tttatcgctg gaagatggcc      240
tcgcgtctga tttgttagtt ggtgaggtaa cggcccacca aggcgacgat cagtagccgg      300
actgagaggt tggccggcca cattgggact gagatacggc ccagactcct acgggaggca      360
gcagtgggga atattgggca atggacgcaa gtctgaccca gcaacgcgcg gtgaagggaag     420
aaggctttcg ggttgtaaac tctttttaag ggggaagagc agaagacggt accccttgaa      480
taagccacgc ctaactacgt gccagcagcc gcggtaatag gtaggtggca agcgttgttc      540
ggatttactg ggtgtaaagg gcgtgcagcc ggagagacaa gtcagatgtg aaatccacgg      600
gctcaaccgc tgaactgcat ttgaaactgt ttcccttgag tgtcggagag gtaatcgga      660
ttccttggtg agcggtgaaa tgcgtagata taaggaagaa caccagtggc gaaggcggat      720
tactggacga taactgacgc tgaggcgcga aagcgtgggg agcaaacagg attagatacc      780
ctggtagtag acgctgtaaa cgatcgatac taggtgtgcg gggactgacc ccctgcgtgc      840
cggagttaac acaataagta tcgcacctgg ggagtacgat cgcaagggtg aaactcaaag      900
gaattgacgc gggccgcgc aagcgggtga ttatgtggtt taattcgaag caacgcgaag      960
aaccttacca gggcttgaca tctgtctaac gaagtagaga tacattaggt gcccttcggg     1020
gaaagcagag acaggtgggt catggtgtgc gtcagctcgt gtcgtgagat gttgggttaa     1080
gtcccgaac gagcgcaacc cctattgtta gttgctacgc aagagcactc tagcgagact     1140
gccgttgaca aaacggagga aggcggggac gacgtcaaat catcatgccc cttatgtcct     1200
gggttacaca cgtaatacaa tggcggttaa caaagggatg caaagccgcg aggcagagcg      1260
aaccacaaaa agcgtcccca gttcggatcg caggctgcaa cccgcctgcg tgaagtcgga      1320
atcgctagta atcgcggatc agcatgccgc ggtgaatacg ttccggggcc ttgtacacac      1380
cgcccgtcac accatgagag tcggggaacac ccgaagtccg tagcctaacc gcaaggaggg      1440
cgcggccgaa ggtgggttcg ataattgggg tgaagtcgta acaaggtagc cgt          1493

<210> SEQ ID NO 55
<211> LENGTH: 1498
<212> TYPE: DNA
<213> ORGANISM: Clostridium coccoides
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1)..(1498)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
        strain 35

<400> SEQUENCE: 55
agagtttgat cctggctcag gatgaacgct ggcggcgtgc ctaacacatg caagtcgaac      60
gggtgtacag aagggaagat tacggtcgga aggtctgtgc atgagtggcg gacgggtgag      120
taacgcgtgg gcaacctggc ctgtacaggg ggataaact tagaaatagg tgctaatacc      180

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gcataacggg ggaagccgca tggcttttcc ctgaaaactc cgggtggtaca ggatgggccc	240
gcgtctgatt attttttttg tcagggtaac ggcctaccaa agcgacgatc agtagccggc	300
ctgagagggc ggacggccac actgggactg agacacggcc cagactccta cgggaggcag	360
cagtggggga tattgcacaa tggggggaac cctgatgcag cgacgccgcg tgggtgaaga	420
agcgctcgg cgcgtaaagc cctgtcagca gggaagaaaa tgacggtacc tgaagaagaa	480
gccccggcta actacgtgcc agcagccgcg gtaatacgta aggggcaagc gttatccgga	540
tttactgggt gtaaaggggg cgacagcggc gatgcaagcc aggagtgaag gcccggggcc	600
caaccccggg actgctcttg ggaactgcgg tggctggagt gcaggagggg caggccggaa	660
ttcttggtgt agcggtgaaa tgcgtagata tcaggaggaa caccggtggc gaaggcgccc	720
tgctggactg caactgacgt tgaggcccg aagcgtgggg agcaaacagg attagatacc	780
ctggtagtca cgccgtaaac gatgattact aggtgtcggg gagcagagac tgcccgtgct	840
cgcagccaac gcattaagta atccacctgg ggagtagctt cgcaagaatg aaactcaaa	900
gaattgacgg ggacccgcac aagcgggtga gcatgtggtt taattcgaag caacgcgaag	960
aaccttacca ggccttgaca tccccctgga tggcccgtaa cggggctcagc ctttcggggc	1020
aggggagaca ggtggtgcat ggtgtcgtc agctcgtgtc gtgagatgtt ggggttaagt	1080
ccgcaacgag cgcaacccct gcccgagta gccagcattt tagatgggga ctctgcgggg	1140
actgccgggg acaacccgga ggaaggcggg gatgacgtca aatcatcatg ccccttatgg	1200
cctgggctac acacgtgcta caatggcgcc gacagaggga ggcaagcgg cgacgcggag	1260
cgaaccccaa aaacggcgtc ccagttcgga ttgtagtctg caacccgact acatgaagcc	1320
ggaatcgcta gtaatcgcg atcagaatgc cgcggtgaat acgttcccg gtcttgatca	1380
caccgcccgt cacaccatgg gagccgggaa tgcccgaagt ctgtgaccga acccgtaagg	1440
ggaggggcag ccgaaggcag gcccggtgac tggggtgaag tcgtaacaag gtagccgt	1498

<210> SEQ ID NO 56

<211> LENGTH: 1491

<212> TYPE: DNA

<213> ORGANISM: Clostridium leptum

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1) .. (1491)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 36

<400> SEQUENCE: 56

agagtttgat catggctcag gacgaacgct ggcggcaagc ttaacacatg caagtcgaac	60
ggagcgccta tgaaggagat ttcggtcaac ggaataggct gcttagtggc tgacgggtga	120
gtaacgcgtg aggaacctgc ctttcagagg gggacaacag ttggaaacga ctgctaatac	180
cgcataacac ataggtgtcg catggcattt atgtcaaaga tttatcgctg aaagatggcc	240
tcgcgtctga ttagctagtt ggtgaggtaa cggctcacca aggcgacgat cagtagccgg	300
actgagaggt tagccggcca cattgggact gagatacggc ccagactcct acgggaggca	360
gcagtgggga atattgggca atggacgcaa gtctgaccca gcaacgcgc gtgaagggaag	420
aaggctttcg ggtgtgaaac tctttttaag aggggaagagc agaagacggt acctcttgaa	480
taagccacgg ctaactacgt gccagcagcc gcggtaatag gtagtgcaa gcgttggtccg	540
gatttactgg gtgtaaaggg cgtgtagccg ggctgacagt cagatgtgaa attccggggc	600
tcaaccccg acctgcattt gaaactgttg gtcttgagta tcggagaggc aggcggaatt	660

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cctagtgtag cggtgaaatg cgtagatatt aggaggaaca ccagtggcga aggcggcctg	720
ctggacgaca actgacgggtg aggcgcgaaa gcgtggggag caaacaggat tagataccct	780
ggtagtccac gctgtaaacg atggatacta ggtgtgcggg gactgacccc ctgcgtgccg	840
cagttaacac aataagtatc ccacctgggg agtacgatcg caagggtgaa actcaaagga	900
attgacgggg gcccgacaaa gcggtggatt atgtggttta attcgatgca acgcgaagaa	960
ccttaccagg gcttgacatc ctgctaacga ggtagagata cgtcaggatgc ccttcgggga	1020
aagcagagac aggtggtgca tggttgtcgt cagctcgtgt cgtgagatgt tgggttaagt	1080
cccgcaacga gcgcaaccct tattgttagt tgctacgcaa gagcactcta gcgagactgc	1140
cgttgacaaa acggaggaag gtggggacga cgtaaatca tcatgccctt tatgtcctgg	1200
gctacacacg taatacaatg gcggtaaaca gagggatgca atactgcgaa gtggagcgaa	1260
cccctaaaag cgtccccagt tcagattgca gtctgcaact cgactgcatg aagtcggaat	1320
cgctagtaat cgcggatcag catgccggcg tgaatacgtt cccgggcctt gtacacaccg	1380
cccgtcacac catgagagtc gggaacaccc gaagtcgta gcctaaccgc aaggagggcg	1440
cggccgaagg tgggttcgat aattgggggtg aagtcgtaac aaggtagccg t	1491

<210> SEQ ID NO 57

<211> LENGTH: 1493

<212> TYPE: DNA

<213> ORGANISM: Clostridium coccoides

<220> FEATURE:

<221> NAME/KEY: rRNA

<222> LOCATION: (1)..(1493)

<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium strain 37

<400> SEQUENCE: 57

agagtttgat cctggctcag gatgaacgct ggcggcgtgc ctaacacatg caagtcgaac	60
gggtgtacgg ggaggaaggc ttgcggcggg aaacctgtgc atgagtggcg gacgggtgag	120
taacgcgtgg gcaacctggc ctgtacaggg ggataaact tagaaatagg tgctaatacc	180
gcataacggg ggaagccgca tggcttttcc ctgaaaactc cggtggtaca ggatgggccc	240
gcgtctgatt agccagttag cagggtaacg gcctacaaa gcgacgatca gtagccggcc	300
tgagaggggc gacggccaca ctgggactga gacacggccc agactcctac gggaggcagc	360
agtgggggat attgcacaat ggggggaaac cctgatgcag cgacgccgcg tgagtgaaga	420
agtatttcgg tatgtaaagc tctatcagca ggggaagaaa tgacgggtacc tgactaagaa	480
gccccggcta actacgtgcc agcagccgcg gtaatacgtg gggggcaagc gttatccgga	540
tttactgggt gtaaagggag cgtagacggc agcgcaagtc tgaagtgaat tcccattggc	600
taaccatgga actgctttgg aaactgtgca gctggagtgc aggagaggta agcggaattc	660
ctagtgtagc ggtgaaatgc gtagatatta ggaggaacac cagtggcgaa ggcggcttac	720
tggactgtac tgacgttgag gctcgaaaagc gtggggagca aacaggatta gataccctgg	780
tagtccacgc cgtaaacgat gattactagg tgttggggga ccaaggctct cggtgccggc	840
gcaaacgcat taagtaatcc acctggggag tacgttcgca agaagaaac tcaaaggaat	900
tgacggggac ccgcacaagc ggtggagcat gtggtttaat tcgaagcaac gcgaagaacc	960
ttacctggtc ttgacatccc gatgacgagt gagcaaagtc actttccctt cggggcattg	1020
gagacagggt gtgcatgggt gtcgtcagct cgtgtcgtga gatgttgggt taagtcgccg	1080
aacgagcgca acccctatct ccagtagcca gcaggtagag ctgggcactc tggagagact	1140

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ccccgggata accgggagga aggcggggat gacgtcaaat catcatgccc cttatgatca 1200
gggtacaca cgtgtacaa tggcgtaaac aaagggaagc gagacgggtga cgttgagcaa 1260
atccccaaaa taacgtccca gttcggattg tagtctgcaa ctgactaca tgaagctgga 1320
atcgctagta atcgcgatc agaatgtcgc ggtgaatacg ttcccgggtc ttgtacacac 1380
cgcccgtcac accatgggag tcggaatgc ccgaagtcag tgacctaac gaaaggaagg 1440
agctgccgaa ggtgggagcg gtaactgggg tgaagtcgta acaaggtagc cgt 1493

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<210> SEQ ID NO 58
<211> LENGTH: 1493
<212> TYPE: DNA
<213> ORGANISM: Clostridium leptum
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1) .. (1493)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
strain 38

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<400> SEQUENCE: 58

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aaagtttgat cctggctcag gacgaacgct ggcggcgtgc ttaacacatg caagtcgaa 60
ggagaccccc tgaaggagtt ttcggacaac ggatgggaat gcttagtggc ggactgggtga 120
gtaacgcgtg aggaacctgc cttccagagg gggacaacag ttggaaacga ctgctaatac 180
cgcatgatgc gttggagcgc catgactccg acgtcaaaga tttatcgctg gaagatggcc 240
tcgctgtcga ttagctagtt ggtgaggtaa cggcccacca aggcgacgat cagtagccgg 300
actgagaggt tggcgcgcga cattgggact gagatacggc ccagactcct acgggaggca 360
gcagtgggga atattgggca atggacgcaa gtctgaccca gcaacgcgcg gtgaagggaag 420
aaggctttcg ggttgtaaac ttcttttaag ggggaagagc agaagacggt accccttgaa 480
taagccacgg ctaactacgt gccagcagcc gcggtaatac gtagtggaac gcgttgctcg 540
gatttactgg gtgtaaaagg cgtgcagcgc gagagacaag tcagatgtga aatccacggg 600
ctcaaccctg gaactgcatt tgaaactgtt tcccttgagt gtcggagagg taatcggaa 660
tccttggtga cgggtgaat gcgtagatat aaggaagaac accagtggcg aaggcggatt 720
actggacgat aaactgacgg tgaggcgcga aagcgtgggg agcaaacagg attagatacc 780
ctggtagtcc acgctgtaaa cgatcgatac taggtgtgcg gggactgacc ccctgcgtgc 840
cggagttaac acaataagta tcgcacctgg ggagtacgat cgcaaggttg aaactcaaag 900
gaattgacgg gggccgcgac aagcgggtga ttatgtggtt taattcgaag caacgcgaag 960
aaccttacca gggcttgaca tctgtctaac gaagtagaga tacattaggt gcccttcggg 1020
gaaagtagag acaggtgggt catggtgtgc gtcagctcgt gtcgtgagat gttgggttaa 1080
gtcccgaac gagcgcaacc cctattgtta gttgctacgc aagagcactc tagcgagact 1140
gccgttgaca aaacggagga aggcggggac gacgtcaaat catcatgccc cttatgtcct 1200
gggtacaca cgtaatacaa tggcggttaa caaagggatg caaagccgcg aggcagagcg 1260
aaccceaaaa agcgtccca gttcggatcg caggctgcaa cccgcctgcg tgaagtcgga 1320
atcgctagta atcgcgatc agcatgccgc ggtgaatacg ttcccggggtc ttgtacacac 1380
cgcccgtcac accatgagag tcgggaacac ccgaagtcag tagcctaacc gcaaggaggg 1440
cgcgccgcaa ggtgggttcg ataattgggg tgaagtcgta acaaggtagc cgt 1493

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<210> SEQ ID NO 59
<211> LENGTH: 1511
<212> TYPE: DNA

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<213> ORGANISM: Clostridium leptum
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1)..(1511)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
strain 39

<400> SEQUENCE: 59

agagtttgat cctggctcag gacgaacgct ggcggcgtgc ttaacacatg caagtcgaac    60
ggagcacccc tgaaggagtt ttcggacaac ggatgggaat gcttagtggc ggactggtga    120
gtaacgcgtg aggaacctgc cttccagagg gggacaacag ttggaaacga ctgctaatac    180
cgcctgatgc gttggagccg catgactccg acgtcaaaga tttatcgctg gaagatggcc    240
tcgcgtctga ttagctagtt ggtgaggtaa cggcccacca aggcgacgat cagtagccgg    300
actgagaggt tggcgggcca cattgggact gagatacggc ccagactcct acgggaggca    360
gcagtgggga atattgggca atggacgcaa gtctgaccca gcaacgccgc gtgaaggaa    420
aaggctttcg ggtgtgaaac ttcttttaag ggggaagagc agaagacggt accccttgaa    480
taagccacgg ctaactacgt gccagcagcc gcggtaatat gtaggtggca agcgtgtgcc    540
ggatttactg ggtgtaaagg gcgtgcagcc ggagagacaa gtcagatgtg aaatccacgg    600
gtcacaaccc tgaactgcat ttgaaactgt ttcccttgag tgctggagag gtaatcgtaa    660
ttccttgtgt agcggtgaaa tgcgtagata taaggaagac accagtggcg aagcggatta    720
ctggacgata actgacgggt aggcgcgaaa gcgtggggag caaacaggat tagatacctg    780
ggtagtcaac gctgtaaacg atcgatacta ggtggtgcgg gggacttgac ccctgccgt    840
tgccggagtt aacaccaata aagtattcgg caccctgggg agtacgatcg caaaggttga    900
aaactcaaaa gaaatggacg gggggccccc ccccaagcgg gtgggattat gttggtttat    960
ttcgaaagca acgcgaagaa ccctaacagg gcttgacatc ctgctaacga agtagagata   1020
cattaggtgc ccttcgggga aagtagagac aggtggtgca tggttgtcgt cagctcgtgt   1080
cgtgagatgt tgggttaagt ccgcgaacga gcgcaacccc tattgttagt tgctacgcaa   1140
gagcactcta gcgagactgc cgttgacaaa acggagggaag gcggggacga cgtcaaatca   1200
tcatgccctt tatgtcctgg gctacacacg taatacaatg gcggttaaca aagggatgca   1260
aagccgcgag gcagagcgaa ccccaaaaag ccgtcccagt tcggatcgca ggctgcaacc   1320
cgccctgcgt aagtcggaat cgctagtaat ccgggatcag catgcccgcg tgaatacgtt   1380
cccgggcctt gtacacacgg ccgcgtcacac catgagagtc gggaaacccc gaagtcgta   1440
gcctaaccgc aaggagggcg cgccgaagg tgggttcgat aattgggggt aagtcgtaac   1500
aaggtagccg t                                                    1511

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<210> SEQ ID NO 60
<211> LENGTH: 1499
<212> TYPE: DNA
<213> ORGANISM: Clostridium leptum
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1)..(1499)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
strain 40

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<400> SEQUENCE: 60

agagtttgat cctggctcag gataaacgct ggcggcatgc ctaacacatg caagtcgaac    60
ggagcgcctt ggaaggagac ttcggtcaac ggaagaggag gcttagtggc ggacgggtga    120
gtaacgcgtg aggaacctgc ctgagagagg gggataaac accgaaaggt gtgctaatac    180

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cgcataacat atgagagggg catcccttcc atatcaaaga tttattgctt tgagatggcc 240
tcgcgtccaa ttagctagtt ggtgaggtaa cggcccacca aggcgacgat tggtagccgg 300
actgagaggt tgaacggcca cattgggact gagacacggc ccagactcct acgggaggca 360
gcagtgggga atattgcaca atggggggaa ccctgatgca gcaatgccgc gtgaaggatg 420
aagggttttcg gattgtaaac ttcttttcta cgggacgaag aaagtgcagg taccgtaaga 480
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cacaccgccc gtcacaccat gagagtcgga aacacccgaa gcctgtagcc caaccgcaag 1440
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<210> SEQ ID NO 61
<211> LENGTH: 1512
<212> TYPE: DNA
<213> ORGANISM: Clostridium coccoides
<220> FEATURE:
<221> NAME/KEY: rRNA
<222> LOCATION: (1)..(1512)
<223> OTHER INFORMATION: 16S rRNA coding gene sequence of Clostridium
strain 41

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<400> SEQUENCE: 61
agagtttgat cctgggtcag gatgaacgct ggcggcgtgc ttaacacatg caagtcgaa 60
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cggacgggtg agtaacgcgt gggtaacctg ccttgcaactg ggggataaca cttagaaata 180
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cagtagccgg cctgagaggg tgaacggcca cattgggact gagacacggc ccaaactcct 360
acgggaggca gcagtgggga atattggaca atggggggaa ccctgatcca gcgacgccgc 420
gtgagtgaag aagtatttcg gtatgtaaag ctctatcagc agggaagaaa gaaatgacgg 480
tacctgacta agaagccccg gctaactacg tgcacgacgc cgcggtaata cgtagggggc 540
aagcgttatc cggatttact ggggtgtaaag ggagcgtaga cggcgatgca agtctgaagt 600
gaaaggcggg ggcccaaccc ccggactgct ttgaaaactg tatggctgga gtgcaggaga 660

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-continued

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ggtaagtgga attcctagtg tagcggtgaa atgcgtagat attaggagga acaccagtgg 720
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gattagatac ctggtagtagc cgccgtaaac gatgatcacc ggtttcggtg gggtatggac 840
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ccgggtcttg tacacacgcg ccgtcacacc atgggagtcg gaaatgcccg aagtctgtga 1440
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caaggtagcc gt 1512

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<210> SEQ ID NO 62
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<400> SEQUENCE: 62

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<210> SEQ ID NO 63
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<400> SEQUENCE: 63

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tacggctacc ttgttacgac tt 22

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<210> SEQ ID NO 64
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<400> SEQUENCE: 64

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aaatgacggt acctgactaa 20

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<223> OTHER INFORMATION: Artificially synthesized primer sequence

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<400> SEQUENCE: 65

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ctttgagttt cattcttgcg aa 22

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-continued

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<210> SEQ ID NO 66
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<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: Artificially synthesized primer sequence

<400> SEQUENCE: 66

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<210> SEQ ID NO 67
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: Artificially synthesized primer sequence

<400> SEQUENCE: 67

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<210> SEQ ID NO 68
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<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: Artificially synthesized primer sequence

<400> SEQUENCE: 68

gagaggaagg tccccac                                     18

<210> SEQ ID NO 69
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: Artificially synthesized primer sequence

<400> SEQUENCE: 69

cgctacttgg ctggttcag                                   19

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The invention claimed is:

1. A pharmaceutical composition, comprising six or more live bacterial strains belonging to *Clostridium* clusters IV and/or XIVa, wherein the six or more bacterial strains induce proliferation and/or accumulation of regulatory T cells, wherein at least one of the bacterial strains is a human commensal bacterial strain, and wherein the pharmaceutical composition is formulated for delivery to the intestine.
2. The pharmaceutical composition of claim 1, wherein the pharmaceutical composition comprises seven or more bacterial strains.
3. The pharmaceutical composition of claim 1, wherein the pharmaceutical composition comprises eight or more bacterial strains.
4. The pharmaceutical composition of claim 1, wherein the pharmaceutical composition comprises nine or more bacterial strains.
5. The pharmaceutical composition of claim 1, wherein the pharmaceutical composition comprises ten or more bacterial strains.
6. The pharmaceutical composition of claim 1, wherein the pharmaceutical composition comprises eleven or more bacterial strains.
7. The pharmaceutical composition of claim 1, wherein the pharmaceutical composition comprises twelve or more bacterial strains.
8. The pharmaceutical composition of claim 1, wherein the pharmaceutical composition comprises thirteen or more bacterial strains.
9. The pharmaceutical composition of claim 1, wherein the pharmaceutical composition comprises fourteen or more bacterial strains.
10. The pharmaceutical composition of claim 1, wherein the pharmaceutical composition comprises fifteen or more bacterial strains.
11. The pharmaceutical composition of claim 1, wherein the pharmaceutical composition further comprises one or more bacterial strains belonging to a *Clostridium* cluster other than *Clostridium* cluster IV or cluster XIVa.
12. The pharmaceutical composition of claim 1, wherein the pharmaceutical composition is formulated for oral administration.
13. The pharmaceutical composition of claim 1, wherein the pharmaceutical composition further comprises a pharmacologically acceptable excipient.
14. The pharmaceutical composition of claim 1, wherein the pharmaceutical composition is in the form of a capsule.
15. The pharmaceutical composition of claim 1, wherein the pharmaceutical composition further comprises a pH sensitive composition comprising one or more enteric polymers.

16. A method of treating a human subject having an infectious disease, an autoimmune disease or an allergic disease, the method comprising administering the pharmaceutical composition of claim 1.

17. The method of claim 16, wherein the human subject 5 has an autoimmune disease.

18. The method of claim 16, wherein the autoimmune disease is organ transplant rejection, inflammatory bowel disease (IBD), ulcerative colitis, Crohn's disease, sprue, rheumatoid arthritis, Type 1 diabetes, graft versus host 10 disease, or multiple sclerosis.

19. The method of claim 16, wherein the subject has an infectious disease, and wherein the infectious disease is *Clostridium difficile* infection.

20. The method of claim 16, wherein the subject has an 15 allergic disease, optionally wherein the allergic disease is food allergy.

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